

The Swords of Armageddon

U.S. Nuclear Weapons Development since 1945

VOLUME I

Chuck Hansen, ed.

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Series Overview

WHAT IS THE SWORDS OF ARMAGEDDON?

Welcome to the Swords of Armageddon, the world's first updatable nuclear weapons information service. You are embarking on a voyage of discovery through one of the "blackest" military weapons programs ever instituted by the United States.

The Swords of Armageddon: U.S. Nuclear Weapons Development since 1945 is the first and only comprehensive illustrated unclassified history of the post-World War II design, development, and testing of U.S. nuclear weaponry.

This unique 2,500 page CD-ROM and microfiche document will be updated periodically with new information.

Contents include:

- A complete technical glossary
- Description of weapons physics
- Postwar technological innovations in fission weapon design
- A history of American thermonuclear weaponry between 1942 and 1957
- Individual nuclear warhead histories, including gravity bombs, artillery shells, missile and rocket warheads, anti-submarine weapons, and atomic demolition munitions
- A history and description of warhead arming and fuzing techniques and equipment, including radar and barometric fuses, manual and automatic inflight insertion equipment, and Permissive Action Links
- Three detailed appendices summarizing the objectives and results of U.S. nuclear tests between 1945 and 1962; warhead specifications; and typical nuclear weapons accidents between 1950 and 1981

HOW & WHY THIS SERIES CAME TO EXIST

Since 1945, the U.S. nuclear weapons program has included 60,000 warheads of 71 different types for 116 separate weapons systems. Another 29 designs were canceled before they reached production. By the end of 1992, the U.S. had detonated more than 1,000 nuclear explosions on the surface of the earth, underground, underwater, in the atmosphere, and in space over and under the Pacific and Atlantic Oceans, and in several states in the continental U.S.

The cost of this program has been staggering: over \$89 billion in development costs and \$700 billion for delivery systems. These nuclear warheads were fabricated by a vast administrative and materials production complex that spans the west coast from Washington to California and which ranges east across the country to the District of Columbia and south to Florida. Many thousands of persons have been and are employed by this network.

In spite of the massive size and scope of this weapons development effort, its products and activities have remained largely out of the public consciousness. This has been even more true since the Limited Test Ban Treaty of 1963 which sent U.S. nuclear testing and, until just recently, most protest underground.

Secrecy about the American nuclear weapons program, while necessary up to a point, has brought many unpleasant side affects. War plans employing hundreds of nuclear bombs were drawn up by the Joint Chiefs of Staff soon after World War II at a time when the total U.S. nuclear stockpile did not contain more than a few dozen weapons. The JCS made their plans totally ignorant of the most closely-held secret in the country. Only recently, nearly 50 years later, has the Department of Energy begun to declassify and release stockpile sizes after 1948.

The country's readiness for war was hindered by nuclear secrecy: obtaining and reviewing secret and top secret clearances for employees, consultants and armed forces personnel has been a major bottleneck since the Manhattan Project of World War II. At one time soon after the end of World War II, Air Force Strategic Air Command aircrews were not cleared to even see the bombs they would be dropping, let alone service, handle, or arm them.

Aircraft manufacturers were given only the most general estimates of weights and volumes for the weapons their yet-to-be-built aircraft would carry. In his book, Ed Heinemann, retired chief designer for Douglas Aircraft, relates how in 1949 he was visited by agents of the Federal Bureau of Investigation after he speculated about an atomic bomb that was considerably smaller than the five foot diameter, fifteen foot long, 10,000 lb. weapon specified for carriage by the new A3D Skywarrior, then in design.

This obsession with nuclear secrecy has bred a gross public ignorance about the origins and effects of nuclear weapons, to the point that in 1979 a sizable part of the American public was fooled into believing that a multistage hydrogen bomb could be built solely from information in a 10-page long political magazine article. To most people, nuclear weapons

are little more than an abstract concept and difficult to imagine as physical hardware.

The government has always gone to extreme lengths to maintain this monopoly of information. In 1950, copies of an issue of SCIENTIFIC AMERICAN containing an article about the H-bomb by Dr. Hans Bethe were destroyed and printing plates smashed after the Atomic Energy Commission objected to a few words of text in the article. In 1967, the AEC attempted unsuccessfully to gain dominion over all privately-funded research into laser-driven fusion.

After the Department of Energy was created in 1977 and James Schlesinger was named its director, there were instances in three consecutive years when DOE suppressed or tried to suppress articles and papers by college students and a free-lance writer about hypothetical nuclear weapons designs. In these cases, the government asserted sole rights to public-domain information, even though most of it had been previously declassified and released by the Atomic Energy Commission, the Energy Research and Development Administration, and DOE itself.

Ironically, concepts at issue in 1979 were unclassified in the Soviet Union, but remained classified here until late in 1980. Before the PROGRESSIVE case, one DOE spokesman boasted that inquisitive journalists always ceased their questioning when the cover of classification was invoked.

Another major effect of secrecy has been the total absence of a comprehensive unclassified technical history of the U.S. postwar nuclear weapons program. A recent three-volume AEC history is a largely administrative account that was subjected to severe censorship about the postwar weapons program.

To fill this gap, I decided to write an unclassified technical history, in the style of many books about the famous wartime Manhattan Project.

In 1971, I began serious research. Four years later, I had written a 100-page monograph that summed up most of what I had learned; a portion of it appeared in 1976 in REPLICA IN SCALE, a now-defunct Texas-based aeromodeling magazine. In 1981, I signed a contract with AEROFAX, Inc. in Dallas to produce a major work on the subject. The results were published in 1988 by Orion Books in New York as U.S. Nuclear Weapons: The Secret History. It was the only book of its type in the U.S. and quite possibly in the world.

Since 1988, I have been extending and updating the manuscript upon which the book was based. The results to date have included almost a four-fold expansion in text volume, and a

near doubling of source photos. It became apparent by 1994 that new media would be required to not only hold this massive amount of information and images, but also to allow regular updates at a reasonable cost. Hence the use of a CD-ROM and microfiche for this service. The CD-ROM version includes all text and illustrations; the fiche includes text and tables only.

VOLUMES NOW AVAILABLE

The information in Swords of Armageddon is currently divided into eight volumes. Significant advances in postwar U.S. nuclear weapons described in these volumes are listed below.

Improvements in Fission Weapons

Boosting	Linear Implosion
Composition	Non-lensed HE ignition systems
Levitation	External initiators
Penetrating weapons	Sealed-pit weapons
Hollow implosion	Tailored effect weapons
Improved high explosives (HE)	Lightweight primaries
Interchangeability	U-235 implosion
Improved reflectors & tampers	Application of U-233

Improvements in Fusion Weapons

Clean weapons	Lightweight high-yield weapons
Dirty weapons	Radiation implosion
Enhanced radiation weapons	Solid-fueled weapons
Improved secondary designs	Three-stage weapons

Unattainable ideas

Classical Super	Small clean tactical weapons
Pure-fusion devices	Uranium hydride implosion

Volume I includes the front matter and a description of fission and fusion weapon physics, fuels, and weapon design and detonation principles.

Volume II describes the technology of postwar U.S. fission weapons development, comprising the application of specific materials and design techniques. These include improvements in high explosives; levitation and composition; the use of uranium hydride and uranium-233; hollow cores; interchangeability; linear implosion; tamper and reflector improvements; the development of internal and external neutron sources (initiators); and the development of boosted pits and primaries.

Volumes III, IV, and V describe the development and testing of thermonuclear weapons between 1942 and the fall of 1957. These sections are written chronologically, and discuss

several topics in parallel, including design advances in "clean" and dirty weapons; weapons testing; the debate over sharing of information with U.S. allies; and interface problems between weapons and their delivery vehicles.

The first two parts (**Vols. III & IV**) of this U.S. hydrogen bomb history, covering the period between 1942 and mid-1954, describe the search for feasibility, and the development and tests of the first U.S. liquid-fueled and solid-fueled devices. Some of the highlights of these sections include:

- The development of fission and fusion weapons by the Los Alamos Scientific Laboratory
- The development of deuterium, lithium, and tritium production facilities
- The genesis of "dirty" and high-yield thermonuclear weapons
- The origins and early program of the Livermore laboratory, including the controversy surrounding its establishment, and later efforts by LASL to gain control of the new lab
- Early Department of Defense war plans for combat uses of thermonuclear weapons
- The on-again, off-again plans for sharing of weapons information between the U.S. and Great Britain
- Soviet reconnaissance of early U.S. Pacific tests, and efforts made by U.S. forces to thwart them
- The effects of overclassification on the thermonuclear weapons program
- Objectives, conduct, and results of early U.S. Nevada and Pacific nuclear tests, including RANGER, TUMBLER-SNAPPER, GREENHOUSE, IVY, UPSHOT-KNOTHOLE, and CASTLE, including details of the selection of test devices for GREENHOUSE and CASTLE
- The repeated frustration of Atomic Energy Commission efforts to conceal results of the Pacific tests from the American public and the Soviets
- Major decisions in the H-bomb program, including the October 1949 General Advisory Committee meeting and Truman's January 1950 directive
- How steadily-increasing tritium requirements finally doomed the "classical Super"

- The symbiosis of the Teller-Ulam H-bomb design concepts
- The effects of the Korean War on Operations GREENHOUSE and IVY
- The Department of Defense's "emergency capability" thermonuclear weapons deployment
- Problems surrounding the mating of as-yet-undisgned weapons to delivery systems
- Attacks by the Luce publishing empire on LASL and J. R. Oppenheimer
- What the AEC was telling Congress, the public, and the President about the weapons development and testing programs

Volume V, the history of U.S. thermonuclear weapons between 1954 and late 1957, includes detailed descriptions of the objectives, conduct, and results of Operation REDWING, including a lengthy exposition on the development of so-called "clean" weapons.

Volumes VI and VII comprise individual warhead histories, including aircraft-delivered gravity bombs; nuclear artillery shells; missile and rocket warheads; anti-submarine weapons; and atomic demolition munitions.

Volume VIII describes nuclear warhead and arming techniques and equipment, including radar and barometric fuses, manual and automatic inflight insertion equipment, and Permissive Action Links.

Also included in the latter volume are three detailed appendices which summarize the objectives and results of U.S. nuclear tests between 1945 and 1962; list warhead specifications; and describe typical nuclear weapons accidents between 1950 and 1981.

PLANS FOR THE FUTURE

Future expansion of this U.S. thermonuclear weapons history will detail Operation HARDTACK Phases I and II; weapons development during the 1958-1961 test moratorium; preparations for a return to atmospheric and underground testing in late 1961; Operation NOUGAT in Nevada; and Operation DOMINIC Phase I in the Pacific in 1962.

This extended history will also contain a description of the continuing attempts to develop "clean" and high-yield bombs, including an account of what I call the "great clean weapons fraud," wherein the Atomic Energy Commission deliberately misled the President, Congress, and the public about the

state-of-the-art of small "clean" tactical weapons as an excuse to continue nuclear testing in the late 1950s and early 1960s.

As more information becomes available, the thermonuclear weapons history will be extended into post-1962 testing, including the objectives, conduct, and results of early U.S. underground weapons development tests. Ultimately this hydrogen bomb history will be extended into the 1970s and 1980s.

Another forthcoming section will detail the history of delivery methods and equipment for airborne gravity bombs between 1944 and the present.

Future changes and growth are somewhat dependent upon declassification actions by various U.S. government agencies, and upon the number and quality of new books, articles, theses, reports, and other secondary sources which deal with subjects of interest. Since 1987, I have been reviewing declassified documents at a rate of between 10,000 and 50,000 pages per year. My present documentation archives and library now hold enough information for this service to be maintained for the indefinite future.

SELECTED SOURCE DOCUMENTS AVAILABLE

Selected source documents cited in this series of volumes are available. Materials referenced in footnotes range in size from a single page to series of volumes several thousand pages in length. Please include the volume and footnote number, and complete title, author, addressee, and date of the document. Prices of documents include costs of reproduction, labor, and shipping. For cost estimates of reproducing either partial or complete sources, write to me at 663 S. Bernardo Avenue, Ste. 226, Sunnyvale, CA 94087, USA, or fax requests to 1-408-730-0148, or send e-mail to hansenc@attmail.com or sobernardo@aol.com.

I value your comments about the current and future content and direction of this information service. If you have questions, comments, criticisms, changes of address, etc., fax them to 1-408-730-0148, or send e-mail to hansenc@attmail.com or sobernardo@aol.com, or write to the Swords of Armageddon Information Service, 663 S. Bernardo Avenue, Ste. 226, Sunnyvale, CA 94087, USA.

You will automatically be informed of release dates of forthcoming updates and the scope of additional information to be provided by those updates.

Chuck Hansen
Editor

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PREFACE (1987)

To whom you tell your secrets,
to him you resign your liberty.
-- Spanish proverb

How can we expect another to keep our secret
if we have been unable to keep it ourself?
-- La Rochefoucauld,
Maxims (1665)

There are no secrets except the secrets that keep
themselves.
-- George Bernard Shaw,
Back to Methuselah (1921)

If used in numbers, atomic bombs not
only can nullify any nation's military
effort, but can demolish its social and
economic structure and prevent their
re-establishment for long periods of
time. With such weapons, especially if
employed in conjunction with other
weapons of mass destruction as, for
example, pathogenic bacteria, it is
quite possible to depopulate vast areas
of the earth's surface, leaving only
vestigial remnants of man's material
works.

-- from "The Evaluation of the
Atomic Bomb as A Military
Weapon," Final Report of the
Joint Chiefs of Staff
Evaluation Board for
Operation CROSSROADS, 30
June 1947

The Manhattan District bore no relation
to the industrial or social life of our
country; it was a separate state, with
its own airplanes and its own factories
and its thousands of secrets. It had a
peculiar sovereignty, one that could
bring about the end, peacefully or
violently, of all other sovereignties.

-- Herbert S. Marks, quoted in
THE MAKING OF THE ATOMIC
BOMB, Richard Rhodes, Simon
& Schuster, 1986.

Since the end of World War II, a vast empire has arisen
largely unnoticed in the United States. Conceived in secrecy

during the war, its scope and products have remained beyond the public consciousness, except when its exploits or blunders have brought it widespread national or international attention.

This secret empire has cost taxpayers dearly: \$89 billion in development costs since 1940, and \$700 billion for delivery systems for its products.¹ The sheer volume and number of these products is mind-boggling: between 1945 and 1986, the nuclear weapons production complex in the U.S. manufactured approximately 60,000 warheads of 71 types for 116 different weapons systems. Of these warheads, 29 types remain in the current inventory.

Since 1945, the U.S. Army has deployed 21 types of nuclear warheads; the U.S. Navy/Marine Corps, 34; and the U.S. Air Force, 43. Another 29 "candidate" warhead types were canceled before reaching production, and an unknown number of other warhead designs have never progressed beyond paper studies. By the end of 1988, the U.S. had detonated more than 930 nuclear devices and weapons on the surface of the earth, underground, underwater, in the atmosphere, and in space during tests in, over, and under the Pacific and Atlantic Oceans, and in several states in the continental U.S.²

The U.S. government has always gone to extreme lengths to keep this orgy of nuclear self-indulgence hidden from public view. Even though the weapons labs, research centers, and production complexes and their artifacts are well-known to the Soviet government, they remain a mystery to most of the citizens of the United States.³

Postwar U.S. nuclear weapons are usually perceived as wondrous and mysterious achievements in the black arts that grew out of the wartime Manhattan Project. How, where and when these warheads were developed; exactly what they are; and how they perform are generally unknown by the average citizen. Literally tens of millions of documents chronicling this vast "black project" remain locked in vaults, well-protected behind a formidable wall of secrecy,

¹NUCLEAR WEAPONS DATABOOK, Volume II: U.S. NUCLEAR WARHEAD PRODUCTION, Thomas B. Cochran, William M. Arkin, Robert S. Norris, and Milton M. Hoenig, Ballinger Publishing Company, Cambridge, Massachusetts, 1986, p. 2.

²NUCLEAR WEAPONS DATABOOK, Vol. II, p. 5. This total includes an estimated 20 to 30 nuclear weapons tests (announced and unannounced) conducted by the U.S. from the beginning of 1986 to mid-1987.

³Soviet "spy" satellites, like their American counterparts, now routinely survey nuclear facilities and test sites.

and hidden in perpetuity by one of the largest permanent classification establishments in the entire U.S. government.

The U.S. Department of Energy's willingness to take absurd steps to protect this status quo was illustrated quite dramatically a few years ago when an obscure political magazine in Madison, Wisconsin attempted to publish an article about the American nuclear weapons complex. The article described the products of the secret empire by means of an illustrated account of the operation and design of a hypothetical thermonuclear weapon.

The U.S. Department of Energy, specifically, James R. Schlesinger, its director at the time, requested the Department of Justice to seek an injunction to prevent republication of a collection of information that had been in the public domain for many years (much of this data had been released by DOE and its predecessor agencies).⁴

A compliant judge was found, and a preliminary injunction against publication was issued quickly. The battle to overturn this injunction lasted for six months in 1979 (the longest prior restraint on publication in the history of the country) until the government, faced with the strong possibility of a precedent-setting unfavorable court ruling, dropped the case in the fall.

I was a key participant in the case: a letter I wrote to a U.S. senator, analyzing the government's misbehavior and probable motives for bringing suit against publication, finally forced an end to the original injunction in Wisconsin and a second injunction against a student newspaper in California. Since 1979, the government has maintained an embarrassed silence about the case.

One of the purposes of this book is to shed more light on the history and products of the secret empire, and to provide at least a partial history and description of some aspects of the U.S. nuclear weapons development and testing programs since the end of World War II. As has been the case with all of this writer's previous articles on this subject, all of the information in this book is republished or derived from unclassified documents (including some very informative government reports newly declassified specifically for this monograph).

⁴The move against THE PROGRESSIVE was the culmination of a three-year long personal campaign by Schlesinger to suppress privately-generated nuclear weapons data and speculation. His first three victims in 1977 and 1978 had not contested his actions.

Extensive footnotes in each chapter cite specific sources for many of the points discussed (footnotes are gathered together at the end of each chapter). All conclusions and opinions are those of the author (except where noted) and have not been reviewed, edited, verified, or approved by any agency of the United States government.

Chuck Hansen
July 1987

Preface (1995)

Some day I hope that the requirements of security will permit the full hydrogen bomb program to be told, with the benefit of official information. When this happens, I fervently hope it will be told with dignity, accuracy, and balance.

---- Statement of Senator Clinton P. Anderson, Member of the Joint Committee on Atomic Energy with reference to THE HYDROGEN BOMB by James R. Shepley and Clay Blair, Jr. published in 1954 by David McKay Co.

When the first edition of U.S. NUCLEAR WEAPONS: THE SECRET HISTORY was published in the spring of 1988, it was the most comprehensive single-volume history ever printed about the postwar technical development of American nuclear weaponry U.S. Based almost entirely on recently-declassified government documents, it quickly became a standard against which subsequent books on its subject were judged. However, despite its voluminous details, it was still incomplete in many respects.

Since 1988, many thousands of formerly Secret and Top Secret reports, personal and interagency correspondence, press releases, memoranda, meeting minutes, histories, congressional testimony and briefings, and telex messages produced by the Atomic Energy Commission, the AEC's General Advisory Committee, the Department of Defense's Defense Nuclear Agency, Military Liaison Committee, Armed Forces Special Weapons Center, the Joint Congressional Committee on Atomic Energy, the National Security Council, Office of the President of the United States, and the nuclear weapons laboratories at Los Alamos and Livermore have been declassified and released to the public.

In addition, many documents used as sources for the first edition have also been recently re-issued in more complete form.

The results of this flood of new material are widespread and dramatic. These primary sources, written when the U.S. nuclear weapons program was in its heyday, represent "history in the raw" and they have been carefully correlated and analyzed before inclusion in this book. Several thousand of these documents have been used as references for The Swords of Armageddon. This edition is as great an improvement over my 1988 book as the book was over anything else about its subject

that was available when it was published. The major changes and differences from the book are as follows:

- Part 1:** More information about nuclear and thermonuclear fuels has been included.
- Part 2:** Additional data on high explosives, including a description of the evolution of multipoint lensless HE spheres, has been incorporated.

Many more details about the postwar refinement of U.S. fission weapons, including new historical and technical information about the development of external fission initiators for sealed pit weapons and primaries, is also included.

- Part 3:** Further technical data has been added about early American thermonuclear weapons between 1951 and 1954, including weights, dimensions, and predicted and actual yields of both devices built and tested and devices that never got off the drawing board.

The important relationship between lithium fuel enrichment and bomb size is explained fully for the first time here.

The origins and early activities of the University of California Radiation Laboratory at Livermore (now the Lawrence Livermore National Laboratory) have been compiled and are included.

Voluminous details of the development of lightweight and "clean" American hydrogen weapons for use as bombs and missile warheads between 1955 and the present have been declassified and incorporated into this section.

Extensive new information on many of the nuclear test series conducted by the U.S. in Nevada and in the Pacific between 1951 and 1962, including Operations RANGER, GREENHOUSE, IVY, UPSHOT-KNOTHOLE, CASTLE, TEAPOT, REDWING, PLUMBBOB, HARDTACK Phase I, NOUGAT, and DOMINIC Phase I supplements previously-published data.

Preliminary information on early U.S. underground nuclear tests conducted after 1962 has also been added.

A history of the development of a monstrous 100-megaton aerial bomb designed by the U.S. in 1965 is presented in this chapter.

Part 4: Additional facts about individual American nuclear warheads, including significant new historical and technical details, have been included.

Newly-declassified information on the development of nuclear warheads for air defense (air-to-air and surface-to-air) missiles has been incorporated.

Heretofore-undisclosed details of post-deployment warhead problems supplements previously-published information on this subject.

New data on the one-point safety tests of sealed-pit weapons conducted during Projects 56, 57, and 58 at the Nevada Test Site has been added.

New details of the weapons programs of the University of California Radiation Laboratory (later the Lawrence Radiation Laboratory and now the Lawrence Livermore National Laboratory) and the Sandia Laboratories at Albuquerque and Livermore have been released and are presented here together for the first time.

Extensive new information about the development of first- and second-generation ICBM and IRBM warheads has been added.

Part 5: Newly-declassified U.S. nuclear weapons fuzing and arming systems data, including a thorough account of the development and deployment of Permissive Action Links, is presented.

Material new to this edition also includes a chapter on delivery methods for aerial bombs and three appendices which summarize the purposes and results of U.S. nuclear weapons development tests between 1945 and 1962; detail in tabular form the physical characteristics of every nuclear device tested during that period and the specifications of all W- and MARK-numbered nuclear warheads deployed by the U.S. since the end of World War II; and list details of many heretofore unreported U.S. nuclear weapons accidents.

New illustrations include previously unpublished photographs and diagrams of U.S. nuclear weapons, test devices, test sites, and delivery vehicles. All new photographic and information sources are noted.

In addition to new data and photographs, fragmented information previously scattered throughout the book and in footnotes has been consolidated wherever possible. Where practical, text previously located in footnotes has been moved into the body of the book to make reading and reference easier. A number of typographical and factual errors which crept into the first edition have been corrected in this volume. The lengthy glossary formerly located in Chapter 1 has been broken out into a separate section.

In 1983, in a special 40th anniversary issue of LOS ALAMOS SCIENCE, a quarterly publication of the famous New Mexico nuclear weapons laboratory, the section on weapon design was subtitled "We've Done a Lot, But We Can't Say Much." Secrecy surrounding U.S. nuclear weaponry was specifically cited as impeding morale and recruiting and causing the creation of a "nuclear priesthood:"

Over the past forty years the task of recruiting talented staff and maintaining a high sense of mission has not always been easy. The strict controls over nuclear secrets continue to be a fact of life. Very little substantive information has been declassified concerning the detailed physics and hydrodynamics of nuclear weapons, much less any of the design principles.

Thus, the scientists and engineers who specialize in nuclear weapons research and development spend their entire careers working within a "closed" technical society with little recognition of their work beyond that of close associates.

This edition (once briefly and humorously entitled "Now It Will be Told") illuminates the work of these tongue-tied scientists and also sheds much more light on a long and expensive but still little-known era of American history.

Additional acknowledgments for assistance and support go to Drs. J. Carson Mark, Herbert York, and Hans Bethe; Priscilla Johnson McMillan; Lee Barker; Ben McCarty, Charles Demos, Gwen Schreiner, and Dave Jackson of DOE/Albuquerque; Mike Binder; Don Brandt; Gregg Herken and Stanley Goldberg of the Smithsonian Institution's National Air and Space Museum; David Downes, John Flyger, and Martin Schneiderman of the law firm of Steptoe & Johnson in Washington, D.C.; Elliot

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Chuck Hansen
Sunnyvale, CA
Fall 1995

FOREWORD

The nuclear age was born in secrecy. That was perfectly natural and understandable. It was wartime, after all, and the scientists and technician assembled in the Manhattan Project were certain that they were engaged in a life and death struggle. They were building a new and terrible superweapon (though most had no inkling of how terrible it would turn out to be) and they were sure that the enemy -- Nazi Germany -- was making a comparable effort. They feared, in fact, that the enemy was ahead in the race to develop atomic weaponry.

As it turned out, the enemy was far behind. By the time atomic weapons were detonated in the summer of 1945 -- first a test explosion in the New Mexico desert, then the bombs dropped on Hiroshima and Nagasaki -- Germany had surrendered and total U.S. victory was in sight. The atom bomb was no longer the desperate recourse of an embattled nation fighting to preserve the values of Western democracy; it had become something else.

President Truman, upon returning from the Potsdam conference aboard the USS AUGUSTA, was informed of the bombing of Hiroshima and exclaimed, "This is the greatest thing in history." What made it so great, in his mind and the minds of other policymakers, was that the United States now had a monopoly on the most frightful weapon ever devised by human ingenuity.

Armed with that weapon, this nation would be able to fashion a postwar world to its liking. In particular, the other great superpower to emerge triumphant from World War II -- the Soviet Union -- would be held in check by America's nuclear monopoly. It was generally agreed that it would take many years -- perhaps decades -- for the Soviets to develop nuclear weapons of their own.

The new role of nuclear arms -- to maintain and extend American hegemony around the world -- meant that the wartime secrecy that had surrounded the Manhattan Project would have to be sustained in peacetime as well. In fact, there could be no more peacetime: We were entering the era of the Cold War, and wartime secrecy would apply -- especially in the atomic realm.

This was not the wish of most of the atomic scientists. Albert Einstein, whose letter to President Roosevelt warning of a German nuclear effort had launched the Manhattan Project, wrote on January 22, 1947:

Through the release of atomic energy, our generation has brought into the world the most revolutionary force since prehistoric man's discovery of fire. This basic force of the universe cannot be fitted into the outmoded concept of narrow nationalisms. For there is no secret and there is no defense; there is no possibility of control except through the aroused understanding and insistence of the peoples of the world. We scientists recognize our inescapable responsibility to carry to our fellow citizens an understanding of the simple facts of atomic energy and its implications for society. In this lies our only security and our only hope -- we believe that an informed citizenry will act for life and not death.

Many joined Einstein in urging the broadest possible public understanding of nuclear technology. Only an informed people, they reasoned, would be able to engage in democratic decision-making on nuclear policy. Henry D. Smyth, the Princeton University physicist who compiled the government's official report on the Manhattan Project, wrote:

Here is a new tool for mankind, a tool of unimaginable destructive power. Its development raises many questions that must be answered in the near future. These questions are not technical questions; they are political and social questions, and the answers given to them may affect all mankind for generations. In a free country like ours, such questions should be debated by the people.

But other views prevailed. In the highest reaches of Truman administration and in the foreign-policy establishment that was beginning to take control of U.S. affairs around the world, democratic decision-making about nuclear weaponry was considered "visionary" and "impractical." Atomic arms were to be screened from public understanding, wrapped in utmost secrecy, and entrusted to the care of a new elite -- a nuclear priesthood that had a monopoly on information and would therefore exercise a monopoly on decision-making.

In effect, the nuclear age was to be exempted from the normal workings of the democratic process. A new law -- the Atomic Energy Act of 1947 -- codified this decision by adopting the most draconian secrecy provisions ever enacted by Congress: All information about nuclear matters, whether or not it originated within the government, was declared to be "restricted data" subject to government control.

It is difficult to calculate the full consequences of nuclear secrecy on American society in the last half of the 20th century. The impact has been sweeping and profound, reaching into areas far removed from nuclear weaponry. When the Soviet Union stunned the world by developing its own nuclear capability years sooner than had been anticipated, it was easy to foster the notion that spies were to blame; someone had given our secrets to the Russians. And with the spectacular spy scares of the late 1940s and early 1950s, the protection of nuclear secrets took on a truly sinister form: It became the rationale for political persecution and repression in the United States.

Atomic secrecy was, after all, the ostensible reason for the compilation of lists of "subversives" and the imposition of loyalty oaths among government employees and many others. Any letter carrier, any school teacher, any file clerk in a Social Security office, might be an atomic spy funneling our precious secrets to the Russians. Nuclear secrecy was at the root of hysteria that swept the nation. Schools and colleges were purged, scientists were hounded, and ordinary citizens had their lives disrupted when their neighbors reported them to the authorities for harboring "suspicious" views. Better to be safe than sorry when atomic spies were in our midst.

Even these, however, were not the most serious consequences of nuclear secrecy. Something even more sinister emerged in the atomic age: the notion that the world was so complex and incomprehensible that our government's most important decisions had to be left to the experts --not just in matters of atomic arms but in all matters affecting what the government was pleased to call "national security." For the first time in American history, the government said to the people, "Trust us. We can't tell you what we know, but if we could tell you, you'd agree that we're doing the right thing." It was that kind of rhetoric that led the United States into the quagmire of Vietnam and into countless other misadventures around the world, some of which have not yet come to public notice.

Atomic secrecy became the model and rationale for pervasive secrecy in all aspects and branches of government -- especially those involving so-called "national security." The circle of decision-makers has grown smaller and smaller -- to the point where, if testimony in the Iran/contra arms scandal can be believed, even the President has at times been excluded. The most crucial decisions on matters of public policy are now often made by a small cabal accountable to no one.

What makes this all the more remarkable - and depressing -- is that it is based on a hoax. The secret that lies at the core of our policy of nuclear secrecy is that there are no

secrets -- none, at least, that are of more than minor technical significance. Nuclear secrecy is a fraud. There is no better witness to this reality than D. Edward Teller, often described as "the father of the H-bomb" and now the principal promoter of President Reagan's Strategic Defense Initiative. In Teller's judgment, all classified information should be declassified within six months -- the maximum time that any scientific or technical information can be kept secret.

Teller once served on a nine-member pentagon "Task Force on Secrecy" which concluded:

With respect to technical information, it is understandable that our society would turn to secrecy in an attempt to optimize the advantage to national security that may be gained from new discoveries or innovations associated with science and engineering. However, it must be recognized, first, that certain kinds of technical information are easily discovered independently, or regenerated, once a reasonably sophisticated group decides it is worthwhile to do so.

In spite of elaborate and very costly measures taken independently by the U.S. and the U.S.S.R. to preserve technical secrecy, neither the United Kingdom nor China was long delayed in developing hydrogen weapons. Also, classification of technical information impedes its flow within our own system, and may easily do far more harm than good by stifling critical discussion and review or by engendering frustration. There are many cases in which the declassification of technical information within our system probably had a beneficial effect and its classification has had a deleterious one.

Such serious and well-reasoned critiques of our system of secrecy have had no impact on at all on official practice: In fact, government secrecy grows constantly more burdensome. What's worse, most Americans seem to have bought whole-hog the notion that there are many things they are simply better off not knowing. I've had the experience more than once, when talking about specifics of the nuclear arms race, of having a listener extend an arm as if to fend me off, while saying, "Don't tell me about that. I don't want to know." And so secrecy thrives while our democracy founders.

Fortunately, the author of this book is a different kind of American -- one who wants to know. The very idea of secrecy offends Chuck Hansen, and as this book testifies, when he's

offended he does something about it. His wholesome mistrust of authority, his insistence on knowing the facts so that he can make up his own mind, makes him the kind of citizen the founders of this republic had in mind when they embraced the assumption of popular self-government.

I don't know enough about the technology of nuclear arms to judge whether Chuck Hansen has all his facts straight. But I know he's made a conscientious effort to tell the truth -- all of it. And that's more than I can say for any member of our government's nuclear priesthood. Because he made that effort, we're all in Chuck Hansen's debt.

Erwin Knoll
Madison, Wisconsin
August 1987

(Erwin was editor of The Progressive magazine between 1973 and 1994; he passed away on November 2, 1994 at the age of 63.)

A NOTE ABOUT SOURCES

To prepare this series of volumes, I have drawn upon newly-released material whenever possible. The watchword of my work has been the credibility of my sources. Perhaps the best counsel I ever received concerning sources came to me in March 1971, at the very beginning of my research, when R. B. Jewell, then Vice President and Chief Engineer of the Mason & Hanger - Silas Mason Company, an AEC contractor, advised me "that any unclassified information with regard to (nuclear devices) which may have occurred in the press or in periodicals is highly unreliable and sometimes conjectural." As the years went by, I came to appreciate the truth in his words: there is very little about nuclear weapons and testing in most publications that is accurate.

For this reason more than any other, I have used official U.S. government documents as primary references. These include many documents declassified for me since 1981 following Freedom of Information requests to several branches and subcontractors of the U.S. Department of Energy; divisions of the U.S. Air Force including Aerospace Defense Command, Systems Command, Strategic Air Command, Tactical Air Command, Logistics Command, and the Office of USAF History; the U.S. Navy's Naval Air Historian at Naval Air Systems Command; and the Defense Nuclear Agency. More than 500 official USDOE, USAF, USN and DNA reports -- never before made available in unclassified form to a historian -- were sources for this book.

This history relies almost exclusively on contemporary primary source documents. In very few instances have later personal comments from participants or principal characters been included. After-the-fact recollections, especially those remembered 40 to 50 years later, are generally not very reliable. Time alters memories and tends to warp them to conform to current beliefs.

Subsequent knowledge and experience mix with and become inseparable from remembrances of earlier events. In addition, in some cases, those who participated in the episodes described in this book cannot recall them at all or are deceased. Mainly for these reasons, this history is primarily documentary rather than anecdotal.

For these reasons, I have chosen to use reports, letters, and memoranda written during the time that events were occurring, unless the source bears an unmistakable or pronounced bias to a particular point of view that distorts its accuracy. Since there were many controversies associated with the postwar development of U.S. nuclear weaponry, ranging in significance from national policy issues (such as whether or not to build thermonuclear weapons) down to military debate over optimal

weapon sizes and mixes in the national stockpile, advocacy documents are unavoidable. In the very few places where such material was a sole source, the prejudicial nature of the commentary has been noted.

Most of the documents used as sources for this information service were originally classified Secret Restricted Data; many were Top Secret Restricted Data at the time of their creation. The apparent accessibility of information in this information service can be very misleading: almost all of it was very tightly held in limited circulation at the time of its creation and was not widely known beyond the highest echelons of the U.S. government. Most of the time, even the President of the United States did not know all there was to know about the nuclear weapons program, including, on occasion, even how many weapons were stockpiled. The U.S. military services often made war plans while entirely ignorant of how many warheads and bombs were available at the time.

Forty years ago, even simple letter-number warhead designations were secret. Now, almost two generations later, the U.S. government still retains much information currently classified as Formerly Restricted Data, which it apparently intends to withhold for eternity. While some of this secrecy can be legitimately defended on grounds of nuclear nonproliferation, much of it is downright silly and unnecessary.

The only apparent justification for most remaining secrecy is the "we've always done it that way" mentality and resistance to change shown by those in the U.S. government charged with maintaining the security of the information (to say nothing of the vested interests of career classification officers in holding onto their jobs). As can be seen from this information service, much of this "secret" data is in fact easily uncovered.

By far, in terms of volume alone, the major source for this book was the Reynolds Electrical and Engineering Company's Coordination and Information Center (CIC) in Las Vegas, Nevada. Between 1987 and 1995, I reviewed more than 15,000 documents from the REECO collection, ranging in length from a single pages to several hundred pages.

These documents comprise a rich source of contemporary primary documentation on the U.S. nuclear weapons program between 1943 and the early 1970s. However, they are often so heavily censored and rarely include vast compendia of data, so "some assembly is required" in terms of analysis and correlation with other sources.

By the end of 1995, the CIC archives contained more than a quarter million documents, most relating to atmospheric

testing in Nevada and the Pacific between 1945 and 1962, and were growing at the rate of 20,000 documents annually.

Several hundred more declassified documents were provided by the Defense Nuclear Agency's Nuclear Test Personnel Review (NTPR) program, which has resulted in the issuance of 45 volumes detailing atmospheric and underground nuclear testing by the U.S. since 1945. Many of the source documents used to prepare these reports are now unclassified, and several are cited in this book.

I also drew heavily from the Natural Resource Defense Council's NUCLEAR WEAPONS DATABOOK volumes. I was a contributor to a number of titles in this series. The NRDC has also issued many excellent working papers regarding nuclear testing. Some of these are cited here.

The three volumes of the official U.S. Atomic Energy Commission history, as inadequate as they are in their descriptions of weapons development (they are primarily administrative and political records), are nonetheless a valuable source of historical background and a useful catalogue of sources. Most of the documents from the Historian's Office of the U.S. Atomic Energy Commission (now the U.S. Department of Energy) used in the writing of this series have now been at least partially declassified, as well as most of the sources used for a 1955 AEC Secretariat thermonuclear weapons chronology.

Transcripts of U.S. Senate and House Armed Services Committees were another valuable source of technical weapons data. Several of these documents are cited. They are tedious reading, but they are also authoritative.

Court documents -- affidavits, transcripts, and legal briefs -- filed by both sides in the PROGRESSIVE case in 1979 were also a valuable reservoir of technical information, frequently pointing to other sources. Looking through them, I often wondered if the Energy and Justice Departments gave any consideration in February 1979 to how much information about atomic and thermonuclear weapons design might become public during and after the case, especially information far more accurate and specific than Howard Morland's initial speculations. Most of the federal court record was declassified and made available to the public between September 1980 and January 1981.

Literally scores of popular books published between 1945 and 1987, as well as hundreds of magazine and newspaper articles, were also reviewed as possible sources for this book. Although some of the historical data in these publications is accurate, most of the technical data they present is terribly incomplete or incorrect since the massive revelations during the

PROGRESSIVE case (even a number of books about the PROGRESSIVE case are inaccurate and incomplete).

In places where I have cited secondary sources, it is generally because corroborative information is available from other, usually official, sources. Most of the authors of these earlier books and articles suffered from a lack of access to accurate unclassified information, or, if their writers possessed active or inactive Department of Energy or Department of Defense security clearances, they were legally bound not to tell all or any of what they knew. This is one reason this book had to be written by an uncleared author.

In the absence of credible accurate information, erroneous data has propagated quickly and widely from one book to another. This has been especially true of the mythology surrounding discovery of the Teller-Ulam H-bomb design concepts. That mythology is dispelled completely here for the first time: the information was always available. No author before now had ever taken the time or made the analyses necessary to determine the complete story.

Before the PROGRESSIVE case, most of these authors did not even know what the Teller-Ulam principles were. Only since 1980 have some credible and reliable accounts of postwar U.S. nuclear strategy and weapons development been written which finally allow detailed examination of historical issues such as the discovery of radiation implosion in the United States.

Another long-lived historical inaccuracy dispelled by this edition is the notion that Edward Teller was the sole or major discoverer of the so-called "three concepts" behind every successful U.S. thermonuclear weapon. For more than 40 years, Stan Ulam's contributions have been either dismissed, mischaracterized, or minimized.

For far too long, Teller has refused to deal honestly with his collaboration with Ulam. It is, nonetheless, this author's opinion that Teller has rightly gained the sobriquet of "father of the H-bomb:" even if he had not made one technical contribution, his unflagging -- though often hyperbolic and overstated -- enthusiasm and salesmanship have properly earned him his title (and his many enemies).

A primary source for the first edition of this book was an unpublished nuclear weapons monograph I assembled in 1975. A small portion of that document appeared in 1976 in REPLICAS IN SCALE, a now-defunct Texas aeromodelling journal. The skeleton of that monograph survives in the overview sections and chapter headings in this book.

A major disheartening change in circumstances since the publication of U.S. Nuclear Weapons: The Secret History in

1988 and the present has been a distressing reversal in attitude on the part of a number of U.S. government agencies, most notably the U.S. Air Force, towards the Freedom of Information Act. Increasingly, many obstacles, mainly in the forms of undue delays, prohibitive processing fees, and improperly-applied legal decisions, have been thrown in the path of this author and others.

The reasons for this shift were and are not apparent; they may have been linked to a recent series of conservative government administrations that have become hysterically inclined toward paranoiac secrecy. It took 11 years, including two spent in federal court in Washington, D.C., to obtain just one key Air Force study for this history.

In addition, until the ascension of Hazel O'Leary to its directorship, several offices of the U.S. Department of Energy had assumed very intransigent postures toward release of historical data, sometimes bordering on outright hostility. Ironically and inexplicably, at the same time, other DOE offices and those of their contractors continue to pour declassified documents into the Reynolds archives.

Even under the relatively enlightened stewardship of Ms. O'Leary, the DOE is still capable of major silliness. Another source for this information service, originally requested via the Freedom of Information Act in late 1989, was finally released in skeletal form in early 1995, with several chapters concerning some of the most basic and well-known aspects of the design, development, and testing of U.S. atomic and thermonuclear weapons removed in their entirety.

Unlike a number of recent books and graduate student theses, one source which I have avoided religiously is anonymous interviews or alleged government "weapon experts." While it may seem necessary at times to rely upon such sources who will not volunteer information unless they are cloaked with obscurity, I have found it neither prudent nor credible to resort to hidden authorities. I have also had the advantage (as a result of the expenditures of much money, time, and work) to have had access to many papers that other authors had not yet discovered.

Photographs were supplied by a number of sources. The major ones are credited in the photo captions as follows:

AFM	U.S. Air Force Museum, Dayton, Ohio
AWST	AVIATION WEEK & SPACE TECHNOLOGY, New York City
DAVA	Defense Audio-Visual Agency, Washington, D. C.

DNA	Defense Nuclear Agency, Washington, D. C.
LANL	Los Alamos National Laboratory, Los Alamos, New Mexico
LLNL	Lawrence Livermore National Laboratory, Livermore, California
NA	National Archives, General Services Administration, Washington, D. C.
NAM	National Atomic Museum, Albuquerque, New Mexico
NASM	National Air & Space Museum, Washington, D. C.
NHC	Naval Historical Center, Washington, D. C.
NRDC	Natural Resources Defense Council, Washington, D. C.
PANAM	Pan Am-DNA Photo Agency, Las Vegas, Nevada
SAND	Sandia National Laboratories, Albuquerque, New Mexico, and Livermore, California
USA	United States Army
USAF	United States Air Force
USDOE	United States Department of Energy, Washington, D. C. and Las Vegas, Nevada
USN	United States Navy

Original artwork is credited to its creator(s).

In this manuscript, deletions from, or revisions or addition to, quotes from sources follow specific conventions. Material in the source that is present but not quoted for one reason or another is indicated by ellipses (...). Specific deletions are indicated by [deleted]. These deletions have not always been made for "national security" purposes, since I have collected many different versions of several source documents, and deletions present in one version are often not in another.

It is impossible to determine why the implementation of classification guidelines is so non-uniform. In some cases, later reviews of a document have resulted in fewer deletions because of liberalized classification rules; in most cases, many deletions appear to be entirely whimsical.

Author's notes and emphasis are indicated where added or in footnotes; other explanatory or re-phrased text is enclosed in parentheses.

In a few cases, original verbiage has been replaced in quoted text by simpler and clearer English; in isolated cases, additional explanatory or identifying text has been added to clarify an original quotation. A frequent change for the quotation of lengthy passages of original material was the separation of long original paragraphs into several shorter ones. The text in those instances is otherwise unaltered.

ACKNOWLEDGMENTS (1987)

Dr. Edward Teller has written that the development of the American hydrogen bomb was "the work of many people." In a similar fashion, this book is the result of many contributions from many sources. A number of people offered assistance which was, in several cases, absolutely indispensable. Donnie Martin of the U.S. Department of Energy's Albuquerque Operations Office was a big help in getting parts of a number of DOE weapons histories declassified in 1986, as well as many other documents requested by me under the terms of the Freedom of Information Act.

Richard Ray, Linnie Grace, and Carroll Canfield of the National Atomic Museum supplied photographs, information and inspiration during the many years it took to prepare this book. Thomas Mehas, Martha Demarre, and the staff of the Coordination and Information Center at the Las Vegas, Nevada office of the Reynolds Electrical and Engineering Company filled my countless requests for Atomic Energy Commission historical documents and nuclear test histories.

In the Department of Defense, Bill Armstrong, Naval Air Historian with Naval Air Systems Command in Washington, D. C. declassified many aircraft and rocket test and development histories for me. Robert J. Smith, until recently historian for the U.S. Air Force's Logistics Command, provided many unclassified and declassified aircraft and weapon system histories. Sue Ladd and Cheri Abdelnour of the Public Affairs Office of the Defense Nuclear Agency supplied periodic updates of lists of source documents for the DNA's excellent Nuclear Test Personnel Review (NTPR) radiological histories.

Among noted nuclear weapons historians, Barton Bernstein of Stanford University and David Rosenberg of the Naval War College offered moral support and many useful documents. Dave was almost single-handedly responsible for getting one major nuclear weapons history declassified, and I am grateful for his assistance. Tom Cochran, Wayne Nail, and Stan Norris of the Natural Resources Defense Council furnished weapons data, photographs, and advance copies of two volumes of the NUCLEAR WEAPONS DATABOOK.

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Among other contributors, James Rowe allowed reprinting of some of the photos of early atomic weapons from his book about his wartime exploits in Wendover, Utah.

I owe a special debt of gratitude to Erwin Knoll of THE PROGRESSIVE magazine and Dr. Hugh E. DeWitt of the Lawrence Livermore National Laboratory for their guidance and assistance in obtaining court documents during and after the PROGRESSIVE case in 1979. Erwin also very graciously provided the thought-provoking foreword to this book.

Ray Wagner, of the San Diego Aerospace Museum, also indirectly influenced this book. I have closely modeled my work after his comprehensive work, AMERICAN COMBAT PLANES (Doubleday, 1960, 1968, 1982). I hope that U.S. NUCLEAR WEAPONS will someday be as well regarded as a source on its subject as his book is on U.S. military aviation.

Finally, I would especially like to thank three former U.S. Department of Energy officers, James R. Schlesinger, Duane C. Sewell, and John A. Griffin, for the wonderful education they gave me in 1979, and for the excitement they brought me that year. I can state emphatically that this book probably would not have been written without their remarkable intervention in my life.

Chuck Hansen
June 1987

GLOSSARY

Like all technological "priesthoods," the nuclear weapons labs have over the years developed their own terminology of unique terms, as well as those borrowed from biology, automotive engineering, and rocketry. Many of the words below have never appeared in the open literature, due to the closed nature of the labs and their work.

Since the PROGRESSIVE case in 1979, the labs have loosened up a little, and some of these "buzzwords" have turned up in congressional hearing transcripts and in other public documents. Most of these terms will be used throughout the remainder of this series of volumes; many of them relate to weapon systems in general, and not just to their warheads.⁵

Many of the following terms were originally coined during World War II to preserve secrecy and to speak of complex topics in a shorthand terminology. There was also a fraternal reason for the language:

This special jargon arises as a means of abbreviated and efficient communications, specially tailored to the unique requirements of the profession. But it also serves as a form of member identification, similar in nature to the secret handshakes once practiced by fraternities: if one speaks and understands the special cant of the organization, one is an accepted member.⁶

Ablative material The special protective coating on the nose cone of a missile re-entry vehicle (RV) which prevents the RV from burning up during re-entry into the earth's atmosphere.

ABM Acronym for Anti-Ballistic Missile, a weapon designed to destroy intercontinental ballistic missile warheads.

⁵Some definitions are from GLOSSARY OF TERMS, Joint Training Division, Nuclear Training Directorate, Field Command, Defense Atomic Support Agency, Sandia Base, Albuquerque, New Mexico, 15 May 1967.

⁶Review by Lt. Cdr. Thomas J. Cutler, USN (Ret.) of IN THE FIELD: THE LANGUAGE OF THE VIETNAM WAR (Dr. Linda Reinberg, Facts on File, New York, 1991) and WORDS OF THE VIETNAM WAR (Gregory C. Clark, McFarland, Jefferson, North Carolina, 1990) in Naval Institute Proceedings, Vol. 118/7/1073, July 1992, p. 102.

Active material	Fissionable material, such as uranium-235, plutonium-239, or the thorium-derived isotope uranium-233, which is capable of supporting a chain reaction. In military terms, active material refers to the nuclear components of atomic or thermonuclear weapons exclusive of the natural uranium parts.
ADM	Acronym for Atomic Demolition Munition, a nuclear land mine designed to be detonated on or below the surface of the earth, or under water, to block, deny and/or channel enemy forces.
AFAP	Acronym for Artillery-Fired Atomic Projectile, or nuclear artillery shell.
AIFI	Automatic In-flight Insertion. See IFI.
Airburst	A nuclear explosion in the atmosphere. Specifically, a nuclear explosion in the air, at a height greater than the maximum radius of the resultant fireball.
Airdrop	To release or let fall from an aircraft in flight.
Air Zero	The exact center or location of an airburst.
Alarm Clock	Autocatalytic, or self-sustaining. A nuclear assembly configured with thin layers of thermonuclear fuels, alternating with thin oralloy or tuballoy sheets or foils.
Alpha	A measure of the efficiency of a fission explosion. A count (in generations per microsecond) of fission reactions, used to indicate the rate of fission build-up before core disassembly. Alpha is a measurement of the logarithmic rate of rise of a fission reaction; it is essentially a measure of the rate at which neutrons are compounded in a chain reaction. The rate of increase in nuclear reactions before mechanical explosive and fuel depletion effects become significant is an excellent indicator of the early stages of the efficiency of the explosion, and alpha is a an important measurement of the reaction rate increase. It is one of three "classical" diagnostic measurements for fission devices, transit time and yield being the other two. Alpha can also indicate whether a nuclear reaction begins prematurely or is started at the desired time by an initiator. During nuclear tests, alpha was measured by the

time of conversion of gamma rays by an organic fluor to light whose strength and duration could be detected by a photocell and recorded by fast oscilloscopes.

Alpha Decay	Radioactive decay (transmutation from one element to an element lower on the periodic table) caused by emission of alpha particles. Plutonium decays to neptunium via alpha decay.
Alpha emitter	A radionuclide that undergoes transformation by emission of alpha particles.
Alpha Particle	A helium nucleus, composed of two protons and two neutrons. It has a positive electrostatic charge (due to the protons). Alpha particles may be ejected spontaneously from the nuclei of some radioactive elements, such as plutonium.
Arming	As applied to a nuclear weapon, the changing from a safe (unarmed) condition to a state of readiness for detonation. The preparation of a nuclear weapon so that a fuzing signal will operate the firing system.
ASW	Acronym for Anti-Submarine Warfare or Anti-Submarine Weapon (depending on context).
Atom	The smallest part of an element which can enter into a chemical or nuclear combination. An atom is composed of electrons orbiting a nucleus of neutrons and protons.
Atomic Number	The number of protons in the nucleus of an atom of a given element. Often indicated by symbol Z. The atomic number determines the chemical properties of an element.
Atomic Weight	The sum of the number of neutrons and protons in the nucleus of an atom of a given element. Each of these nucleus particles (nucleons) has an approximate atomic mass unit of 1. The atomic weight is also the relative weight of an atom of an element compared to the weight of an oxygen atom with an atomic weight of 16; hence an atom with an atomic weight of 1 has 1/16th the weight of an atom of oxygen.
Attenuation	The process by which radiation is reduced in intensity while passing through some materials, due to absorption or scattering or both.
Ballistic co-	A measure of the efficiency of a projectile in overcoming air resistance. The higher the

efficient	coefficient, the lower the air resistance and the more accurate the projectile is at long ranges. The ballistic coefficient is the property of a projectile best described as its "carrying power" and is closely related to its ratio of weight to diameter. Too little weight for a given diameter and the "carrying power" falls off, leading to low velocity, and for bombs, wide dispersion around the intended target. The ballistic coefficient is calculated from projectile mass, diameter, and "form factor;" the latter is derived empirically by testing. The ballistic coefficient represents a weight-to-drag ratio, and like air pressure, it is measured in lbs. per square foot.
Ballistic missile	Any missile, guided or unguided, which does not rely upon aerodynamic surfaces to provide lift and consequently follows a ballistic trajectory when thrust is terminated. Ballistic missiles may be of either intermediate or intercontinental range.
Ballistic shape	An inert dummy weapon with an external configuration and ballistic characteristics identical to the weapon with which it is associated. The aerodynamic contour of a weapon.
Bare device	A nuclear test device outside an aerodynamic casing, or a nuclear core without a reflector.
Barge	A floating platform used as the support for the cab, or shelter, in which nuclear devices were both prepared for testing and later detonated.
Barge shot	A detonation of a nuclear device aboard a barge.
Barn	A measure of area, equal to 10^{-24} centimeters ² per nucleus. It is used to measure the probability of a specific nuclear reaction occurring, such as neutron capture. See also Cross-section.
Baro	A pressure-sensitive device (essentially a pressure altimeter) used in some nuclear warheads to actuate arming and firing circuits. The term is a contraction of "barometric switch," also referred to as a "baroswitch."
Beryllium	Element number 4, with an atomic number of 4, and atomic weights between 6 and 11. Used in

	nuclear weapons as a neutron reflector and neutron source.
Beta Decay	Radioactive decay via emission of a beta particle (a high velocity electron) from the nucleus of an atom.
Betatron	A device which accelerates electrons by means of magnetic induction.
Bird cage	A container attached to a cagelike structure which maintains a safe separation distance between fissile weapon cores to prevent inadvertent criticality.
Blackbody	An idealized physical body that absorbs all electromagnetic energy falling upon it. At constant temperature, it emits electromagnetic radiation with a distribution or spectrum characterized by its temperature.
Boosted Weapon	A nuclear weapon to which a gas (such as tritium and/or deuterium) or a solid (such as lithium deuteride) has been added to produce significant increase in fissioning and consequent increases in both yield and efficiency.
Boosting	Enhancement of a fission reaction by thermonuclear neutrons. A term coined by Dr. Edward Teller in 1947 to describe a technique of improving the efficiency and explosive yield of fission devices by the injection of small amounts of tritium and deuterium gases into the hollow center of the fissile weapon core before the core is compressed and begins to fission. When the gases are compressed inside the collapsing hollow shell of an implosion system and subsequently heated by fission in the shell of active materials, a fusion reaction occurs in the gas, releasing large quantities of high-energy neutrons. If a large supply of neutrons is suddenly introduced into a fissioning system, while neutron multiplication is still occurring and before appreciable core expansion has taken place, the total number of nuclei fissions, and the resultant efficiency and energy yield, is greatly increased. Also called "gas boosting."
Boron	Element number 5, used as a source of lithium and as a neutron absorber. During the Manhattan Project, crystalline boron of normal composition was known as "Product 89" and boron enriched in the neutron-absorbent boron-10 isotope was known as "Vitamin B."

Burn Efficiency	In a thermonuclear device, the percentage of nuclei that fuse. It is proportional to the product of fuel density and fuel confinement time.
Cab	The shelter which covers a nuclear device being readied for test. A cab may be located on a tower, the earth's surface, or a barge.
Capsule	The fissile core of a fission weapon, or a fusion fuel stage in a thermonuclear weapon. In early atomic weapons, the capsule consisted of the fissionable material and its associated components, which, when inserted into the weapon, warhead, or primary, completed the nuclear arming of the device.
CEP	Acronym for Circular Error Probable, a statistical measure of a hypothetical circle within which at least half of the projectiles targeted on its center will fall. Also called aiming error, this factor is used to determine the probability of damaging a target. In terms of ballistic missile accuracy, CEP is defined as an ellipse whose major and minor axes describe the maximum miss distances along and across a missile's trajectory, respectively, at the target. A CEP can be measured in any distance unit from feet to miles.
Chain Reaction	A self-sustaining nuclear reaction, where the energy and neutrons produced by the fission of the nucleus of an atom cause the fission(s) of other nuclei, which in turn cause other nuclei to fission. This process multiplies at an exponential rate until all the fissile material is expended (converted by fission to energy, subatomic particles, and other elements), or until the explosive assembly physically disintegrates or expands to a point where fission can no longer occur.
Class A, B, C, D, E, F weapon	Terms commonly applied to one of six general classes of thermonuclear weapons (by weight and yield), specified by the Joint Chiefs of Staff for development and deployment. Class "A" weapons are the heaviest and Class "F" weapons are the lightest. These categories encompass warhead weights from 42,000 lb. down to 700 lbs. Class "A" weapons weighed 50,000 lbs. or less, with a minimum yield of 15 to 20 megatons (MT) of TNT-equivalent energy; Class "B" weapons weighed between 23,000 and 15,000 lbs. and

yielded a minimum of 5 to 10 MT; Class "C" weapons weighed 8,500 lbs. or less, with a minimum yield of two to 10 MT; Class "D" weapons weighed 3,000 to 4,000 lbs., with a minimum yield of a quarter megaton to three megatons; Class "E" weapons weighed from 1,000 to 1,500 lbs.; and Class "F" weapons weighed less than 1,000 lbs.

Clean Nuclear Weapon

A nuclear weapon so designed that upon detonation, the amount of contaminating material, in the form of either fission fragments or irradiating material (or both), is relatively low by comparison with other possible design variations of the same Mark-numbered weapon. A weapon may also be "clean" by comparison with different mark-numbered weapons of the same general yield.

CNWDI

Acronym (pronounced "sin-widdy") for Critical Nuclear Weapons Design Information, i.e., that Top Secret Restricted Data or Secret Restricted Data which reveals the theory of operation or design of the components of a thermonuclear or implosion-type fission bomb, warhead, demolition munition, or test device. Specifically excluded from CNWDI is information concerning arming, fuzing, and firing systems; limited-life components; total quantities of high explosive, fissionable, and fusionable materials contained within the weapon or device; and weapon components which are set, maintained, operated, tested, or replaced by military personnel.

Collimate

To align radiant outputs, via the use of baffles, from a nuclear explosion within a specified angle to enhance diagnostic measurements.

Collimator

A device or structure of special materials which confines a beam of particles or radiation within a specified narrow angle.

Confinement time

The period during which a fusion explosion can be contained to burn an appreciable fraction of its fuel. Confinement time can be increased by surrounding the explosion with a heavy material, such as lead or steel.

Contamination

The deposition of radioactive material onto the surfaces of structures, areas, objects, and personnel following a nuclear detonation. This material generally consists of fallout in which

fission products and other device debris have been physically bound up with dust particles, vaporized device platform components, etc. Contamination can also arise from radioactivity induced in certain substances by the bombardment of the substance by neutrons emanating from a nuclear explosion.

Conventional Nuclear Weapon	A nuclear weapon which is designed without special effort to enhance or minimize the fission products or other radioactivity resulting from its detonation.
Core	The fissile center of a fission weapon or thermonuclear primary. May refer to plutonium and/or or alloy components only, or those materials and the initiator. Occasionally called the "globe." The core could usually be divided into two "hemispheres."
Crater	The depression formed in the earth's surface by a nuclear explosion detonated underground or on the surface. Craters may result from subsidence, vaporization of surface material, by throwout of surface and subsurface material, or by the scoring effect of a low-altitude airburst.
Critical Mass	The minimum mass of a fissionable material that will just maintain a fission chain reaction under precisely specified conditions; these conditions include the particular material and its purity, the amount and type of tamper or neutron reflector used, the density or degree of compression of the fissionable material, and its physical shape and geometry. The critical mass of a bare, untamped and unreflected sphere of or alloy (uranium enriched to contain 93.5% or more of the uranium-235 isotope) is 112 lbs. (51 kg). A critical mass is the quantity of fissionable material within which the average number of neutrons remains constant over time. A neutron inserted into it will always exist, either as the original neutron or as a replacement neutron spawned by a fission reaction. At Los Alamos, an early measurement of criticality was the "Christy Crit," which was the amount of fissionable material required to sustain a critical mass when it was immediately surrounded by an effectively infinite tub alloy tamper. A critical mass is also sometimes called a "crit."
Cross Section	A measure of the statistical likelihood (probability) of a given event or reaction, such

as neutron capture, nuclear disintegration, particle repulsion, or element conversion within the nucleus of an atom of a given element when struck by a particle or photon. A measure of the probability of an encounter between particles that will result in a specified effect. Scattering cross section, either elastic or inelastic, depending upon preservation of energy or energy exchange, is the probability of a particle collision resulting in a rebound (repulsion). Capture cross section is the probability of a particle collision resulting in the capture (absorption) of the particle. Fission cross section is the probability of a particle collision resulting in the splitting of a nucleus into two or more fragments. Cross section is measured in barns and is dependent upon the kinetic energy of the impacting particle, such as a neutron.

- Cruise missile** A guided missile, the major portion of whose flight path to its target is conducted at approximately constant velocity, and which depends upon the air for lift and upon propulsion forces to overcome drag.
- Cryogenics** The study of the behavior and properties of materials at low temperatures, usually below -189°F . (-123°C . or 150°Kelvin). Cryogenic equipment is the equipment used to generate extremely low temperatures, usually in liquefied gases.
- Cryogenic bomb fuels** Liquefied deuterium and tritium maintained at extremely low temperatures.
- Delivery System** An aerospace vehicle considered as a whole, with all associated components required to transport, launch, guide, and deliver nuclear weapon(s) to their assigned targets.
- Depletalloy** Uranium enriched from 99.85% to 99.97% in the uranium-238 isotope. Natural uranium consists of about 99.3% U-238 and about 0.7% U-235. Depletalloy is natural U-238 after the U-235 isotope has been almost completely extracted via gaseous diffusion, or centrifugal, laser, or electromagnetic separation. Also known as depleted uranium. Depletalloy is often designated by the symbol D-38, which arose from the fact that at one time it contained 0.38% U-235; greater depletion has since been achieved. Depletalloy was once called Q-metal,

but this name is no longer in common use.

Detonator	As used in atomic weapons, an electro-explosive device designed to start a detonation in response to an applied electric current.
Deuteride	A combining form for compounds containing deuterium, such as lithium deuteride.
Deuterium	A heavy isotope of hydrogen, with an atomic weight of 2, an atomic number of 1, and a nucleus composed of a proton and a neutron. Deuterium is often used as a thermonuclear fuel constituent. In the complete transformation of deuterium to helium, the energy release is about 150 tons per gram, or about 68 kilotons per pound. One cubic meter of gaseous deuterium, when completely fused to itself, yields about 10 megatons of TNT-equivalent energy. One liter of liquid deuterium has the explosive potential of 39 kilotons of TNT. Deuterium is often represented by the symbols D or ${}_1\text{H}^2$.
Deuteron	The nucleus of a deuterium atom, composed of a proton and a neutron.
Device (nuclear)	Nuclear fission and fusion materials, together with their arming, fuzing, firing, chemical high explosive, and effects-measuring components, that have not yet reached the development status of an operational weapon. A system designed to produce a nuclear explosion for purposes of testing the design, verifying nuclear theory, or gathering information on system performance. Many U.S. nuclear devices were (and are) designed for diagnostic purposes and not as bombs or weapons.
Dirty Weapon	A nuclear weapon so designed that, upon detonation, the amount of contaminating material in the form of either fission fragments or irradiating material (or both), is <u>relatively high</u> by comparison with other possible design variations of the same Mark-numbered weapon. A weapon may also be "dirty" by comparison with different Mark-numbered weapons of the same general yield.
Drop Sequence	The prescribed order of events which take place in the arming, fuzing, and firing systems of a nuclear weapon from the time of release to detonation.

Efficiency	The ratio of the energy actually developed when a nuclear weapon or device explodes (the energy yield) to the total energy available in the fissile and/or fusion fuels in the weapon or device. Efficiency is the fraction of energy available which is released in an explosion. In the case of a fission weapon, this is equal to the number of nuclei which fission, compared to the total number of fissionable nuclei in the weapon or device core. Efficiency is calculated by dividing actual yield, determined mainly by radiochemical or photographic means, by the predicted yield, which is based on the amount of fissile material in the device, degree of core compression, and behavior of similar designs. Also called nuclear efficiency. See also Burn Efficiency and Implosion Efficiency.
Electron Volt	The amount of energy gained by a neutron in passing from a point of low electrical potential to a point one volt higher in potential. Abbreviated eV. Neutrons involved in nuclear reactions typically have kinetic energies on the order of several million electron volts (MeV).
Element	A fundamental substance consisting of atoms of only one specific atomic number, with consistent chemical properties, and which cannot be further decomposed chemically. A part of the universal matter of the universe. An entry in the periodic table.
EMP	Acronym for Electro-Magnetic Pulse, a strong surge of electromagnetic energy induced by nuclear explosions, typically occurring within a hundredth of a microsecond (a shake). EMP can cause great damage to electronic equipment and can have effect at long distances when it is initiated by a high-altitude nuclear explosion.
Enriched material	Material in which the relative amount of one or more isotopes has been increased.
Enriched uranium	Natural uranium from which the uranium-238 isotope has been extracted via gaseous diffusion, or centrifugal, laser excitation, or electromagnetic separation. Also called or alloy or uranium-235. Weapons-grade or alloy is usually enriched to a content of 93.5±1% uranium-235.
ENS	Acronym for External Neutron Source. See Neutron Generator.

Equilibrium Thermo-nuclear reaction A thermonuclear reaction in which materials and temperatures are in thermodynamic equilibrium, i.e., there is not a wide spectrum of temperatures (particle kinetic energy levels) throughout the reacting system, and the thermonuclear reaction has reached a semblance of a steady state. See non-equilibrium thermonuclear reaction.

ESD Acronym for Environmental Sensing Device, a component of a nuclear weapon arming system that is activated by an environmental condition unique to a warhead's flight or trajectory. ESDs are programmed to respond to accelerations, decelerations, thermal energy variations, barometric pressures, or spin-induced centrifugal forces encountered during the latter stages of weapon's stockpile-to-target sequence.

Fallout Precipitation of radioactive particles from smoke and vapor produced by the explosion of a nuclear weapon. After detonation of the weapon, bomb debris and particles of dirt and/or droplets of water become contaminated with radioactivity and then gradually fall to earth, sometimes after being carried considerable distances downwind. Particles in fallout radiate alpha particles, gamma rays, beta rays, and neutrons. Fallout reaches the stratosphere and is deposited world-wide throughout the earth's atmosphere.

Fireball The luminous sphere of hot gases that forms a few millionths of a second after a nuclear explosion as a result of the absorption by the surrounding medium (air, water, earth) of the thermal x-rays emitted by the extremely hot (in the range of several tens of millions of degrees) nuclear device residues.

Fissile Material Nuclear material that can be made to fission by neutrons of any energy. Used interchangeably in this text with the term "fissionable material." Common fissile materials include uranium-233, uranium-235, and plutonium-239.

Fission The splitting of an atomic nucleus into two more-or-less equal-weight fragments, accompanied by release of neutrons, x- and gamma rays, and energy (expressed as the kinetic energy of the fission products). This energy derives from the fact that total weight of the nuclei of the fission products is less than the total weight

of the original unfissioned nucleus. The "lost" weight is converted into kinetic energy of the fission products in accordance with Einstein's law of the conservation of mass and energy, i.e., total kinetic energy = change in mass multiplied by the speed of light squared. Approximately 80 different fission fragments result from roughly 40 different modes of fission of uranium-235 and plutonium-239.

Fissionable Material	Nuclear material that can be made to fission by neutrons of a specific energy. Used interchangeably in this text with the term "fissile material."
Fission products	Radioactive and non-radioactive elements or compounds resulting from nuclear fission.
Fizzle	A failed or incomplete nuclear detonation, which yields little or no nuclear explosive force (although the high explosives detonate). A "fizzle" is frequently caused by the initiation of a self-sustaining chain reaction as soon as the fissile assembly becomes just critical. If this occurs, there will be little or no explosion, since the neutron density will not increase rapidly, and energy resulting from fission will not be produced rapidly enough to cause a significant detonation.
Flyaround	A flight of an aircraft mated with a bomb or missile casing containing instrumentation instead of a warhead. The instruments measure accelerations, temperatures, air pressures, and "g" loads imposed on a nuclear warhead while it is aboard a maneuvering delivery aircraft during a typical simulated strike mission.
FUFO	Acronym for Full-Fuzing Option. When applied to a nuclear bomb, it provide capabilities for airburst, contact (surface) burst, and delayed detonation following a low-impact laydown.
Fusion	The combination of smaller atomic nuclei into larger nuclei, with a release of energy from mass transformation. The weight of the final larger nucleus is slightly less than the combined weights of the original smaller nuclei, and the "missing" mass is converted into kinetic energies of particles. Extremely high temperatures and pressures, resulting in highly-energetic, fast-moving nuclei packed closely together, are required to initiate fusion reactions.

Fusionable material	Nuclear material that can be made to fuse at high temperatures, such as lithium, deuterium, and tritium.
Fuzing system	Those components of a nuclear weapon system which signal the warhead to detonate at a particular point in its trajectory, e.g., such as above, on, or below its target.
Ganex	Abbreviation for gamma-neutron experiment, a diagnostic means of determining the rate of production of neutrons as a function of time by measuring the gamma radiation flux produced by the neutrons in a "converter" close to the neutron source. These gamma rays arrive at a scintillator-detector system where an electrical current is produced. This current is proportional to radiation flux and is displayed as a signal on an oscilloscope which can then be photographed for a permanent record.
Gun-type Weapon	A nuclear weapon consisting of a propellant charge which drives a projectile of fissionable material into a tamped fissionable target to form an uncompressed supercritical mass. Also called gun assembly weapon.
Half-life	The time required for a given radioactive isotope to decay (transmute to another element) so that only half as much original radioactive material remains. The time required for a radioactive mass to lose 50% of its radioactivity. Every radioactive isotope (radio-isotope) has a unique half life. For example, the half-life for the conversion of plutonium to uranium-235 is 20,000 years.
Harden	To design and fabricate a nuclear weapon or its delivery vehicle so as to be resistant to one or more types of nuclear detonation effects, e.g., blast, heat, radiation, EMP, etc.
Hard target	A military target that is difficult to damage (compared to other targets). Targets may be hardened in a number of ways against a variety of weapon detonation effects.
HE	Abbreviation for high explosives, especially the conventional high explosives used in nuclear weapons.
Heavy Water	Deuterium oxide (D ₂ O), i.e., water containing

deuterium in place of natural hydrogen. Electrolysis of heavy water results in the release of oxygen and deuterium gas, which can be used to "boost" a fission device, or combined with metallic lithium to form lithium deuteride, a thermonuclear explosive.

Height of burst	The vertical distance in feet above a target at which a nuclear detonation occurs or is to occur.
Helium	Element number 2, with an atomic number of 2 and atomic weights between 3 and 8. A by-product of nuclear fusion reactions.
HEU	Highly enriched uranium. See oralloy.
High-altitude burst	An arbitrary term for a nuclear detonation at an altitude in excess of 100,000 feet above the surface of the earth. Above this altitude, the partition of energy between blast, thermal, and nuclear radiation changes appreciably.
Hydride	A combining form for a compound containing hydrogen, for example, lithium hydride.
Hydro-dynamics	The behavior of continuously deformable media, including the compressibility of materials by shock waves. The cores of nuclear weapons behave like fluids under compression resulting from explosion of their chemical high explosives.
Hydro-dynamic test	A full-scale test of the implosion system in a nuclear weapon primary containing no fissile isotopes; there is no nuclear yield generated by a hydrodynamic test. Also called hydrotest. Hydrodynamic tests are used to develop the required levels of implosion symmetry, peak compression, and fission reaction initiation timing.
Hydrogen	Element number 1, with an atomic number of 1 and atomic weights between 1 and 3. The three isotopes of hydrogen are protium, deuterium, and tritium, with atomic weights of 1, 2, and 3, respectively.
Hydro-nuclear test	A full-scale test of the high explosive implosion system in a nuclear weapon primary containing small amounts of fissile uranium and/or plutonium isotopes and which typically yields no more than a few pounds TNT equivalent of explosive force and which is limited to subcritical, or slightly supercritical, neutron multiplication.

ICBM	Acronym for Intercontinental Ballistic Missile, a ballistic missile with a range between 3,000 and 8,000 nautical miles.
IFE	Abbreviation for In-flight Extraction, an early manual or automatic nuclear weapon safing technique, usually performed after a mission had been aborted and weapons were to be disarmed.
IFI	Abbreviation for In-flight Insertion, an early manual or automatic nuclear weapon arming technique, usually performed after launch or takeoff.
Ignition temperature (fusion)	The temperature at which energy lost from a surrounding plasma cloud is exactly replaced by energy put into it by fusion reactions taking place within it, thereby maintaining a self-sustaining thermonuclear reaction.
Implosion	A violent collapsing inward or compression of a substance. Fissile and fusionable fuels in nuclear weapons are imploded by shock waves generated by either high explosives or radiation. An implosion is the opposite of an explosion.
Implosion efficiency	In a thermonuclear weapon, the overall energy efficiency with which x-rays from the primary stage couple to the secondary stage and compress the fusion fuel.
Implosion warhead	That part of a missile warhead installation which includes the nuclear components, sphere case, high explosive system, detonators, detonator cables, firing element, firing switch, internal electrical circuits, in-flight insertion mechanism (if any), deuterium-tritium gas-boosting system (if any), and hardware required to join these parts.
Implosion weapon	A type of atomic weapon in which a subcritical configuration, either spherical or linear, of fissionable material is compressed radially into a supercritical state by a centrally-directed radial shock wave, either spherical or cylindrical, to produce an atomic explosion. The implosive shock wave may be generated by an HE explosion, or by other means, such as radiation.
Induced radio-activity	Radioactivity produced in a substance as a result of bombardment of the substance by neutrons or other subatomic particles.

Inertial Confinement	A technique for maintaining the physical integrity of a mass of thermonuclear fuel by compression, which also increases the likelihood and rate of fusion. A thermonuclear weapon is a large, inertially-confined explosive device.
Inertial Confinement Fusion	Retention of thermonuclear fuel by inertial forces within a reaction volume long enough for fusion reactions to occur. Abbreviated ICF.
Initiator	The bimetallic or high-voltage hydrogen isotope neutron generator used to initiate fission in a compressed critical mass of fissile fuel. Also called urchin (in early weapons) or nuclear capsule. See Neutron Generator.
Inverse Compton effect	A physical effect that causes photons, gamma, and x-rays to lose energy to longer-wavelength radiation. This was a major problem in early thermonuclear weapon designs.
Ion	An electrically charged atom or group of atoms. Atoms become ionized when stripped of their electrons.
IRBM	Acronym for Intermediate Range Ballistic Missile, a ballistic missile with a range between 1,500 and 3,000 nautical miles.
Isentropic	A physical or chemical process which occurs with no change in energy of the reactants. Isentropic compression is the compression of a substance without an increase in its temperature.
Isotope	An atom with a unique atomic weight. Each element can have more than one isotope, all with the same atomic number, but with different atomic weights. The difference in atomic weights between isotopes of a given element is caused by the presence of different numbers of neutrons in their nuclei. Isotopes have nearly identical chemical properties; however, differences in atomic weights can sometimes be exploited to separate different isotopes of an element.
Kiloton	A measure of explosive force equivalent to that of 1,000 tons (two million pounds) of TNT, e.g., 10^{12} calories or 4.18×10^{12} joules. Abbreviated KT or kt.
Kinetic energy	Energy associated with the motion of matter and its atoms.

LABS Acronym for Low Altitude Bombing System, an early automatic nuclear bomb release control system (with associated delivery aircraft maneuvers) for low-altitude bombing. Also called "toss bombing." Weapons delivered by LABS were usually not parachute-retarded.

LADD Acronym for Low Altitude Drogued Delivery, a technique to deliver parachute-retarded bombs at low altitudes and high speeds. LADD weapons are usually laydown (delayed detonation) bombs, to allow the delivery aircraft time to escape the resultant blast.

Laydown A delayed surface burst. The surface-level explosion of a nuclear weapon after the delivery aircraft has escaped. A laydown weapon is usually parachute-retarded, with some type of impact-absorption mechanism in its nose, such as a spike or frangible nose cone.

Levitation A technique of suspending the heavy fissile nuclear core of an atomic weapon or thermonuclear weapon trigger within the tamper-high explosives assembly surrounding the core. This arrangement results in better core compression and more complete fissioning of the nuclear fuel.

Lithium Element number 6, with an atomic number of 6 and atomic weights between 5 and 9. Lithium is usually compounded with deuterium when used as a thermonuclear fuel constituent. When bombarded by neutrons, lithium is transformed to tritium, which fuses with the deuterium to release energy, helium, and neutrons. When used as a thermonuclear fuel, lithium is usually enriched to 95% in the lithium-6 isotope, which comprises about 7.5% of naturally-occurring lithium.

MADM Acronym for Medium Atomic Demolition Munition, an atomic land mine emplaced by engineer teams and carried by jeeps and helicopters.

Mark-Mod-Alt An abbreviation for Mark - Modification - Alteration, the basic system of the U.S. Atomic Energy Commission for nomenclature of major assemblies of nuclear weapons. This naming convention indicates the overall design of a major assembly and changes to it. This system was adopted in July 1950 when each weapon was separated into three major assemblies -- bomb, fuse, and nuclear core -- each with its own

Mark-Mod-Alt designation.

Marriage program	The development effort conducted to fit atomic warheads to guided missiles, or nuclear weapons to aircraft.
Mass discontinuity	A change in the density of adjacent or adjoining materials. As applied to nuclear weaponry, this term often refers to the density variations in materials surrounding a nuclear warhead, such as an artillery shell case, missile re-entry vehicle, or bomb casing. The nature and geometry of material surrounding a nuclear warhead can, by reflecting neutrons back into the expanding fissioning weapon pit, affect the weapon's behavior and explosive yield, compared to its behavior and yield in an unreflected or "bare" geometry without surrounding materials.
Megaton	A measure of explosive force equivalent to that of 1,000,000 tons (two billion pounds) of TNT. Abbreviated MT or Mt.
Microsecond	A measure of time equivalent to one millionth (1/1,000,000th) of a second. 100 shakes or 1,000 nanoseconds.
Mixing	The intermingling of non-reactive materials with nuclear fuels. The foreign matter reduces or ends nuclear reactions. The interaction of an exploding fission weapon core with its non-fissionable tamper is an example of mixing. Another example is the unwanted mingling of fission reaction products with "unburned" fusion fuels in the core of a boosted fission device, or in the fuel capsule(s) of a thermonuclear device. The presence of impurities or foreign substances in the fusion fuel interferes with, and dampens, thermonuclear reactions.
Mod Designator Number	Modifications made to the major assembly design of a nuclear weapon. Mod "0" is the first version of a weapon design, with subsequent modifications numbered consecutively.
Mushroom cap	The top of the cloud formed by the fireball of a nuclear explosion.
Nanosecond	A measure of time equivalent to one billionth (1/1,000,000,000th) of a second. 1/1,000th of a microsecond, 1/10th of a shake, or 1,000 picoseconds.

Nautical mile	A nautical mile is equivalent to 1.1508 statute miles, or 6,076 feet. At the equator, a nautical mile is equivalent to one minute (1/60th of a degree of latitude
Neutron	An atomic particle with no electrostatic charge, and an approximate atomic mass of 1. A neutron adds mass to an atom's nucleus with negligible effect on chemical properties, but with a significant effect on its nuclear properties. Neutrons interact with matter predominantly by means of collisions; these collisions may result in the neutron either rebounding or being absorbed. "Fast" neutrons possess kinetic energies in the million electron volt (MeV) range, while thermal ("slow") neutrons have kinetic energies of one electron volt (1 eV) or less. Neutrons are required to initiate fission, and large numbers of neutrons are produced by both fission and fusion reactions in nuclear explosions.
Neutron Flux	The intensity of neutron radiation. The product of neutron density (number per cubic centimeter) and velocity, equal to the total distance traveled in one second by all the neutrons in a unit volume. Neutron flux is a measure of neutron concentration in a given volume and is quantified in terms of neutrons per square centimeter per second. Neutron fluence is the time integral of neutron flux, expressed as neutrons per square centimeter of a material subjected to neutron irradiation.
Neutron Generator	A high-voltage vacuum tube used in contemporary nuclear weapons to furnish neutrons of a specific energy level at a precise moment, to begin fission reactions in composite plutonium/oralloy fissile cores. Also called ENS (External Neutron Source). Neutron generators replaced bimetallic "golf ball" internal neutron sources (see Urchin).
Nominal Atomic Bomb	An obsolete term formerly used to describe an atomic weapon with an energy release of 20 kilotons of TNT-equivalent explosive power.
Non-equilibrium thermo-nuclear reaction	A thermonuclear reaction in which materials and temperatures are not in thermodynamic equilibrium, i.e., there is a wide spectrum of temperatures (energy levels) throughout the reacting system, and the thermonuclear reaction is not in a steady state. See equilibrium thermonuclear reaction.

Nuclear Efficiency	See Efficiency.
Nuclear Hardness	A measure of the ability of a nuclear warhead to withstand exposure to one or more of the environments and associated effects produced by an external near or distant nuclear detonation without suffering an unacceptable change in the warhead's performance. Weapons may be hardened against effects of electromagnetic pulse (EMP), blast, thermal radiation, shock, nuclear radiation, and/or debris produced by a nuclear explosion external to the attacking warhead.
Nuclear Reaction	A physical reaction involving the particles in the nuclei of atoms. Nuclear reactions usually release or absorb large amounts of energy, and result in the transformation of elements either up or down the periodic table.
Nuclear Transit Time	For multistage thermonuclear weapons, the time interval between high-order detonation of the primary and the high-order detonation of the secondary, or the time between high-order detonation of the secondary and high-order detonation of the tertiary. See also transit time.
Nuclear Weapon	A thermonuclear (fusion) or atomic (fission) weapon, or a weapon which derives its explosive force from nuclear fission or nuclear fusion reactions.
Nucleon	A particle in an atomic nucleus. For the purposes of this series of volumes, these are limited to protons and neutrons.
Nucleus	The dense central core of an atom, composed of protons and neutrons (nucleons). The nucleus comprises almost all the weight of an atom of an element.
One-Point Safe	A description for a nuclear weapon that will not produce a measurable nuclear yield if just one of its explosive charges is detonated. Gun-type weapons are inherently one-point unsafe because of their design: they have only one ignition point. Technically, a nuclear weapon is one-point safe if in the event of a detonation started at any one point in the HE system, there is not more than a one-in-a-million chance that it will result in a nuclear yield equivalent to more than four pounds of TNT explosive force.

Opacity	A measure of the resistance of a substance to light, x-rays, and neutrons or other atomic and subatomic particles. The more opaque a substance, the more impermeable it is to nuclear radiation. Opacity is a very important property of materials used in nuclear weapons; it is of basic importance in calculating the transport (movement) of radiation within a nuclear weapon. The opacity of a material is dependent upon its temperature and density.
Oralloy	Oak Ridge Alloy, or enriched weapons-grade uranium metal, containing 93.5±1% uranium-235 isotope, and 5.5% uranium-238 and 1% uranium-234. Commonly represented by letters Oy or by symbols U-235, U^{235} , or ${}_{92}U^{235}$. One kilogram (2.2 lbs.) of oralloy, if completely fissioned, yields the equivalent of 17 kilotons (17 KT) of TNT-equivalent explosive force. Oralloy can be fissioned by neutrons with a wider range of kinetic energy levels than those which are required to fission tuballoy. Also called HEU, or highly enriched uranium. Oralloy is a moderately stable uranium isotope with a half-life of about 710 million years.
Over-initiation	A condition resulting from an over-exposure of a fissioning nuclear core to neutrons, usually resulting in a reduced yield caused by fission occurring in too many places at once in the core, creating an asymmetrical explosion.
PAL	Acronym for Permissive Action Link, a coded switch that serves as a mechanical supplement to the arming control of a nuclear weapon. When installed, it prevents the weapon from being armed without a specific code. Modern PALs are microcomputer chips. There are six separate types of PALs, Categories A through F.
Penetration Weapon	A nuclear weapon designed to penetrate the earth or an armored surface before exploding.
Periodic Table	A table of elements arranged by their atomic numbers and chemical combining powers (valences).
Phase	A stage in the life of a nuclear weapon. There are no time limitations for each phase. Phase 1 - weapon concept definition; Phase 2 - program feasibility study; Phase 2A - cost and design data definition; Phase 3 - development

engineering; Phase 4 - production engineering; Phase 5 - first production; Phase 6 - quantity production; Phase 7 - retirement and salvage of reusable materials and components.

Physics package	The nuclear components of an atomic or thermonuclear weapon. The physics package is limited to only those components which participate in nuclear reactions to produce nuclear energy yield.
Picosecond	A measure of time equivalent to 1/1,000,000,000,000th of a second. 1/10,000th of a shake.
Pit	The portion of either an atomic weapon, or the primary stage of a multistage thermonuclear weapon, enclosed by high explosives, exclusive of the fissionable material and those portions of the pit which are attached to the fissionable material. In an implosion weapon, the pit together with the surrounding high explosives comprise the implosion system.
Pit Liner	A light metal (copper) liner installed inside hollow weapon cores (pits) to retain boost gases such as deuterium and tritium. This liner prevents gas loss via diffusion through the core metal.
Plasma	The fourth state of matter (solid, liquid, gas, plasma). A very hot, very dense, very energetic ionized "gas," composed primarily of atomic nuclei stripped of their electrons. A plasma has a positive electrostatic charge due to the protons in it.
Plutonium	Element number 94, with an atomic number of 94 and atomic weights between 232 and 246. Plutonium is a radioactive, artificial man-made element with a half-life of 24,360 years (it decays via alpha particle radiation). An atomic weapon fuel (in metallic form). One kilogram (2.2 lbs.) of plutonium, if completely fissioned, releases the explosive equivalent of 18 to 20 kilotons (20 KT) of explosive force. Plutonium is sometimes also called "ploom" and is commonly represented by letters Pu, or as plutonium-239, Pu-239, Pu^{239} , or $_{94}\text{Pu}^{239}$.
Polonium	Element number 84, with atomic weight 210. Polonium is radioactive with a half-life of only 138 days. When mixed mechanically with

beryllium, polonium reacts and releases neutrons. During Manhattan Project days, the code name for polonium was "Postum."

Postdetonation	The undesirable delayed initiation of a fission or fusion explosion or reaction after optimum conditions have been achieved. Postdetonation usually results in a partial or fizzle yield. Also called postinitiation.
Predetonation	The undesirable premature initiation of a fission or fusion explosion or reaction before optimum conditions are achieved. Predetonation usually results in a partial or fizzle yield. Also called preinitiation. Predetonation is also defined as the premature explosion of the high explosive charge or propellant in a nuclear weapon before the weapon has reached the predesignated point of burst in its trajectory.
Preheating	The undesired nuclear bombardment of a fusion fuel capsule, including fuel, pusher, and sparkplug, by radiation or particles from the primary or another exploding fusion stage, before compression and ignition of the fusion fuel within the capsule. This bombardment results in nuclear transmutation, physical expansion, and chemical degradation of the thermonuclear fuel constituents, usually resulting in lower-than-predicted energy yield(s) from the thermonuclear fuel charge(s).
Primary	The fission trigger or first stage of a multistage thermonuclear weapon or device. The term is derived from "primary bomb," originally called an "auxiliary bomb" in multistage thermonuclear weapons.
Protium	A hydrogen isotope with atomic number and atomic weight of 1, and a nucleus composed of a single proton. The most common form of hydrogen.
Proton	A electrostatically-positive charged atomic particle with an approximate atomic weight of 1. The nucleus of an atom of common hydrogen.
Pusher	A tuballoy or orally tube or hollow foil cylinder encasing the fusion fuel in secondary and tertiary stages of multistage thermonuclear weapons or devices. This material initially protects the sparkplug and fusion fuel from disintegration or predetonation caused by the blast and radiation from preceding stages (the primary or secondary). It next becomes a

circumferential piston, driven by plasma to axisymmetrically compress the fusion fuel and sparkplug within the capsule to a volume between 5% and 7% of their original volume. It then tamps the thermonuclear explosion, and fissions when bombarded by high-energy neutrons radiating outward from the fusion reactions within the fusion fuel capsule. The fission of this jacket, when used, contributes significantly to the total explosive yield, its own yield proportionate to its or alloy content, its uncompressed thickness, and extent of its compression by radiation from the explosion(s) of preceding weapon stages.

The term pusher was originally used to describe the aluminum hydrodynamic stabilizer around early implosion fission bomb pits. This stabilizer prevented the formation of Taylor instabilities in the implosive spherical shock wave.

"Q" clearance A security clearance granted by the Atomic Energy Commission (and its successor agencies, the Energy Research and Development Administration and the Department of Energy), following and based upon an investigation by the Federal Bureau of Investigation. A "Q" clearance permits access to certain types of nuclear weapons design and effects information. "P" and "L" clearances were lower-level AEC nuclear weapons data access clearances.

Radiation A method of energy transmission, including any electromagnetic wave and any moving electron or nuclear particle, charged or uncharged, emitted by a radioactive substance.

Radiation coupling Use of the radiant energy from a fission explosive to compress and heat fusion fuel(s). Also called radiation channeling or ducting.

Radiation implosion A technique of using radiation from a nuclear explosion to implode subsequent nuclear weapon stage(s) (capsules) containing thermonuclear fuel. The basis of modern U.S. multistage thermonuclear devices and weapons. See staging.

Radio-activity The physical process whereby certain elements undergo spontaneous atomic disintegration in which energy is liberated, usually resulting in the formation of different elements which may or may not be radioactive. The process is accompanied by the emission of one or more types

of radiation, such as alpha particles, beta particles, and x- and gamma rays (high-frequency, short-wavelength radiation). Ultimately, as a result of one or more stages of radioactive decay, a stable nonradioactive element isotope is formed.

Re-entry Vehicle	An integral portion of a ballistic missile which re-enters the atmosphere and which may include a warhead section, nose cone, flared section (skirt), decoys, arming and fuzing devices and other missile components. Frequently abbreviated RV. Also known as a Re-entry Body (RB).
Reflector	A dense light or heavy metallic shield surrounding a fissile weapon core which reflects escaping neutrons back into the reacting (exploding) core. In the case of a boosted fission device, a beryllium reflector may also act as a neutron source, following the initiation of fusion reactions in the center of the imploded fissile core.
Restricted Data	All data (information) concerning (i) design, manufacture, or utilization of atomic weapons; (ii) the production of special nuclear material; and/or (iii) the use of special nuclear material in the production of energy. (This definition does not discriminate between the use of special nuclear material to produce energy for military purposes, and its use for peaceful purposes, e.g., the fission of uranium or plutonium in reactors to produce electricity for commercial use.) Restricted data is defined by the Atomic Energy Act of 1954, as amended.
Rollup	The orderly dismantling and transfer to other areas of equipment and facilities after a nuclear test series.
Runaway burn	A self-sustaining, uncontrolled thermonuclear fusion reaction (explosion).
SADM	Acronym for Special Atomic Demolition Munition, a small nuclear land mine, components of which are man-portable.
Safing	As applied to nuclear weapons, the changing from an armed condition (ready to fire) to a state of being unable to fire (unarmed or disarmed).
Salted weapon	A nuclear weapon which contains, in addition to its normal components, certain elements or

isotopes which capture neutrons at the time of its explosion and produce radioactive products in amounts significantly in excess of the usual amount of radioactive debris which would be produced by the weapon if the certain elements or isotopes were not present in its design.

- Salvage fuse** A passive fuse actuated by an effect of a defensive nuclear explosion, or by contact with the earth or water.
- Sealed pit** An assembly, located inside a nuclear weapon implosion system, in which the nuclear components are integrally contained and sealed. When gas-boosting is used, the gas-boosting components are located outside the implosion system. No further positioning, such as in-flight insertion, of the nuclear components is required for operation of the implosion system.
- Sealed Pit weapon** A nuclear weapon containing a sealed pit.
- Secondary** The thermonuclear explosives stage of a two-stage thermonuclear weapon or device, or the first thermonuclear explosives stage of a three-stage weapon or device. The term is derived from "secondary bomb," originally called a "principal bomb" in multistage thermonuclear weapons. In a multistage weapon, the secondary and subsequent stages (if included) are physically separate from the primary stage.
- Shake** A measure of time equivalent to 1/100,000,000th of a second. 10 nanoseconds or 1/100th of a microsecond. The time required for a uranium nucleus to fission.
- Shine** Slang term for detectable nuclear radiation emanating from the earth, sea, air, or an object. A common source of airborne "shine" is the mushroom cloud and its downwind fallout generated by a nuclear explosion.
- Sigma** An AEC/ERDA/DOE system used to identify, categorize, compartmentalize, and permit or deny access to various levels of classified weapons design, stockpiling, utilization, and production information. These categories have ranged from Sigma 1 to Sigma 15 and have been applied to both information and the clearance levels required for access to that information. The following definitions of Sigma categories and subcategories are shown for both 1957 and 1993.

- Sigma 1** 1957 & 1993: Classified information which describes thermonuclear weapons internal warhead design, hydrodynamic principles of theory, and nuclear principles of theory.
- Sigma 1.1** Sigma 1 information which is limited to a designated weapon or weapons, and/or designated class or classes of components. This category had been eliminated by 1993.
- Sigma 2** 1957: Classified information which describes warhead design and theory for all types of fission weapons, including boosted fission weapons and gun-type weapons. This category includes the high explosive system with its detonators and firing unit, pit system, and nuclear installation system, as they pertain to warhead design and theory. 1993: Theory of operation or complete design of fission weapons or their unique components, including the high explosive system, and nuclear initiation system as they pertain to weapon design and theory.
- Sigma 2.1** Same as 1957 Sigma 2 category, except limited to a designated weapon or weapons, and/or designated class or classes of components. This category had been eliminated by 1993.
- Sigma 3** 1957: Classified information which describes the general external weapon configuration for all thermonuclear and fission weapons, including size, shape, weight, environmental behavior, fuzing, ballistics, handling and storage and operational requirements, yields, effects, and related information. This category specifically includes all information, internal as well as external, required for normal stockpile and strike operations. 1993: Manufacturing and utilization information which does not reveal comprehensively the theory of operation or design of the physics package. This category also includes complete design and operation of non-nuclear components but only information as prescribed below for nuclear components. Also included is utilization information as required to support the stockpile-to-target sequence. The latter includes:
- (a) General external weapon configuration, including size, weight, and shape.
 - (b) Environmental behavior, fuzing, ballistics, yields, and effects.
 - (c) Nuclear components or subassemblies which do not reveal theory of operation or

- significant design features.
- (d) Production and manufacturing techniques relating to nuclear parts or subassemblies.
- (e) Anticipated and actual strike operations.

Sigma 3.1 Same as 1957 Sigma 3 category, except limited to a designated weapon or weapons, and/or designated class or classes of components. This category had been eliminated by 1993.

Sigma 4 1957: Classified information which describes tests of nuclear weapons or devices, including logistics, administration, other agency participation, special test construction, special test effects, safety, and general nature of weapons or devices tested, including expected or actual yield and conclusions derived from such tests, excepting design features. 1993: Information inherent in preshot and post-shot activities necessary in the testing of atomic weapons or devices. Specifically excluded from this category are the theory of operation and the design of such items. Sigma 4 information includes:

- (a) Logistics, administration, or other agency participation.
- (b) Special construction and equipment.
- (c) Effects and safety.
- (d) Purpose of tests, and general nature of nuclear explosive tested, including expected or actual yields and conclusions derived from tests, except design features.

Sigma 4.1 Same as 1957 Sigma 4, except limited to a designated test or test series, and/or designated aspects of the test or series. This category had been eliminated by 1993.

Sigma 5 1957: Classified information which describes non-nuclear production rates, specifically, any figure identified, or identifiable when associated with unclassified information, as to the total production for war-reserve or all purposes combined of any type, or all types combined, of non-nuclear weapons (complete weapons less nuclear components) for any period. 1993: Production rate and/or stockpile quantities of nuclear weapons and their components.

Sigma 5.1 Same as 1957 Sigma 5, except limited to a designated weapon or weapons, and/or designated production source or sources. This category had been eliminated by 1993.

- Sigma 6** 1957: Classified information which describes nuclear production rates, specifically, any figure identified, or identifiable when associated with unclassified information, as the total production for war-reserve or all purposes combined of any type, or all types combined, of nuclear component assemblies (weapons units) for any period. This category had been eliminated by 1993.
- Sigma 6.1** Same as 1957 Sigma 6, except limited to a designated nuclear type or types and/or designated production source or sources. This category had been eliminated by 1993.
- Sigma 7** 1957: Classified information which describes non-nuclear stockpile quantities, specifically, any figure identified, or identifiable when associated with unclassified information, as to the total stockpile for war-reserve or all purposes combined of any type, or a subtotal by storage site location(s) of any type or all types combined, of non-nuclear weapons (complete weapons less nuclear components) for any period. This category had been eliminated by 1993.
- Sigma 7.1** Same as 1957 Sigma 7, except limited to a designated weapon or weapons and/or designated storage site or sites. This category had been eliminated by 1993.
- Sigma 8** 1957: Classified information which describes nuclear stockpile quantities, specifically, any figure identified, or identifiable when associated with unclassified information, as to the total stockpile for war-reserve or all purposes combined of any type, or all types combined, of nuclear component assemblies (weapon units) for any period. This category had been eliminated by 1993.
- Sigma 8.1** Same as 1957 Sigma 8, except limited to a designated nuclear type or types and/or designated storage site or sites. This category had been eliminated by 1993.
- Sigma 9** 1957 & 1993: Classified information which describes general studies not directly related to weapon design and/or performance, e.g., reliability studies, fuzing studies, damage studies, aerodynamic studies, etc.
- Sigma 10** 1957 & 1993: Classified information which

describes chemistry, metallurgy, and processing or production of materials peculiar to the field of atomic weapons or nuclear explosive devices.

Sigma 10.1 Same as Sigma 10, except limited to a designated material, e.g., plutonium, uranium, or tritium. This category had been eliminated by 1993.

Sigma 11 1957: Classified information which describes non-weapons data, including reactor theory, design, development and operation; non-weapon nuclear physics; nuclear propulsion; biology, medical and health problems, including research instruments, etc. except data on projects listed under Sigma categories 15, 16, and 17. 1993: Information concerning inertial confinement fusion which reveals or is indicative of weapon data.

Sigma 12 1957: Information concerning visits to AEC San Francisco field office and any Sandia facility in connection with administration, construction, inspection, audit, safety, fire protection, "cook's", "VIP," museum, and lecture tours, etc., with purpose of visit to be clearly defined, e.g., audit of XYZ company contract; fire protection survey of XYZ company plant, etc. 1993: Complete theory of operation, complete design, or partial design information revealing either sensitive design features or how the energy conversion takes place for the nuclear energy converter, energy director, or other nuclear directed-energy weapon systems or components outside the envelope of the nuclear source but within the envelope of the nuclear directed-energy weapon (such as an x-ray laser).

Sigma 13 1957: Information concerning attendance at designated training courses, such as bomb commander's course, weaponeer's course, staff officer orientation course, weapons orientation course, nuclear supervisor's training course, etc. The course to be attended must be stated when clearance is requested. 1993: Manufacturing and utilization information and output characteristics for nuclear energy converters, directors, and other nuclear directed-energy weapon systems or components outside the envelope of the nuclear source, and not comprehensively revealing the theory of operation, sensitive design features of the nuclear directed-energy weapon or how the energy conversion occurs. This information

includes:

- (a) General external weapon configuration and weapon environmental behavior characteristics, yields, and effects.
- (b) Component or subassembly design that does not reveal theory of operation or sensitive design features of nuclear directed-energy weapons categorized as (1993) Sigmas 1, 2, or 12.
- (b) Production and manufacturing techniques for components or subassemblies of nuclear directed-energy weapons that do not reveal information categorized as (1993) Sigmas 1, 2, or 12.

Sigma 14 1957: An AEC clearance category which allowed access to detailed information in all 1957 Sigma categories except Sigma 5, 6, 7, 8, 15, 16, 17, and their subcategories. This category had been eliminated by 1993.

Sigma 15 1957: Classified information which describes non-weapons data similar to 1957 Sigma 11 data for Project Sherwood, the early U.S. study of controlled thermonuclear energy for peaceful purposes. This category had been eliminated by 1993.

Sigma 16 1958: Non-weapon data similar to Sigma 11 for Project Pluto and Project Rover. This category had been eliminated by 1993.

Sigma 16.1 1958: Limited to Project Pluto. This category had been eliminated by 1993.

Sigma 16.2 1958: Limited to Project Rover. This category had been eliminated by 1993.

Sigma 17 1957: Reserved for future use. This category had been eliminated by 1993.

Sigma 18 Limited to project Plowshare, the peaceful use of nuclear explosives. This category had been eliminated by 1993.

SLADD Acronym for Strategic Low Altitude Drogued Delivery. Like LADD, except that high-yield weapons are delivered.

Soft target A military target which is easily damaged (compared to other targets).

Soft x-rays X-rays with an energy of about 1,000 electron volts (1 KeV).

Sparkplug	An oralloy or plutonium rod used as the igniter of a compressible fusion fuel capsule. The sparkplug fissions when compressed and bombarded at one end by neutrons, heating the compressed fusion fuel, and furnishing neutrons for tritium generation via lithium transformation in the compressed fusion fuel mass.
Stage	A fission or fusion explosive capsule in a multistage fusion weapon. A fission stage consists of a fissile core, reflector, tamper, high explosives, detonators, and external neutron generator. A fusion stage consists of a sparkplug, fusion fuel, and pusher.
Staged weapon	A nuclear weapon in which energy from the primary initiates the explosion of a secondary and any subsequent stages, such as a tertiary.
Staging	A technique of generating large thermonuclear explosions by means of increasingly-powerful intermediate nuclear explosions. See Teller-Ulam configuration. Also called heterocatalytic explosions.
Stockpile-to-Target Sequence (STS)	The order of events involved in removing a nuclear weapon from storage, and assembling, testing, transporting, and delivering it to its target.
Subcritical mass	A mass of fissile material which is insufficient to maintain a chain reaction. See Critical Mass.
Subsurface burst	A nuclear explosion below ground or sea level.
Supercritical mass	The state of a given fission system when the specified conditions are such that a greater than critical mass of fissionable material is present and the fission neutron production rate exceeds all neutron losses, causing a rapid and uncontrollable exponential increase in the neutron flux or population within the mass, usually leading to an explosion, although low-level detonations and thermal meltdowns are also possible. Once started, nuclear reactions within a supercritical mass usually continue until the heat and radiation generated dissipate the fissionable material to a subcritical mass.
Surface	A nuclear explosion at ground or sea level,

burst	usually on land, island, or reef surface, or on a barge on water.
Surface zero	The point at ground or sea level directly at, above, or below a nuclear explosion. The "ground zero" of a surface burst; the point directly above a subsurface burst; or the point directly below an airburst.
Tamper	A light or heavy, dense metal surrounding either a fissile core and reflector assembly or a charge of thermonuclear fuel. A tamper is used to contain and increase the force of an explosion by increasing the burn time and slowing disassembly. A tamper may also serve as a reflector, bouncing neutrons back into the reacting mass, and, in the case of a boosted fission or thermonuclear device, as an additional fission fuel source. Also called "reflector" or "pusher." Thermonuclear weapon fusion stage tampers may be made from either nonfissionable or fissionable material, the latter including depletalloy, tuballoy, oralloy, or thorium.
Taylor instability	A physical phenomena in which strong eddy currents build up at the boundaries of materials of differing densities moving through each other, causing loss of energy via unwanted transfer to random motion; the instabilities are strongest when light materials are accelerated against heavier materials. When a dense fluid is accelerated by a lighter one, hydrodynamic theory states that any initial perturbation at the interface between the fluids will grow without limit. The resultant eddy currents cause the boundary between the two materials to be unstable and turbulent, leading to unpredictable blending. Taylor instability was a significant factor in the performance of certain types of early U.S. thermonuclear weapons. Taylor instability is named after a British physicist, Dr. G. I. Taylor, who worked at LASL during World War II.
Teller-Ulam configuration	An assembly of staged atomic and thermonuclear explosives packages, permitting theoretically-unlimited explosive power. This configuration requires the use of oralloy or plutonium sparkplugs to ignite the thermonuclear fuel charge(s). The Teller-Ulam configuration is characterized by separate fission and fusion stages, and compression (before ignition) of the fusion stage by ducted (channeled) radiation

emanating from the preceding fusion and/or fission stage(s).

Tenex	Abbreviation for temperature neutron experiment, a diagnostic means of determining the temperatures attained during a thermonuclear reaction. This temperature is directly related to the efficiency of a thermonuclear devices, higher temperatures meaning that a greater amount of the fuel will "burn," and coincidentally, a greater number of high-energy neutrons will be produced. An oscilloscope is used to produce a neutron energy spectrum, which is proportional to the velocity (kinetic energy) of the neutrons.
Terra-ballistics	The description and study of the characteristics of objects moving through soil, rock, or ice. Also known as terradynamics, the study of earth penetration kinetics. The terraballistics of a given object depend in part on its nose shape, angle-of-attack, impact angle and velocity, length-to-diameter ratio, and inherent stability.
Tertiary	The second thermonuclear explosives stage of a three-stage thermonuclear weapon or device.
Thermo-nuclear reaction	A nuclear reaction which occurs only under conditions of temperature, expressed as nuclei kinetic energies, and pressure sufficient to overcome the mutually-repulsive electrostatic charges of nuclei of light elements. Commonly called a TN reaction.
Thermo-nuclear weapon	A nuclear weapon which releases a significant fraction of its yield from fusion processes. Commonly called a TN weapon.
Thorium	Element number 90, with an atomic number of 90 and atomic weights between 223 and 234. Thorium is a source of fissionable uranium-233 via neutron bombardment in reactors.
TNT Equivalent	The weight of TNT explosive which would release the same amount of thermal energy as a specified nuclear explosion or explosive material. One ton of TNT releases approximately 5.1 billion calories, or 5.1 million kilocalories, i.e., 5.1 kilojoules per gram.
Tracer	A selected amount of a specific substance placed either within a nuclear device or in the immediate vicinity of the device to provide diagnostic information on the functioning of the

system or on the distance of dispersal of its products. A tracer has little or nothing to do with the primary nuclear process of an explosion, and serves only as a remote indicating substance, when analyzed. The existence of a tracer substance in or near the exploding device assures that the substance will be bombarded with neutrons and gamma rays. If the proper substance is selected and situated close enough to the device to be thoroughly mixed with the device debris, the radiation bombardment results in easily-detectable radioactivity in every fallout sample subsequently analyzed. Careful positioning of tracer substances can furnish information about the detailed behavior of some particular portion or component of the nuclear device.

Transit time The time between the transmission of firing signals to the detonators in the chemical high explosives in a nuclear device or weapon and the beginning of observable nuclear reactions. The start of nuclear reactions is usually indicated by the emission of gamma radiation, which travels at the speed of light, from the weapon or device casing. Transit time, also known as "implosion time," is the time required for the shock wave generated by the high explosives to propagate through ("transit") the tamper-reflector-pit assembly and compress and "ignite" the nuclear core. Also called HE transit time (HETT). See also nuclear transit time.

Transuranic element An element in the periodic table that has a higher atomic number than uranium, e.g., plutonium, curium, fermium, einsteinium, etc.

Trap Door A removable portion of an atomic weapon tamper - high explosives assembly around a fissile core to allow insertion of nuclear components. Used in conjunction with inflight arming and disarming of early atomic weapons.

Tritide A combining form for a compound containing tritium, such as lithium tritide.

Tritium A heavy isotope of hydrogen, with an atomic number of 1, an atomic weight of 3, and a nucleus containing a proton and two neutrons. Tritium is mildly radioactive via beta decay, with a half-life of 12.3 years. Tritium can be produced artificially via lithium transformation by neutron bombardment in pressurized nuclear reactors, or in the fusion fuel capsules of

thermonuclear weapons. Often represented by letter T. The thermonuclear fusion of tritium plus deuterium yields 134 KT per kilogram of tritium.

Triton The nucleus of a tritium atom, consisting of a proton and two neutrons.

Tropopause The boundary between the troposphere and the stratosphere above it. The troposphere is that part of the atmosphere above the earth in which weather (clouds, rain, snow, etc.) forms, there are convective air currents, and temperature changes rapidly with altitude. The stratosphere is that portion of the atmosphere generally above seven miles high, depending upon season, latitude, and weather, in which temperature changes little with increasing altitude and where clouds of water never form. In tropical latitudes, such as those in which the Pacific Proving Ground was located, the tropopause occurs at about 55,000 feet, although it may vary between 48,000 and 59,000 feet.

Tuballoy Ordinary uranium metal, consisting of 99.28% of the isotope uranium-238, 0.711% uranium-235, and 0.0052% uranium-234. Tuballoy is used mainly as an additional fissile explosive surrounding thermonuclear fuel capsules, where it also serves as a tamper for the thermonuclear explosion. Tuballoy can also be used as a tamper in fissile bomb cores. The name derives from "Tube Alloy," the wartime Manhattan Engineer District code name for uranium, which in turn was derived from the designation Tube Alloys Limited, the cover name of the early wartime British atomic energy research project. Tuballoy is commonly represented by the letter pair Tu, or as U^{238} , U-238, or ${}_{92}U^{238}$. Tuballoy can be only be fissioned by "fast" high-energy neutrons, with kinetic energies in excess of 1 MeV, such as those emanating from thermonuclear fusion reactions. Fission of tuballoy by fast neutrons can make a significant contribution to the total energy release (yield) produced in nuclear weapons. When fissioned, tuballoy yields approximately 18 kilotons per kilogram.

TX Acronym for Test Experimental, a designation assigned to a nuclear warhead as long as it is in test status, or for the duration of tests of a modification to the basic design. A generic prefix to the weapon number which designates the

specific development model of a new nuclear weapon. When the weapon reaches the production stage, the TX designation was changed to a Mark or MK or W prefix (later to a B or W prefix).

Uranium	Element number 92, with an atomic number of 92 and atomic weights between 227 and 240. Two isotopes, uranium-235 and uranium-238, are used in nuclear weapons. Uranium-238 is also used as a source of plutonium (via neutron bombardment in reactors).
Urchin	Bimetallic (polonium and beryllium) neutron generator used to initiate fission explosions in early U.S. nuclear weapons.
W	Abbreviation for Warhead, the basic designation of a U.S. nuclear warhead, regardless of its ultimate use (bomb, missile warhead, ADM, AFAP, etc.).
War-Reserve weapon	A stockpiled nuclear weapon whose yield, reliability, and quality have been determined and certified. War-reserve weapons are immediately available for combat.
Weapon debris	The radioactive debris of a nuclear device after it has been detonated, consisting of fission products, various products resulting from neutron capture, weapon casing and other components, and unfissioned uranium and/or plutonium.
Weapon-grade plutonium	Plutonium containing low, specified quantities of plutonium isotopes other than Pu-239. U.S. weapon-grade plutonium nominally contains 93.5% Pu-239, 6% Pu-240, and 0.5% other Pu isotopes. The high neutron background from spontaneous fission of Pu-240 makes low Pu-240 content highly desirable in plutonium for weapons.
Weaponization	The conversion or modification of a nuclear test device into a combat-ready warhead that can withstand all the harsh environmental and handling conditions to which it will be subjected if stockpiled and used in combat. Weaponization usually includes the design and production of a ballistic casing with structural supports and any required retardation and impact-absorption or shock-mitigation devices, as well as special fuses, power sources, and arming and safing systems or equipment required to make a complete weapon. Weaponization occasionally perturbs the physics behavior of

the nuclear device in the warhead.

Wooden bomb	A largely-hypothetical nuclear weapon with absolute reliability, an infinite storage life, and which requires no special handling, storage, or surveillance while in stockpile. The term derives from the expression "as trouble-free as a block of pine."
X rays	Penetrating electromagnetic radiations with wavelengths very much shorter than those of visible light.
X-unit	The firing set in a nuclear weapon, consisting of fuses, detonators, and associated electrical and electronic equipment.
XW	Acronym for Experimental Warhead, the designation applied to a specific experimental model of a nuclear warhead. The designator applied to a warhead before it goes into development engineering, i.e., while it is in Phases 1 or 2 of its development lifecycle.
Yield	The total effective energy released in a nuclear explosion. It is usually expressed in terms of the equivalent tonnage of TNT required to achieve the same energy release, primarily blast, in a conventional explosion. The total energy yield is manifested as nuclear radiation, thermal radiation, and blast and shock energy, the proportional distribution of which depends upon the medium in which the nuclear explosion occurs and the type of weapon or device. Also called energy yield.
Yield-to-weight ratio	A measure of nuclear weapon efficiency. Commonly expressed as the total yield of a given weapon divided by total weapon weight, it is properly defined as the total yield of the weapon divided by the total mass of its fissile and/or thermonuclear fuel charge(s). Because the fuel mass is usually small compared to weapon mass, properly-computed yield-to-weight ratios are typically much higher than those often cited in public literature. Total yield for a given total weapon weight is usually of interest to the military service(s) which deliver the weapon to its target; total yield for a given fuel mass is usually of interest to the weapon designers. In many cases, especially for aerial bombs, combined weights of parachutes and arming and firing systems often exceed the total weight of the nuclear system.

Zipper

Generic nickname for electronic external initiator used to provide neutrons to initiate fission in primaries and fission warheads.

WEAPONS PHYSICS

Overview

A nuclear weapon is a complex device. The overall design of a nuclear weapon is an intricate operation involving a cooperative effort among mathematicians, scientists, and engineers. A nuclear device is a system of subsystems, all interrelated in their design and operation.

Inside the casing of an atomic or hydrogen bomb resides an incredible array of precision instruments, sophisticated electronic equipment, exquisitely machined and plated chemical and metal parts, expertly cast and shaped high explosives, and a core of fissionable material resembling an ingenious Chinese puzzle.

From their earliest development in the 1940s and 1950s, nuclear weapons have always been some of the most complex pieces of movable technology on earth.

For example, the following terms describe the major components of a vintage-1960 BOMARC air-defense missile warhead:

(1) Nuclear System - That portion of the warhead which provides the nuclear reaction. This system normally consists of all the nuclear and non-nuclear material contained in the capsule, pit, pit liner, tamper, and trap door.

(2) Nuclear Safing System - That portion of a weapon that integrally contains all the apparatus, which, on receipt of proper signals from the arming system or by manual operation, functions so as to place the nuclear system in an armed or safe condition.

(3) Implosion System - That portion of the atomic weapon which provides the compression necessary to cause the nuclear system to function. This system will normally consist of the high explosive, detonators, and the necessary structure to combine these parts into an entity.

(4) Firing System - That portion of the weapon, which, upon signals from the arming system, transforms and stores electrical energy, and, upon signal

from the fuzing system, discharges this stored electrical energy to detonate the implosion system. This system will normally consist of the firing set, firing switch, load coils, load plates, detonator cables, and other interconnecting cables and structures.

(5) Warhead - Includes the nuclear system, nuclear safing system, implosion system, firing system, plus the hardware required to hold these parts together, cabling to interconnect the internal electrical circuits and the necessary electrical connectors to join the warhead system to the adaption kit. The power supply, derivation of arming and firing signals and special hardware needed to mount the warhead in its compartment are specifically excluded.

(6) Arming System - That portion of the weapon which derives (originates) the signals required to arm, safe, or re-safe the firing system and the fuzing system. The arming system will normally consist of pullout plugs, arming bars, or similar components.

(7) Fuzing System - That portion of the weapon which derives the signals that discharge the firing system. This system normally consists of such components as pressure, proximity, time, or acceleration-sensing elements.

(8) Power Supply - This is the basic source of power which provides the electrical energy needed for the operation of an atomic weapon.

(9) Adaption Kit - Those items peculiar to the warhead installation less the warhead; namely, the arming and fuzing systems, power supply, and all hardware, adaptors, etc., required by a particular installation.

(10) Warhead Installation - Consists of the warhead and the adaption kit.⁷

⁷"Glossary of Terms Used in Military Characteristics for a 22

A modern nuclear bomb, the MK 61, contains over 4,000 components.⁸

The design and operation of these complicated devices are, however, based on certain fundamental physical principles and on the physical and nuclear characteristics of certain special materials, some of which occur naturally and others that are man-made and that have been present on earth for only a little longer than the last 50 years.

Nuclear weapons design is a largely empirical science: even now, after more than 50 years of U.S. nuclear weapons development, there are still not a set of "first principles," i.e., completely known, expressed, and rigorous laws and equations, of nuclear weapons physics.⁹

U. S. scientists have been designing nuclear weapons since 1943, and thermonuclear weapons since 1951. Along the way, many designs have proven to be failures, and many nuclear tests have brought surprising results.

The use of nuclear energy for explosive purposes involve several fields of knowledge:

Inch Outside Diameter Atomic Warhead for Air Defense," U.S. Air Force Air Defense Command Historical Study No. 21, BOMARC AND NUCLEAR ARMAMENT, 1951 - 1963, Thomas W. Ray, 1963, p. 44.

⁸Department of Energy Fiscal Year 1984 National Defense Programs Authorization, Hearing before the Subcommittee on Strategic and Theater Nuclear Forces of the Committee on Armed Services, United States Senate, Ninety-Eighth Congress, First Session, on S. 1106 and S. 1107, 12 April 1983, p. 88.

⁹A good overview of the development of principles of design for U.S. nuclear weapons is AN INTRODUCTION TO NUCLEAR WEAPONS, WASH-1037 (revised), Samuel Glasstone and Leslie M. Redman, U.S. Atomic Energy Commission, Division of Military Application, Washington, D.C., June 1972. This 166-page document provides a description of nuclear weapons operating principles, and describes how the ideas used in weapons design have changed over the years. It contains sections describing the principles of nuclear energy release; the fission process in weapons; fission weapons development; implosion systems; nuclear systems; boosting; thermonuclear weapons development; weapon vulnerability and hardening measures; and weapons safety and safeguards. While not intended as a technical guide to weapons design, the Glasstone and Redman paper is nonetheless one of the most comprehensive yet concise overviews ever written about its subject.

1. The nuclear characteristics, especially with respect to neutrons, of all materials which are employed;
2. The physical, chemical, and metallurgical characteristics of these materials;
3. The mechanics and dynamics of methods of initiating the nuclear energy release; and
4. The behavior of supercritical systems in which the energy generation per unit volume is very large.¹⁰

What happens in a nuclear explosion occurs very rapidly, is difficult to measure or capture on a moment-by-moment basis, and can, in many cases, only be inferred or estimated from the by-products and effects of the explosion that are collected by diagnostic experiments.

Processes of interest in a nuclear explosion include high explosives detonation; implosion of materials to high densities; intermixing of fluids (in the form of plasmas) by instabilities; ignition of thermonuclear burn; transport (movement) of neutrons, photons (thermal radiation), and charged particles; and intermixture of neutronic and thermonuclear reactions.

Even the most up-to-date computer simulations, based on hundreds of atmospheric and underground nuclear tests, can only approximate what happens.¹¹ Nuclear weapon operation is intrinsically complex. Implosion asymmetries and the instabilities of material interfaces can lead to complex geometries for imploded nuclear fuel. These geometric details are difficult to predict and model mathematically.

Many uncertainties about nuclear weapon operation remain even today. Over broad ranges of temperature and pressure, weapon

¹⁰Report of the Manager, Santa Fe Operations, U.S. Atomic Energy Commission, July 1947 to July 1950, Book Two, p. 12.

¹¹RELIABILITY OF THE NUCLEAR STOCKPILE UNDER A CTB, RDA-TR-122100-001-Rev. 1, Jack W. Rosengren, R & D Associates, Arlington, VA, revised December 1982, pp. 23, 123, 124, 131-133. For a good summary of the limits on simulating nuclear weapons explosions, see "Nuclear Tests Mean New Weapons," by Josephine Anne Stein in BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 42 No. 9, September 1986, pp. 8-11.

designers still do not have reliable data for equations of state and opacities for different materials. The designers cannot model in detail many basic processes in the operation of a weapon. Various processes are known only poorly; some major examples are:

- Growth of instabilities at material interfaces
- Deuterium-tritium ignition and burn during boosting
- Thermal-radiation transport

The first two items above make yield predictions of boosted weapons somewhat unreliable; the latter item makes multistage weapon modeling difficult. Modeling the operation of a secondary (thermonuclear) stage has many uncertainties, due mostly to a lack of diagnostic information. There are currently no meaningful laboratory experiments, short of a full-scale explosion, to study the implosion and burn of a secondary; while recent inertially-confined fusion (ICF) experiments have provided some data, it is applicable only on a microscopic scale.¹²

The reliable modeling of radiation flow between the fission trigger and fusion fuel capsule(s) in multistage weapons has also proved difficult. There has been an almost universal but variable discrepancy between the calculated and measured energy transfer from primary (fission stage) to secondary (fusion stage) in radiation-coupled devices.

Some of the greatest difficulties in calculating radiation generation and flow have been revealed by calculations associated with ICF experiments; for both ICF and weapons, several possible explanations of the disagreement between computations and experiments have been proposed, including more elaborate radiation transport (movement) models and computer simulations. The weapons laboratories have also been improving their experimental data base.¹³

The computer programs used by the weapons laboratories simulate weapon component geometry as a mesh of discrete spatial elements, and the advance of time is treated in discrete steps. Limitations in the speed and memory of available computers have historically limited the degree of detail that can be routinely included in computer-based weapon and explosion models. Only very recently, with the advent of new generations of "supercomputers" have the labs

¹²Rosengren, RDA-TR-122100-001-Rev. 1, p. 156.

¹³Rosengren, RDA-TR-122100-001-Rev. 1, pp. 147-149.

been able to model explosions in three dimensions. (In one case, three-dimensional modeling of the W-79 artillery-fired artillery projectile revealed potential one-point safety problems.)

Within these computer simulations (or "codes," as they are known in the weapons labs), times, distances, temperatures, and pressures are of paramount importance.

In order to understand weapons materials and components and their functions and configurations, and to better comprehend the goals and problems of post-World War II U.S. nuclear weapons designers, we must begin with an understanding and appreciation of the physical forces and phenomena involved in the detonation of nuclear weapons.

This educational process must start at the lowest level of matter, in the structures and units of atomic nuclei. In addition, the jargon of the weapons builders must be defined and understood (see the Glossary).

Units - A Frame of Reference

There are four key parameters of nuclear weapons design and performance: time, distance, temperature, and pressure.

The science of the behavior of materials at extremely high temperatures and pressures involves both thermodynamics and hydrodynamics, two of the scientific disciplines vital to the design and development of nuclear weapons. An exploding nuclear warhead creates the highest pressures, most extreme temperatures, and greatest densities ever made by man on earth, within some of the shortest times ever measured.

This all occurs within very small confines: the casing of the weapon, before the resultant expanding fireball breaks through. An understanding of these concepts of time, distance, temperature, pressure and density is required before reading the following sections which describe nuclear fission, nuclear fusion, and nuclear weapons design and operation.

Time

Perhaps one of the best and most evocative delineations of the concept of time, humankind's arbitrary division of eternity, is given in Robert Ardrey's AFRICAN GENESIS:

I suspect that the first and last difficulty afflicting the student of pre-human affairs is the length of the yard-stick he must carry about with him. If a million years were as easy to grasp as a breakfast tray, then much of

our history would be as digestible as orange juice. It is the yard-stick, not evolution, that sticks in our throats; and with very good reason, since for a creature whose life span is three score and ten, any real comprehension of a million years is a bit too much to ask. But if we cannot truly understand the term, then we may at least gain a certain respect for it. The task is not too difficult, since it is more one of the spirit than of the mind. What is demanded is humility, not mathematics.¹⁴

Ardrey then goes on to compare the distant past to a long beach, starting in California and sweeping south along the Pacific Ocean to the tip of South America. The problem in grasping units of time involved in nuclear reactions is very similar to that of appreciating the prehistoric eons, except that the times used in nuclear weapons design are the inverse of the eons -- a microcosm, instead of a macrocosm.

Those persons who make their livings in the computer industry are usually at least passingly familiar with these microscopic units of time: computer program instructions are executed in microseconds (millionths of seconds), and new, state-of-the-art cryogenic computers are routinely performing operations in picoseconds (trillionths of seconds). Most modern large-scale computers can process instructions and data at speeds measured in nanoseconds (billionths of seconds).

Within human perception, such fleeting moments can never be sensed directly. These units of time are as difficult to grasp for their ephemerality as the ages are to grasp for their longevity. These fractions of seconds and millions of years represent diametrically-opposed concepts: the infinitesimal versus the infinite. If Ardrey's Pacific coast beach is the span of history, then a second is but a grain of sand, and a microsecond is only a silica molecule in the grain.

A second of time is relatively easily observed: a movement of a watch hand; a beat of the heart; or a number of yards driven on a freeway. Let us then consider the ideas of microseconds, nanoseconds, and picoseconds. If our perception of time could be slowed by a factor of one million so that every click of a watch that counted each second marked off a

¹⁴AFRICAN GENESIS, Robert Ardrey, 1961. Dell paperback edition June 1967, pp. 211-212.

microsecond instead, and the earth continued rotating at its normal rate, then a full second (a million microseconds) would not pass for 11 and a 1/2 days!

If each click were a nanosecond in this frame of reference, then a full second would last 31 and 2/3 years; if each click were a picosecond, a second would stretch over 316 centuries. Yet many complicated nuclear reactions, involving astronomical numbers of atomic nuclei, occur in an exploding three-stage thermonuclear weapon in a few hundred microseconds -- just a few hundred clicks of the watch in our microsecond frame of reference.

The weapons scientists at Los Alamos devised their own unit of time during the early days of the Manhattan Project in World War II. A "shake" was one one-hundred millionth (1/100,000,000th) of a second (or 10 nanoseconds, 10^{-8} seconds) -- the time required for a uranium atom to fission (split apart). It was supposedly given this name because it was "faster than a shake of a lamb's tail."¹⁵ The full explosive process of a large hydrogen bomb occurs in a few hundred "shakes" -- a force comparable to all the energy that has ever gone into building human civilization can be unleashed for destruction in an almost-immeasurable moment.

In terms of other physical phenomena, a beam of light covers one foot in a nanosecond (the speed of light is 3×10^8 meters per second, or about 10^9 feet per second). An x-ray moves 10 feet in the time required for a uranium nucleus to fission (the distance between stages in multistage thermonuclear weapons is usually on the order of a foot or less). Large fission fragments travel only about four inches in the same time, if they do not collide with anything. Sound, by comparison, is very slow: only 1,000 feet in a second. Light is a million times faster: it can circle the earth 7 1/2 times in a second.

Distance

A similar contrast can be made for the distances involved between the parts of atomic nuclei in the molecules of nuclear explosives. Although most fission and fusion fuel stages in a nuclear warhead are located within inches or at the most, feet, of each other, this distance is enormous when compared to the spaces between particles in an atom.

A classic illustration of the size of atoms, and the distances between their constituent particles, is the often-

¹⁵THE CURVE OF BINDING ENERGY, John McPhee, Farrar, Straus and Giroux edition, New York, New York, 1974, p. 115.

cited analogy between an atom and a planetary system. In this frame of reference, if a hydrogen atom (the simplest atom, with a single electrically-positive charged proton in the nucleus, orbited by a single negatively charged electron) were to be enlarged 25 trillion (25,000,000,000,000) times, its nucleus would be just one inch in diameter, and its single electron "planet" would revolve in an orbit with a radius of 1,400 feet.

If our solar system were to be reduced in scale to the same proportions, so that the sun had a diameter of one inch, the 3,670,000,000 miles to Pluto's orbit (Pluto is the outermost known planet in our solar system) would be a little less than 353 feet¹⁶ -- just a quarter of the distance to the electron in our hydrogen "planetary" system. Thus a hydrogen atom is mostly empty space, and the apparent solidity of matter is only an illusion caused by the enormous numbers of atoms in a small volume.

Atoms themselves are so small that if a drop of water were enlarged to the size of the earth, the individual atoms in it would be smaller than oranges. Yet these "oranges," in particular, hydrogen atoms, are capable of releasing virtually incomprehensible amounts of energy through the conversion of matter to energy when very large numbers of them fuse (join) in an exploding thermonuclear bomb.

The weapons scientists devised their own measure of distance, just as they had for time. A "fermi," named after Italian scientist Enrico Fermi, who headed the team of engineers and physicists who produced the world's first controlled nuclear chain reaction at Chicago in December 1942, is about the diameter of an electron, or 10^{-13} centimeters.¹⁷ There are over 300 trillion (300,000,000,000,000) fermis in a foot.

¹⁶A HISTORY OF THE AIR FORCE ATOMIC ENERGY PROGRAM, 1943 - 1953, Lee Bowen and Robert D. Little, et. al., 1959, Vol. I, p. 7. This 5-volume, 9-part history, classified TOP SECRET until 1980, served as a major source for this information service. It will be cited often, and for that reason, will be referred to from here on as AF ATOMIC ENERGY PROGRAM, followed by volume and page citation. Its declassification in 1980 was due largely to the efforts of Dr. David A. Rosenberg, and I owe him a large debt of gratitude. To write this history, I stood on the shoulders of a number of giants, and Dave was one of the tallest.

¹⁷McPhee, p. 117.

Temperature

Another concept required to understand postwar nuclear weapons development is temperature. We normally tend to associate temperature with our perception of it, usually as radiation from warm objects felt by our skin, such as that from hot water, fire, sunlight, etc.. However, for the purposes of this discussion, temperature should be considered primarily as a measure of the activity of atoms or nuclei of atoms within a substance.

In this context, temperature is proportional to the kinetic energy, i.e., the energy due to motion or velocity, of an atom or its nucleus. The "hotter" a substance, i.e., the greater its heat content, the more vigorous the movement of its individual atoms. As materials heat, their molecules absorb energy and become agitated.

This molecular kinetic energy is very powerful at temperatures of several hundred million degrees. It is sufficient to strip a hydrogen atom of its electron; give the nucleus a velocity sufficient to overcome the repulsive electrostatic force of another hydrogen atom nucleus (both are positively charged, and like charges repel one another); and smash the two nuclei together, forming a helium nucleus and releasing energy.

The mass of the helium nucleus is slightly less than the combined mass of the two hydrogen nuclei; the missing mass is converted into energy according to Einstein's famous equation, $E = mc^2$ (energy = change in mass multiplied by the speed of light squared).

Because this collision can occur only at extremely high temperatures after the individual nuclei have gained large kinetic energies, this type of reaction is known as a thermonuclear reaction. These fusion reactions between hydrogen nuclei are the source of the explosive force of a thermonuclear warhead, and are the reason that such a weapon is commonly called a "hydrogen bomb."

Pressure

Concurrent with the requirement for high temperatures in atomic and thermonuclear weapons is the requirement for high pressures. Fission and fusion reactions function more efficiently under high pressures.

A metric unit of pressure is the "bar," which is equal to 98.69% of one atmosphere, or 14.5 pounds per square inch (psi) or 29.53 inches of mercury at sea level. Typical daily air pressure fluctuations from sea level normal (14.7 psi, or 29.92 inches of mercury) are measured in millibars (one

millibar = 1/1,000th of a bar). These are the numbers seen on daily weather maps.

Pressure in the context of nuclear weapons refers mainly to the effects, primarily recoil forces, generated by the bombardment of a substance, such as a fission bomb core or a "tuballoy" jacket around a fusion fuel charge, by particles or shock waves. These pressures are almost unimaginable: in an exploding thermonuclear weapon, pressures are as high as a trillion atmospheres -- about 8,000,000,000 tons per square inch.¹⁸

Pressures within exploding nuclear devices are typically measured in megabars (one megabar = one million bars, or approximately one million times normal sea level air pressure). A trillion atmospheres is about a million megabars, which is also one thousand gigabars, or one terabar. In an exploding fissile core, pressures on the order of 10 to 1,000 megabars are generated in its center.¹⁹

The result of such unearthly pressure on a substance is an increase in its density, which is a measure of the compaction or packing of its constituent molecules and atoms. High densities in fission and fusion weapon fuels are especially important: the more closely crowded the atoms of a substance, the greater the likelihood of collision between highly-agitated atoms, and the higher the frequency or rate of such collisions.

In a fissile material, a higher compaction results in a shorter path that neutrons must travel between nuclei to cause more fission. Also, the greater the density of a fissile or fusionable material, the better it can withstand outwardly-directed forces which seek to disassemble it and end the nuclear reactions occurring within it. Pressures in the center of an exploding hydrogen bomb can generate plasmas (ionized gases) that are twenty times denser than solid lead.²⁰

Another factor entering into nuclear reactions is the tremendous amount of energy released by the conversion of mass to energy. According to Einstein's law, energy = mass multiplied by the speed of light squared, the conversion of one gram (about 0.035 ounces) of nuclear material to energy

¹⁸"The H-Bomb Secret," Howard Morland, THE PROGRESSIVE, Vol. 43 No. 11, November 1979, p. 21.

¹⁹Glasstone and Redman, WASH-1037 (Revised), p. 30.

²⁰McPhee, p. 164.

yields 85 billion (85,000,000,000) British Thermal Units (BTUs), or 33,000,000 horsepower-hours, or 25,000,000 kilowatt-hours (25,000 megawatt-hours or 25 gigawatt-hours).

This is enough energy to fulfill the electrical needs of a large city for many weeks: the annual electrical consumption of the entire United States -- about 600,000 megawatt-hours -- could be met by the complete fission of only 24 grams of fissile material. This is less than one ounce! One kilogram (2.2 lbs.) of plutonium, if completely fissioned, releases energy equivalent to 20,000 tons of exploding TNT.

To further illustrate the potential energy of an atomic weapon, comparison with an explosion of ordinary high explosive, such as TNT, is instructive. The energy released when a pound of TNT explodes would, if converted entirely into heat, raise the temperature of 36 lbs. of water (just over a half-cubic foot, or 4.3 gals.) from freezing temperature (32° F, or 0° C) to boiling temperature (212° F, or 100° C). The thermal energy released by a pound of completely-fissioned uranium would produce an equal temperature rise in over 200,000,000 lbs. of water (3,205,128 cubic feet, nearly 24 million gals.).

This is the environment of nuclear reactions: short times, high temperatures, huge pressures, great densities, and tremendous energies. Given these conditions, measuring the effectiveness and performance of a particular weapon design is extremely difficult. It is a remarkable tribute to nuclear weapons engineers and instrument designers that these parameters were ever measured at all.

Fission Physics

A nuclear explosion, like a conventional chemical explosion, is caused by the release of a large amount of energy in a very short time. The energy of a TNT explosion, released in about a millisecond, is created by a rearrangement of the hydrogen, oxygen, carbon, and nitrogen atoms that comprise the TNT.

On the other hand, the energy of a nuclear explosion, released in about a microsecond, is produced by rearrangement of the components of the atomic nuclei; hence the expression "nuclear energy." In this discussion, "nuclear" includes both "atomic" and "thermonuclear."

Nuclear explosions are created by either uncontrolled nuclear fission or uncontrolled nuclear fusion. To understand these processes, let us examine the atom itself. All elements in the universe are made up of atoms.

An atom of an element is the smallest particle of that element which retains its characteristic physical and chemical properties. An atom can be broken up into even smaller components, electrons and nucleons, but these particles have entirely different characteristics. A nuclear explosion is created by putting these components to work.

An atom consists of a nucleus and a number of electrons revolving (orbiting) around the nucleus. The nucleus is small and dense and made up of protons and neutrons. Protons have a positive electrostatic charge, but neutrons, as the name implies, are electrostatically neutral. These particles of a nucleus, both protons and neutrons, are known as nucleons.

Let us take hydrogen as an example. The simplest of all elements, it has only one nucleon (a proton) and one electron orbiting around it. A much more complex element, natural uranium, may have between 227 and 240 nucleons, and 92 electrons.

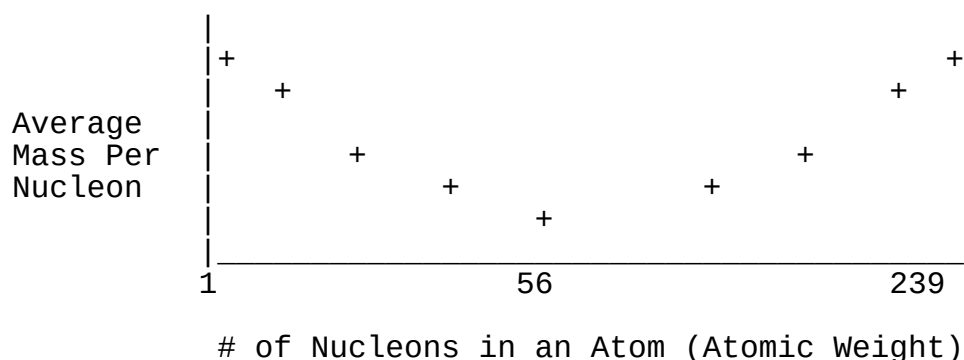
With this information, we are now ready to see what makes a nuclear warhead release enormous amounts of energy. The nuclei of all atoms consist of varying arrangements of protons and neutrons. An isotope of an element is an atom of the element that has the same number of electrons and protons (the atomic number) as all other isotopes of the element, but which also has a different and unique number of neutrons, hence a different atomic weight, than other isotopes of the element.

The word "isotope" derives from the Greek expression meaning "in the same place," as in the same place on the periodic table.

It is a fact of nature that a proton by itself or a neutron by itself always has more mass than when that same proton or neutron is a constituent of an atomic nucleus. Furthermore, the average mass per nucleon, protons and neutrons, differs for each atomic nucleus. In a nucleus of about 56 nucleons (the element barium), the mass per nucleon is less than in any other combination.

Figure I-1 shows this graphically. Fission and fusion can occur among elements at the high and low ends of the periodic table, respectively, because nature always tries to drive the mass per nucleon of elements toward the low point around 56 nucleons. This is the most stable region of nuclei. Elements at the far right side of the chart in Figure I-1, including many man-made, short-lived elements, are always radioactive, and decay spontaneously to more-stable elements. Nuclear fission, especially a nuclear explosion, is a method of speeding up this decay.

Figure I-1



The change in the average mass per nucleon with the number of nucleons in an atom. The mass per nucleon is least in a nucleus of 56 nucleons.

Another fact of nature is that mass and energy are interchangeable under certain conditions. If, for example, a nucleus of uranium-235 containing 235 nucleons is fragmented into two fairly equal parts, the mass per nucleon before fragmentation is at the high right-hand end of the scale in Figure I-1. The mass per nucleon after fragmentation is at a lower point in the scale. This "lost" mass per nucleon is converted instantaneously into energy, primarily kinetic energy of the fission products, according to Einstein's law, $E = mc^2$ (energy = mass change multiplied by the speed of light squared).

A heavy nucleus fragments when it is bombarded by neutrons. The resultant particles are smaller atomic nuclei called fission products. Uranium and plutonium release two or three neutrons, as well as large fission products, when bombarded by neutrons. These newly-released neutrons, moving very rapidly, collide with other uranium or plutonium nuclei in the vicinity and may cause these nuclei to fission, in turn releasing other neutrons, as shown in **Figure I-2**.

Although they appear to be dense and heavy metals, plutonium and uranium, like all matter, are actually composed of nuclei spaced widely apart (at least 2,000 nuclei diameters from one another -- see discussion under "Distance" above). These nuclei are orbited by many small electrons, which do not affect the bombarding neutrons (an electron is virtually massless compared to a neutron or proton).

If only a small mass of uranium or plutonium is available, the neutrons released by fission can escape from the material without causing further fission. When a mass of fissile material is increased until just as many neutrons escape per unit time as are released by fission, then a critical amount

of material exists and a chain reaction is barely sustained. If this amount of material is increased still further, the neutrons build up in number and many more fissions occur in each successive generation of neutrons, creating a supercritical mass and a nuclear explosion.

This neutron flux becomes quite large. The mass of fissioning plutonium, uranium or thorium required to produce just one kiloton (1 KT) of weapon yield causes approximately 10^{23} atoms to fission, each atom releasing between 2.5 and 3 neutrons per generation.

There are over 100 different ways that uranium, plutonium, and thorium can fission, creating several dozen possible combinations of by-products, both radioactive and inert (non-radioactive).

In summary, nuclear explosions are like chemical explosions in that both are the result of rearrangements. In the case of nuclear explosions, a rearrangement of nucleons in atomic nuclei occur, while in chemical explosions, a rearrangement of atoms within the chemical occur. Further, while radioactive by-products result from nuclear explosions, they are not produced by chemical explosions.

Fusion Physics

The fusion process is the opposite of fission: nuclei of light elements combine with each other and form heavier nuclei. Since extremely light elements have more mass per nucleon than heavier elements (up to element 56: see Figure I-1), mass is converted into energy and released as kinetic energy of the resultant products.

Fusion explosions caused by thermonuclear weapons force hydrogen or hydrogen isotopes to fuse (join) into heavier elements. The process requires extreme temperatures and pressures. So far, the only practical way to obtain the necessary temperatures of hundreds of millions of degrees to start a large, uncontrolled fusion explosion is to use a fission explosion. U.S. thermonuclear weapons actually use two fission explosions, as explained in "Fusion Explosion Principles" below: one to compress the fusion fuel, and a second to "ignite" the compressed fuel.

Before proceeding, an explanation of notation is in order. Isotopes of elements (and some subatomic particles) will be represented occasionally in the form ${}_m^X$. In this notation, X is the chemical atomic element symbol; those pertinent to the following discussion are shown below:

<u>Symbol</u>	<u>Element</u>
H	Hydrogen
He	Helium
Li	Lithium
Be	Beryllium

The subscript m represents the number of protons in the isotope (the atomic number). The superscript n represents the number of nucleons -- the atomic weight of the isotope. Both protons and neutrons have an atomic weight of one per nucleon.

Because of the extremely elevated temperature at which fusion reactions occur, in the hundreds of millions of degrees, the electrons normally associated with an element have acquired sufficient energy to tear themselves loose from the electrostatic attraction of the protons in the nucleus. As a result, the ionized nuclei are stripped of all electrons and are positively charged and moving about in a dense gaseous cloud, known as a plasma. Their tendency to repel one another because of identical electrostatic charge is overcome by their high kinetic energies.

Lithium and hydrogen isotopes are very important in fusion reactions. Lithium has several isotopes, ranging from ${}^5_3\text{Li}$ to ${}^9_3\text{Li}$. The two isotopes most important in hydrogen bomb design are the isotopes lithium-6 and lithium-7, or ${}^6_3\text{Li}$ and ${}^7_3\text{Li}$. Lithium is the lightest of all metals, with a density only about half that of water. It does not occur free in nature; it is found combined in small amounts in nearly all igneous rocks, and in the waters of mineral springs.

Lithium-7, a natural isotope, is usually produced electrolytically from lithium chloride. Lithium-6 makes up about 7.5% of all naturally-occurring lithium (about 10 times the fraction of uranium-235 in natural uranium) and is separated by electrolysis, distillation, chemical exchange, or electromagnetic methods.²¹ The other three isotopes of lithium do not occur in nature and are artificially-produced

²¹The largest, most difficult, expensive, and time-consuming part of a country's preparations to "go thermonuclear" and produce hydrogen weapons is the design, construction, operation, and maintenance of an enormous industrial base to produce special weapons components and materials. Building up such a capacity typically takes many years, and the appearance of certain characteristic structures is often a sure sign of thermonuclear intent, long before actual weapons testing.

radioactive isotopes. Since the difference in weight between lithium-6 and lithium-7 is on the order of 15%, while the difference in weight between U-235 and U-238 is about one percent, the separation of lithium isotopes is much easier and cheaper than the separation of uranium isotopes.

Hydrogen, the most abundant and lightest element in the universe, has three isotopes: protium, deuterium, and tritium. These isotopes can be represented as P, D, T, or ${}_1\text{H}^1$, ${}_1\text{H}^2$, and ${}_1\text{H}^3$, respectively. The nucleus of protium is composed of a single proton; the nucleus of deuterium is a proton and a neutron; and the nucleus of tritium is a proton and two neutrons. Tritium does not occur naturally; it is an artificial radioactive isotope produced by bombarding lithium with neutrons in nuclear reactors.

Deuterium can be found in a ratio of about three parts per 20,000 (0.015%) in all naturally-occurring hydrogen. Protium comprises the remaining 99.985% of natural hydrogen. A protium nucleus is called a proton; a deuterium nucleus is a deuteron; and a tritium nucleus is a triton.

Another useful element in both nuclear fission and fusion weapons is beryllium. Beryllium isotopes range from ${}_4\text{Be}^6$ to ${}_4\text{Be}^{11}$; only ${}_4\text{Be}^9$ is nonradioactive, and is 100% of all naturally-occurring beryllium. The other five isotopes are artificially-produced. Beryllium has two applications in fission weapons: when bombarded by alpha particles (helium nuclei, two protons and two neutrons) from radium or polonium, for instance, neutrons are released in the ratio of about 30 neutrons per million impacting alpha particles. Because of this property, beryllium can be used in neutron sources (fission initiators) for atomic bomb cores.

In addition, beryllium has a low thermal neutron absorption cross section (a thermal neutron is a relatively-low energy, i.e., low velocity, neutron released at early stages of atomic explosions). This means that when beryllium is used as a reflector around a critical mass of uranium or plutonium, impacting neutrons radiated from the nuclear reaction are bounced (reflected) back into the mass, speeding up the reaction rate and "burning" more of the fissile material. Beryllium has been employed as a substitute for gold or natural uranium reflectors in early atomic weapons, thereby saving much weight and money (see "Advances in Reflectors and Tamperers" in Volume II).

In a fusion weapon, when struck by a high-energy neutron released by a thermonuclear reaction, beryllium releases two high energy neutrons of its own. This is a convenient method for increasing the neutron flux (neutron density) in an exploding hydrogen bomb (see note 4 below).

Lithium and hydrogen are combined in nuclear weapons in various forms of lithium hydride, as shown below.

<u>Lithium Isotope</u>	<u>Hydrogen Isotope</u>	<u>Hydride Formula</u>
Lithium-6	Protium	Li^6P (${}_3\text{Li}^6_1\text{H}^1$) (lithium-6 protide)
Lithium-6	Deuterium	Li^6D (${}_3\text{Li}^6_1\text{H}^2$) (lithium-6 deuteride)
Lithium-6	Tritium	Li^6T (${}_3\text{Li}^6_1\text{H}^3$) (lithium-6 tritide)
Lithium-7	Protium	Li^7P (${}_3\text{Li}^7_1\text{H}^1$) (lithium-7 protide)
Lithium-7	Deuterium	Li^7D (${}_3\text{Li}^7_1\text{H}^2$) (lithium-7 deuteride)
Lithium-7	Tritium	Li^7T (${}_3\text{Li}^7_1\text{H}^3$) (lithium-7 tritide)

In brief, the H-bomb mechanism used in two- and three-stage weapons works as follows: inside the bomb, lithium serves as a source of instantly-made tritium, the heaviest isotope of hydrogen, and one of the two hydrogen isotopes used as fuel in thermonuclear reactions; it is generated when the fissile "sparkplug" in the middle of the fusion fuel mass explodes (fissions) and releases a great swarm of high-energy neutrons.

These neutrons strike the lithium-6 (or lithium-7; both can be used), producing helium and tritium, and another neutron, if lithium-7 is present. The tritium then reacts with deuterium in a fusion reaction which generates helium and releases vast amounts of energy. Even more energy is released by the formation of tritium, and other fusion reactions when tritium fuses with tritium and deuterium fuses with deuterium.

If every atom in a pound of lithium-6 deuteride split and every tritium atom thus formed fused to a deuterium atom, then this single pound of salt would release energy equivalent to 29,000 tons of TNT (29 kilotons TNT equivalent), twice the power of the LITTLE BOY bomb that

leveled Hiroshima in World War II.²² In practice, such complete fusion of lithium-6 deuteride cannot be realized, but it is probable that between five and 15 kilotons of TNT-equivalent energy is yielded by each pound of lithium-6 deuteride in U.S. thermonuclear weapons.

The main thermonuclear sequence in a multistage weapon is shown below. D represent deuterium (${}_1\text{H}^2$), and T represent tritium (${}_1\text{H}^3$). The symbol ${}_0\text{n}^1$ represents a neutron. Below the nuclear equation, the rearrangements of protons (p) and neutrons (n) is shown. The symbol MeV is described following the lists of reactions. Note that in all cases, the total number of neutrons and protons after the reaction is always the same as it was before the reaction.

- Fission (of the sparkplug) releases neutrons and heat (expressed as thermal kinetic energy of the neutrons).
- ${}_3\text{Li}^6 + {}_0\text{n}^1 \text{ -----} \rightarrow {}_2\text{He}^4 + \text{T} + 4.7 \text{ MeV}$
 $(3\text{p}, 3\text{n}) + \text{n} \text{ -----} \rightarrow (2\text{p}, 2\text{n}) + (\text{p}, 2\text{n})$
- $\text{D} + \text{T} \text{ -----} \rightarrow {}_2\text{He}^4 + {}_0\text{n}^1 + 17.58 \text{ MeV}$
 $(\text{p}, \text{n}) + (\text{p}, 2\text{n}) \text{ -----} \rightarrow (2\text{p}, 2\text{n}) + \text{n}$
- ${}_0\text{n}^1 + {}_{92}\text{U}^{238} \text{ -----} \rightarrow \text{More neutrons (second fission}$

explosion in pusher) + radioactive

by-products + energy

Additional fusion reactions include:

- (A) ${}_3\text{Li}^6 + \text{D} \text{ -----} \rightarrow 2({}_2\text{He}^4) + 22.4 \text{ MeV}$
 $(3\text{p}, 3\text{n}) + (\text{p}, \text{n}) \text{ -----} \rightarrow 2(2\text{p}, 2\text{n})$
- (B) ${}_3\text{Li}^6 + \text{T} \text{ -----} \rightarrow 2({}_2\text{He}^4) + {}_0\text{n}^1 + 16 \text{ MeV}$
 $(3\text{p}, 3\text{n}) + (\text{p}, 2\text{n}) \text{ -----} \rightarrow 2(2\text{p}, 2\text{n}) + \text{n}$
- (C) ${}_3\text{Li}^6 + {}_1\text{H}^1 \text{ -----} \rightarrow {}_2\text{He}^4 + {}_2\text{He}^3 + 4.0 \text{ MeV}$
 $(3\text{p}, 3\text{n}) + \text{p} \text{ -----} \rightarrow 2(2\text{p}, 2\text{n}) + (2\text{p}, \text{n})$
- (D) ${}_3\text{Li}^7 + {}_1\text{H}^1 \text{ -----} \rightarrow 2({}_2\text{He}^4) + 17.3 \text{ MeV}$
 $(3\text{p}, 4\text{n}) + \text{p} \text{ -----} \rightarrow 2(2\text{p}, 2\text{n})$
- (E) ${}_3\text{Li}^7 + \text{D} \text{ -----} \rightarrow {}_3\text{Li}^8 + {}_1\text{H}^1 + 0.3 \text{ MeV}$
 $(3\text{p}, 4\text{n}) + (\text{p}, \text{n}) \text{ -----} \rightarrow (3\text{p}, 5\text{n}) + \text{p}$

²²Glasstone and Redman, WASH-1037 (Revised), p. 156.

- (F) ${}_3\text{Li}^7 + {}_0\text{n}^1 \rightarrow {}_2\text{He}^4 + \text{T} + {}_0\text{n}^1 - 2.5 \text{ MeV}$
 $(3\text{p}, 4\text{n}) + \text{n} \rightarrow (2\text{p}, 2\text{n}) + (\text{p}, 2\text{n}) + \text{n}$
- (G) ${}_3\text{Li}^7 + \text{D} \rightarrow {}_4\text{Be}^8 + {}_0\text{n}^1 + 15.1 \text{ MeV}$
 $(3\text{p}, 4\text{n}) + (\text{p}, \text{n}) \rightarrow (4\text{p}, 4\text{n}) + \text{n}$
- (H) $\text{D} + \text{D} \rightarrow \text{T} + \text{p} + 4.036 \text{ MeV}$
 $(\text{p}, \text{n}) + (\text{p}, \text{n}) \rightarrow (\text{p}, 2\text{n}) + \text{p}$
- (I) $\text{D} + \text{D} \rightarrow {}_2\text{He}^3 + {}_0\text{n}^1 + 3.265 \text{ MeV}$
 $(\text{p}, \text{n}) + (\text{p}, \text{n}) \rightarrow (2\text{p}, \text{n}) + \text{n}$
- (J) $\text{D} + \text{D} \rightarrow {}_2\text{He}^4 + 23 \text{ MeV}$
 $(\text{p}, \text{n}) + (\text{p}, \text{n}) \rightarrow (2\text{p}, 2\text{n})$
- (K) $\text{T} + \text{T} \rightarrow {}_2\text{He}^4 + 2({}_0\text{n}^1) + 11.4 \text{ MeV}$
 $(\text{p}, 2\text{n}) + (\text{p}, 2\text{n}) \rightarrow (2\text{p}, 2\text{n}) + 2\text{n}$
- (L) ${}_2\text{He}^3 + \text{D} \rightarrow {}_2\text{He}^4 + \text{p} + 18.8 \text{ MeV}$
 $(2\text{p}, \text{n}) + (\text{p}, \text{n}) \rightarrow (2\text{p}, 2\text{n}) + \text{p}$
- (M) $\text{D} + {}_0\text{n}^1 \rightarrow \text{T} + 6.2 \text{ MeV}$
 $(\text{p}, \text{n}) + \text{n} \rightarrow (\text{p}, 2\text{n})$
- (N) $5 \text{ D} \rightarrow \text{p} + {}_2\text{He}^3 + {}_2\text{He}^4 + 2({}_0\text{n}^1) + 25 \text{ MeV}$
 $5 (\text{p}, \text{n}) \rightarrow \text{p} + (2\text{p}, \text{n}) + (2\text{p}, 2\text{n}) + 2\text{n}$
- (O) ${}_2\text{He}^3 + {}_0\text{n}^1 \rightarrow \text{T} + \text{p} + 0.766 \text{ MeV}$
 $(2\text{p}, \text{n}) + \text{n} \rightarrow (\text{p}, 2\text{n}) + \text{p}$

The following notes apply to the preceding reactions:

- (1) The symbol MeV represents a million electron volts, a unit of energy. One atomic mass unit, converted to energy by the formula $E = mc^2$, is equivalent to 931 MeV.²³ It takes about 10^{16} MeV to raise the temperature

²³One atomic mass unit is approximately 1.66×10^{-27} kilograms; when this is multiplied by the speed of light squared and converted from joules to MeV (1 joule = 6.242×10^{12} MeV), the result is approximately 931 MeV. (PHYSICS FOR STUDENTS OF SCIENCE AND ENGINEERING, Combined Edition, David Halliday and Robert Resnick, John Wiley & Sons, Inc., New York, 1962, p. 148; ATOMIC ENERGY FOR MILITARY PURPOSES, Henry DeWolf Smyth, Princeton University Press, Princeton, New Jersey, 1945, p.

of one pound of water one degree Fahrenheit. One kiloton (1 KT) of explosive bomb energy yield from a D + T reaction produces about 2.6×10^{25} MeV, or about 2.6×10^{28} MeV per megaton (which is equivalent to 2.6×10^{12} BTU). Most of the energy released in fusion reactions is in the kinetic energy of neutrons. In the D + T reaction, the helium (${}^4_2\text{He}$) is formed with a resultant kinetic energy of 3.5 MeV, but the neutron released by the reaction has a kinetic energy of 14.1 MeV, about four times larger.

The advantage of a fusion explosion over a fission explosion can be shown as follows: 200 MeV are released per fission of ${}^{235}_{92}\text{U}$, which has an atomic weight of 235, with 2.55×10^{24} atoms per kilogram, and 7.5×10^{10} BTU per kilogram (200 MeV is enough energy to make a visible grain of sand jump). In the fusion of protium (ordinary hydrogen) with lithium-7, 17.3 MeV of energy is produced; ${}^7_3\text{Li}$ has an atomic weight of only 8, with 7.5×10^{25} atoms per kilogram -- about 30 times that of uranium-235.

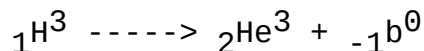
In terms of weight alone, deuterium-tritium fusion yields over four times as much energy as uranium fission. Whereas the fission of a single uranium nucleus yields nearly 12 times as much energy as the fusion of single tritium and deuterium nuclei, the fusion reaction requires only five nucleons. The uranium fission reaction involves a single nucleus of 235 nucleons. For each fusing nucleon, the amount of energy (17.6 MeV/5) is about 3.5 MeV per nucleon; each fissioning nucleon yields only about 0.85 MeV (200/235). Thus fusion is more than four times (3.5/0.85) as efficient as fission.

A modern thermonuclear weapon may contain only a few tens of kilograms of ${}^{235}_{92}\text{U}$ (mostly in the primary, sparkplug and pusher, which may use uranium-238 or uranium-235), but it may also contain several hundred kilograms of LiH.

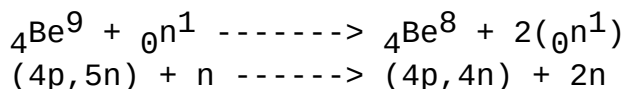
- (2) Most of the fusion reactions shown above are very temperature-dependent. The more heat in the form of kinetic energy of reacting particles that is added, the faster the reaction. The D + D fusion reaction occurs at a temperature of 600 million degrees, but the T + D reaction can occur at "only" 70 million degrees. Even at this "low" temperature, the fusion reaction happens in less than a millionth of a second. In a fusion device

containing both T and D, a T-D reaction is about 150 times more likely to occur than a D-D reaction.²⁴

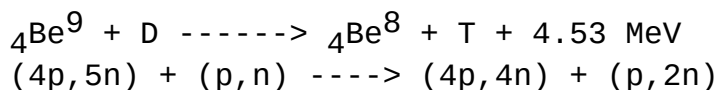
- (3) Tritium decays by releasing a beta particle (a negatively charged particle) from its nucleus:



- (4) The neutron flux (neutron density) in a thermonuclear explosion can be enhanced by the presence of beryllium:



When bombarded by deuterons, beryllium is also a source of tritium:



- (5) A fusion explosion produces about 30 times more neutrons per kiloton of yield than a fission explosion (this is the basis of the so-called "neutron bomb," which is actually a fusion weapon). Assuming the production of one neutron per 10 MeV of energy release, a one-megaton explosion produces 2.5×10^{27} neutrons.

These fusion-derived neutrons cause reactions similar to those resulting from fission, although their higher initial kinetic energies increase the likelihood of further neutron- and proton-releasing fusion reactions. In addition to these neutrons, tritium is generated in relatively large amounts, on the order of 10^{24} atoms of tritium per kiloton of fusion yield. Neutrons released by fission reactions have an energy of about 2.5×10^{-8} MeV and cannot fission ${}_{92}\text{U}^{238}$, but can transform lithium into tritium. Fusion-generated neutrons have an energy in excess of 1 MeV and can fission natural or enriched uranium (tuballoy or oralloy).

²⁴Memorandum to File dated September 12, 1952 by John S. Walker, Counsel to the Joint Committee on Atomic Energy, Subject: Status Report on the Thermonuclear Program (hereafter referred to as September 12, 1952 Walker memorandum), p. 34. In terms of nuclear physics, a T-D reaction has about 150 times the cross-section of a D-D reaction and will "burn" at only about 1/10th the temperature (kinetic energy level) required to initiate the D-D reaction.

- (6) Although the fusion reaction begins with transformation of lithium-6 to tritium, even ordinary lithium-7 is useful, as shown in equations (D), (E), (F), and (G). Equation (F) played a very important (and unanticipated) role in the development of early U.S. thermonuclear weapons. One surprise encountered during thermonuclear device tests was the large amount of tritium (T) produced from ordinary lithium-7. Only after there were unexpectedly high yields (from 68 to 175% more than predicted) from the first solid-fueled thermonuclear devices did scientists at Los Alamos accurately measure the cross section (probability of occurrence) for it. The affinity of Li-7 for neutrons turned out to be much stronger than expected, resulting in greater tritium production, and weapon efficiencies much higher than predicted.²⁵

Equation (F) is endoergic, that is, there is a net energy loss from the reactants on the left side of the equation to the products on the right. This threshold energy must be supplied via acceleration of the neutrons, for the reaction to occur. As shown dramatically in the Pacific in 1954, this was not a problem.

- (7) As can be seen from equations (C) and (D), even natural hydrogen (protium) enters into fusion reactions at sufficiently high temperatures. Hence ordinary lithium hydride (LiH, or ${}_3\text{Li}^7{}_1\text{H}^1$) can be used as fuel in thermonuclear weapons.²⁶
- (8) Reactions (H) and (I) occur with equal probability.
- (9) In a mixture of deuterium and tritium, the main reaction is between a deuteron and a triton which results in the release of a neutron as one of its products. In deuterium alone, a neutron is also a product of one of

²⁵"Nuclear Data: The Numbers Needed to Design the Bombs," Ben C. Diven, et. al., LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, pp. 121, 122.

²⁶That Li-7 was used in early thermonuclear weapons can be verified from "General Information Pertaining to the Handling of Lithium Hydrides" in "Movement of CASTLE Test Materials to the Pacific Proving Grounds," memo to Commander, Military Air Transport Service, Andrews Air Force Base, Washington, D.C., from Major P. W. Clarkson, U.S. Army, dated 4 December 1953. In addition, AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 40, cites a "tritium bonus" verified in 1954. This "bonus" resulted from the bombardment of Li-7 by neutrons, causing the generation of tritium, helium, and neutrons.

the deuteron-deuteron (D-D) reactions.

In another D-D reaction, a triton (T) is formed and then reacts readily with a deuteron to produce another neutron. Both D-D and D-T reactions are employed in weapons to initiate fission in surrounding fusion tampers made from fissionable materials.

High-energy neutrons (14 MeV) released by the D-T reaction are used in many fission weapons to achieve boosting to maintain and enhance the fission chain reaction.²⁷

The reactions shown above are highly relevant to the materials and internal configuration of components used in U.S. thermonuclear weapons. In order to better understand the implementation of these reactions, we must first examine the materials and principles which produce the conditions required for these reactions to occur.

Fissile Explosives

The three primary fissile fuels used or considered for use in U.S. nuclear weapons since the end of World War II are uranium, plutonium, and thorium. The choice of which material to use in a given weapon has occasionally dictated the design of the warhead, due to the unique physical and nuclear characteristics of the chosen explosive, as described below. These materials are generally used in their metallic form as explosives.

The first fuel, uranium, has an atomic number (the number of protons in the nucleus) of 92, and atomic weights between 227 and 240, due to the varying number of neutrons in its nucleus. The isotopes ${}_{92}\text{U}^{235}$ and ${}_{92}\text{U}^{238}$ are most commonly used in nuclear weapons; naturally-occurring uranium contains about 99.3% U-238 and 0.7% U-235 by weight.

Uranium metal is extremely dense; it has a specific gravity of 18.95, and weighs about 0.7 lbs., or 11.2 ounces, per cubic inch. A cubic foot of uranium metal weighs over 1,183 lbs. When a solid core of uranium metal is imploded and compressed at the beginning of a nuclear explosion, this density increases by a factor of 8 to 27, depending upon the chemical high explosives and the degree of compression.

In its refined state, uranium is a heavy, silvery-white metal, and is pyrophoric (self-igniting) when finely divided in powdered or sintered form. It oxidizes rapidly when exposed to the atmosphere; all 14 isotopes are radioactive, most by alpha

²⁷Glasstone and Redman, WASH-1037 (Revised), p. 15.

particle decay, and the most common isotope, ${}_{92}\text{U}^{238}$, has a half-life of over 4 1/2 billion years, the period after which half the original material will have decayed into another element lower on the periodic table. Much of the heat at the center of the earth is believed to be attributable to the radioactive decay of large amounts of uranium and thorium (the earth itself is about 4 1/2 billion years old).

A century ago, before uranium's utility as an explosive was exploited, the metal was believed useful in ordnance applications:

Experiments and tests which have been made by the great steel and gun manufacturers have shown that the addition of a small percentage of uranium to steel increases its elasticity, and at the same time its hardness, to an extent that makes its use in the manufacture of guns, armor plates, etc., most desirable, but the scarcity of the material and especially the great difficulty in reducing the ore to metal makes the price of uranium steel too high.²⁸

Figure I-3 shows an enriched uranium "biscuit."

The second fissile fuel, plutonium, is very similar to uranium in its physical and chemical characteristics. It has an atomic number of 94, and atomic weights between 232 and 246. Unlike uranium, plutonium is an artificial, man-made element, produced in reactors by bombarding ${}_{92}\text{U}^{238}$ with slow neutrons (all commercial nuclear reactors breed plutonium as a by-product of electrical power generation). The resultant uranium decays first into neptunium, then plutonium.

Plutonium has a half-life of about 24,000 years. When used in nuclear weapons, plutonium is usually in its delta phase allotrope, alloyed with approximately one percent by weight of gallium to stabilize the allotrope at room temperature. Weapons-grade plutonium usually contains no more than 6% of the isotope Pu-240.

Metallic plutonium exists in six different allotropic forms between room temperature and their melting points (approximately 641⁰ C, or 1186⁰ F). These different phases of plutonium are stable within separate temperature ranges.

²⁸Quote from July 1891 issue of SCIENTIFIC AMERICAN in "50 and 100 Years Ago," SCIENTIFIC AMERICAN, July 1991, p. 15.

The two forms of interest in nuclear weapons design are the alpha and delta phases. The alpha phase is stable at temperatures below 115⁰ C (239⁰ F) and has a density of about 1,240 lbs. per cubic foot. The high density is advantageous from certain weapon standpoints, since it permits criticality in a smaller mass than other phases of plutonium.

On the other hand, alpha-phase plutonium is brittle and difficult to machine. Early pieces of alpha phase plutonium fabricated at Los Alamos during World War II experienced large volume changes when heated, and then cracked when cooled.²⁹

Alpha phase plutonium is also inherently physically unstable, and weapon components made of nominal alpha-phase plutonium are susceptible to dimensional changes, although not usually at low or stable ambient temperatures.

Delta phase plutonium is stable at temperatures between 310⁰ C (590⁰ F) and 458⁰ C (856⁰ F), and has a density of about 994 lbs. per cubic foot. Delta phase plutonium can be made stable at room temperature by the addition of 1% by weight of gallium.

Most of the plutonium used in stockpiled U.S. nuclear weapons is in the stabilized delta phase. This phase is much less brittle and easier to shape and machine than alpha phase plutonium; in fact, delta phase plutonium is said to resemble copper in its malleability and ductility, whereas alpha phase plutonium is more like cast iron in these respects.

Delta phase plutonium was used in early implosion weapons because it could be fabricated more readily, and because it offered an additional advantage in that its relatively low density permitted the use of a much larger subcritical mass of plutonium, thereby making possible an increase in total energy yield. This latter aspect of weapons design is, however, no longer significant.

In order to maximize the critical mass and thus the overall weight of a weapon, it may be desirable to use high-density plutonium. Alpha phase plutonium was used to some extent, but in addition to its fabrication difficulties, it is more sensitive to preinitiation by neutrons than is the delta phase. For this reason also, its use has been very limited in the U.S. nuclear weapons program.³⁰

²⁹"Plutonium: A Wartime Nightmare but a Metallurgist's Dream," Richard D. Baker, et. al., LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 144.

³⁰Glasstone and Redman, WASH-1037 (Revised), pp. 16, 17.

Plutonium almost always has a smaller critical mass for a given yield than does uranium-235. Even a critical mass of plutonium containing as much as 35% by weight of the undesirable Pu-240 isotope is still considerably less than that of U-235.³¹ Plutonium has a fission cross-section for fast neutrons nearly 3.5 times as large as that of U-238.³²

The third fuel, thorium, is of interest primarily because it can be irradiated with neutrons to form fissionable ${}_{92}\text{U}^{233}$. Thorium itself is fissionable only by high-energy neutrons, such as those released in fusion reactions; it can be used as a substitute for ${}_{92}\text{U}^{238}$ in thermonuclear weapon secondaries and tertiaries. At one time in the early postwar U.S. nuclear weapons program, thorium was proposed as a feed material to reactors to alleviate a critical shortage of uranium.³³ Thorium has a half-life of about 14 billion years.

These fuels are useful because when they fission, they release neutrons as well as radioactive fission products. The number of neutrons released when a nucleus of these fuels undergoes fission varies somewhat with the particular type of fission, but the average number of fission neutrons is well defined. The value depends on the energy of the neutrons which strike the nucleus and cause fission, and increases to some extent with increasing neutron energy.

Table I-1 shows the average number of neutrons produced per fission caused by "slow" neutrons, i.e., neutrons of essentially zero kinetic energy; neutrons of about 0.5 MeV energy, which is roughly the average energy of the neutrons which maintain a chain reaction in a simple fission weapon; and "fast" neutrons of 14 MeV energy which contribute greatly to the fission energy yield of boosted single-stage weapons and many two- and three-stage thermonuclear devices. The isotopes uranium-235, plutonium-239, and uranium-233 are fissionable by neutrons of a wide energy spectrum, from less than an electron volt to many millions of electron volts.

³¹Glasstone and Redman, WASH-1037 (Revised), p. 17.

³²DARK SUN: THE MAKING OF THE HYDROGEN BOMB, Richard Rhodes, Simon & Schuster, New York, 1995, p.70.

³³U-233 was proposed for use in weapons by mid-1951; it was tested successfully in a device during Operation TEAPOT in mid-1955. (Progress Report to Joint Committee on Atomic Energy, Part III, Weapons, December 1950 through May 1951, May 29, 1951, p. 19.)

The numbers in the rightmost column show clearly the utility of fissioning large quantities of uranium and plutonium by the high-energy neutrons radiating from thermonuclear reactions. The many neutrons released by these materials at this energy level can in turn be used to fission even more material, in boosted primaries, and in fissionable fusion tampers around secondaries and tertiaries.

Table I-1 Neutrons Released Per Fission³⁴

<u>Nuclide</u>	<u>Neutron Energy</u>		
	<u>0 MeV</u>	<u>0.5 MeV</u>	<u>14 MeV</u>
Uranium-235	2.43	2.49	4.1
Plutonium-239	2.80	2.85	4.9
Uranium-233	2.45	2.51	4.2

Nuclear weapons are designed so that a fission chain reaction is maintained by fast neutrons, with kinetic energies in the range of approximately 0.1 to two MeV.³⁵

Fission Explosion Principles

A nuclear weapon should be considered as an explosive engine. A multistage thermonuclear weapon can involve from six to nine individual chemical, fission, and fusion explosions before it finishes reacting. An exploding boosted fissile primary consists of a chemical high explosives detonation, a fission explosion, and a fusion (boosting) blast. Each thermonuclear stage can yield up to three separate blasts: a fission reaction (the sparkplug), a fusion explosion, and a second fission explosion, the latter occurring if a tuballoy or oralloy jacket encases the fusion fuel and sparkplug.

In fact, a rough analogy can be made between a thermonuclear weapon and an internal combustion engine: one critical component of a fusion warhead, the "sparkplug,"³⁶ is named after an automobile engine part. The concept of fusion-boosting is analogous in its purpose and results to the injection of nitrous oxide into high-performance racing car and reciprocating aircraft engines. The latter provides additional oxygen for improved combustion, while the former provides additional neutrons for improved fission.

³⁴Glasstone and Redman, WASH-1037 (Revised), pp. 6, 7.

³⁵Glasstone and Redman, WASH-1037 (Revised), p. 11.

³⁶THE SECRET THAT EXPLODED, Howard Morland, Random House, New York, New York, p. 20.

Nuclear weapons -- especially thermonuclear weapons -- borrow terminology from rocketry as well as from the automotive world. Specifically, the term "staging," used to denote the concept of creating powerful explosions by means of successive blasts of increasing magnitude, is drawn directly from rocketry. The term has an opposite meaning when applied to spaceflight, however: each subsequent rocket stage is less powerful than its predecessor, as the vehicle requires less thrust to move a decreasing mass above a planet's atmosphere and gravitational attraction.

Thermonuclear bombs, like rockets, can be either liquid-fueled with liquid deuterium or solid-fueled with lithium-6 deuteride and lithium hydride.

Just as in an internal combustion engine, the compression of the fuel before detonation is perhaps the single most important measure of the performance of either a fission or fusion weapon. In an internal combustion engine, a high compression of the fuel-air mixture in a cylinder of the engine is required to provide a relatively dense medium through which an explosive (burning) wave front can travel more rapidly. This reaction releases more energy to push the piston back down than would be the case in a less highly-compressed mixture.

In fission and fusion warheads or explosive devices, a high degree of fuel compression is required to increase the reaction rates, and to push the fuel atoms closer together, to resist as long as possible the outwardly-directed forces tearing the fuel mass apart. The longer the fuel stays together, the more explosive yield can be produced, and the more fuel will be consumed before the mass disintegrates.

Primarily for this reason, most postwar weapons development has centered on getting better compression of the fuel, often resulting in the need for less active material for the same yield. Critical mass is inversely proportional to the square of its density; if the density is doubled, for example, then the critical mass is just a quarter of its previous value.³⁷

This has very practical applications. For example, the critical mass of unreflected alpha phase plutonium is around 10 kilograms. Surrounding the plutonium with a moderate nuclear reflector reduces critical mass to about 5.5 kilograms. Under just three- or four-fold compression -- very low compression -- this 5.55 kg can be reduced to just 2.75 kg, with a resultant explosive yield of 10 to 20 kilotons.

³⁷LOS ALAMOS PRIMER, Robert Serber, University of California Press, Berkeley, California, 1992, p. 60.

Under high compression, as little as one kilogram of plutonium can produce about a kiloton of explosive yield.³⁸

The "secret" to an efficient nuclear explosion is to crowd as much fissionable material together as fast as possible so that nuclear reactions can proceed and so that the reacting mass is not thermally dissipated before very much energy has been released. In this case, only the amount of reacting material in excess of a critical mass is available for "burning:" as soon as the mass of the material drops below a critical amount, the nuclear reactions will stop.

If fissile material and a surrounding neutron reflector are compressed uniformly, the amount of material required for a particular yield is inversely proportional to the square of the increase in density. If the density of material is doubled, only a quarter as much material is required for the same yield; if the density is tripled, only one-ninth as much material is needed.³⁹

Some of the specific techniques and arrangements used by U.S. nuclear weaponeers to accomplish this are described in subsequent chapters; basic methods of initiating fission and fusion explosions for military purposes are explained in this chapter.

As far as is presently known, there are three methods of initiating a fission explosion in a mass of fissile fuel. The first is the so-called "gun assembly" method, by which two or more subcritical masses of fissionable material, each incapable of starting an uncontrolled fission reaction by itself, are driven together very rapidly with little or no compression to form a supercritical mass.

The best-known example of this method was in the design of the MK I LITTLE BOY bomb detonated over Hiroshima, Japan on August 6, 1945. This weapon used a projectile of uranium-235

³⁸"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 6. The one kiloton of yield from one kilogram of plutonium represents an efficiency of between five and 5.55 percent.

³⁹McPhee, p. 136; "Fuchs and Fallout: New Insights into the history of the H-Bomb," Daniel O. Hirsch and William G. Mathews, Committee to Bridge the Gap, Los Angeles, California, January 1990, p. 23. The latter paper was the basis of an article entitled "The H-Bomb: Who Really Gave Away the Secret?" in the Bulletin of the Atomic Scientists, Vol. 46 No. 1, January/February 1990.

(comprising 60% of the critical mass) in the tail of the bomb. This projectile was shot through a tube in the bomb casing, into a set of colinear rings of uranium-235 (comprising the remaining 40% of the critical mass) in the nose of the weapon. This particular device reportedly fissioned about 700 grams -- about 25 ounces -- of a total fuel mass of 64 kilograms, for an efficiency of about 1.2% and a yield of 15 KT.⁴⁰

The gun-type assembly method is inherently wasteful of nuclear material, due to the lack of any sort of compressive mechanism to reduce the amount of fissionable fuel required: the assembly method usually merely unites the subcritical masses, without compression.

The second method of initiating a fission explosion is the more sophisticated and far more complex spherical implosion technique first used in the MK III FAT MAN bomb and most postwar U.S. nuclear warheads. This type of mechanism works by "squeezing" (imploding) a subcritical mass of fissile material by means of high explosive "lenses," charges specially shaped and positioned to generate and focus a blast wave, which cumulatively produce an inwardly-directed spherical shock wave when all the charges are detonated simultaneously.

This spherical blast wave rapidly increases the density of the fuel by reducing its volume without any loss of mass -- the metal is actually liquefied before it explodes -- and forces the mass to become supercritical. The FAT MAN detonated over Nagasaki, Japan on August 9, 1945 reportedly used 6.2 kilograms (13.6 lbs.) of plutonium for fuel; about 1.3 kilograms (2.8 lbs.) were fissioned, for a yield of 23 kilotons and an efficiency of 21%, about 17.5 times that of the LITTLE BOY Hiroshima bomb, which yielded about 15 kilotons.⁴¹

⁴⁰"The Continuing Body Count at Hiroshima and Nagasaki," Frank Barnaby, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 33 No. 10, December 1977, p. 49; McPhee, pp. 14, 15; Serber, THE LOS ALAMOS PRIMER (1992 edition), p.60. See the LITTLE BOY history in Volume VI for a slightly higher estimate of efficiency.

⁴¹Barnaby, loc. cit.; "The Inheritance of Hiroshima," Tsutomu Hoshima, BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 42 No. 6, June/July 1986, p. 37. The yields are from the latter source and the fissionable masses from the former. These new yield figures -- which represent a 20% upward revision at Hiroshima and a 10% downward revision at Nagasaki -- supersede those officially released by the United States Department of Energy in UNITED STATES NUCLEAR TESTS, JULY 1945 THROUGH SEPTEMBER

The high degree of supercriticality and increased efficiency of the use of fissionable material attainable from compression constitutes the great advantage of implosion-type weapons over gun-assembly weapons. The critical mass of a given fissile material, under specified conditions, is inversely proportional to the square of the ratio of the volume before compression to the volume after compression.

1992, DOE/NVO-209 (Rev. 14), December 1994. These new yield figures are based on a March 1986 joint U.S.- Japanese dosimetry reassessment workshop at Hiroshima. See also "The Yields of the Hiroshima and Nagasaki Nuclear Explosions, " John Malik, LA-8819 (DE86004194), Los Alamos National Laboratory, Los Alamos, New Mexico, September 1985.

Ferenc Morton Szasz in THE DAY THE SUN ROSE TWICE (University of New Mexico Press, Albuquerque, New Mexico, 1984) claims that the TRINITY bomb contained about 12 lbs. of plutonium (p. 58), and yielded 20 kilotons for an efficiency of 18% (p. 117); his estimate of the mass of the core is apparently based on the yield and the efficiency: a yield of 20 KT corresponds to complete fission of 1 kg of Pu; 1 kg is 18% of 5.55 kg, or 12.22 lbs. If the bomb in fact contained 6.2 kg of Pu, and had an efficiency of 18% with total yield limited to the plutonium without contribution from the uranium tamper, then 1.116 kg of Pu fissioned for a yield of 22.32 KT. According to a contemporary source, the TRINITY device yielded a "nuclear explosion of good efficiency." ("Events in Camp Immediately Following Shot - July 16, 1945.")

According to Dr. David Bradley, a radiologist at the CROSSROADS tests in 1946, "half an A-bomb (core)...was about the size of half a tennis ball." (NO PLACE TO HIDE, 1946/1984, University Press of New England, Hanover, New Hampshire, 1984, p. 214.) See also NUCLEAR WEAPONS DATABOOK, Volume I, U.S. FORCES AND CAPABILITIES, Dr. Thomas B. Cochran and William M. Arkin (eds.), Ballinger Publishing Company, Cambridge, Massachusetts, 1984, p. 31; CITY OF FIRE, James W. Kunetka, Prentice-Hall, Inc., Englewood Cliffs, New Jersey, 1978, p. 164; NUCLEAR WEAPONS AND AMERICAN STRATEGY, 1945-1953, doctoral thesis by Mark Bernard Schneider, University of Southern California, 1974, p. 79; MANHATTAN DISTRICT HISTORY, PROJECT Y, THE LOS ALAMOS PROJECT, Vol. I, Inception Until August 1945, David Hawkins, Los Alamos Scientific Laboratory, Los Alamos, New Mexico, LAMS-2532, Vol. 1, 1946, p. 12. A solid sphere of plutonium with a mass of 6.2 kilograms (13.6 lbs.) would have a diameter of 3.3", about the size of a tennis ball, with the weight of a light bowling ball. Since the FAT MAN cores had small polonium-beryllium neutron initiators in their centers, their diameters were in excess of 3.3".

Thus the higher the compression, the smaller the critical mass.⁴²

A later variation of implosion was the use of high explosives to implode cylindrical critical masses.⁴³ This technique of linear implosion is described in more detail in Volume II.

As regards relative usefulness of implosion versus gun assembly weapons, between two and three times as many implosion as gun-type warheads can be obtained from the same amount of fissionable material.⁴⁴ On the other hand, early implosion warheads were bulkier and heavier than early gun assembly type weapons, and consequently required larger delivery vehicles, either bombs or missiles.⁴⁵

In addition, the most outstanding feature of a gun-type weapon is its performance reliability. Other advantages are found in its lesser diameter for the same yield when compared to an implosion weapon, and its adaptability to penetration weapons for employment in subsurface bursts. Almost all U.S. penetrating nuclear warheads have been gun assembly type weapons.

Although an implosion weapon requires an extremely complicated assembly of explosives and pit for positive detonation, a gun-type weapon is horribly inefficient from the standpoint of yield-per-kilogram of fissile material (see figures above contrasting the efficiencies of the LITTLE BOY and the FAT MAN). In addition, because of its single-point detonation system, a gun-type weapon is not as safe as an implosion weapon with its multipoint detonation system.⁴⁶

Both of these methods of starting fission chain reactions require the use of an initiator, such as a small, internal

⁴²Glasstone and Redman, WASH-1037, p. 14.

⁴³"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 4.

⁴⁴AF ATOMIC ENERGY PROGRAM, Vol. II, p. 678.

⁴⁵THE AIR FORCE AND THE NATIONAL GUIDED MISSILE PROGRAM, 1944-1950, Max Rosenberg, USAF Historical Division Liaison Office, June 1964, p. 162.

⁴⁶HISTORY OF THE TACTICAL AIR COMMAND, 1 July through 31 December 1951, Volume VII, Special Weapons Activities. Headquarters, U.S. Air Force Tactical Air Command, Langley AFB, Virginia, May 1952, p. 36.

polonium-beryllium sphere, or an external high-voltage neutron generator, to release a burst of neutrons at the right moment to begin the reaction. Fortunately for mankind, background neutron radiation found on the surface of the earth is inadequate by itself to start a nuclear reaction in fissionable materials.

The initiator, or "urchin," must release neutrons in great abundance, since the number of neutrons released by any given fission generation is directly proportional to the number which start the first generation. The development of neutron sources is discussed in Volume II.

The third method of fission initiation is the bombardment of ordinarily non-fissionable uranium-238 ("tuballoy") with very high energy neutrons released by fusion reactions (${}_{92}\text{U}^{238}$ will not fission spontaneously in any amount: it has no critical mass).

The efficiency of the fission process in "tuballoy" can be improved significantly if the uranium is compressed; however, this compression is not related to the formation of any critical mass: it merely pushes the atoms closer together and gives the material an inwardly-directed momentum, preventing physical disintegration for a longer time. This particular method of fission initiation is discussed in greater detail in "Fusion Explosion Principles" later in this volume.

It should be noted here that the gun assembly method is not generally suitable for use with plutonium, due to the presence in most plutonium of the isotope plutonium-240 (${}_{94}\text{Pu}^{240}$), which has a very high spontaneous neutron emission rate, on the order of one million neutrons per kilogram-second (as opposed to only about 22 neutrons per kilogram-second for Pu-239).⁴⁷ One gram of plutonium-240 emits more than 1,000 neutrons per second.⁴⁸ Weapons-grade plutonium usually contains 6% Pu-240; "super-weapons grade" plutonium contains only 3% Pu-240.

No gun design has been developed with a mass assembly velocity fast enough to preclude predetonation, i.e., fission initiation before optimum conditions have been achieved. A detonation wave moves through high explosives about 20 times

⁴⁷"Explosive Properties of Various Types of Plutonium," paper presented by Richard L. Garwin at the NATO Advanced Research Workshop on Managing the Plutonium Surplus: Applications and Options, Chatham House, London, January 24, 1994, p. 6.

⁴⁸Glasstone and Redman, WASH-1037 (Revised), p. 12.

as fast as a projectile can move through a gun barrel,⁴⁹ so implosion of plutonium completely obviates the predetonation problem. Plutonium-239 with an appreciable proportion of Pu-240 can be used in an implosion system.⁵⁰

However, linearly-imploded plutonium has been used in a number of U.S. artillery-fired atomic projectiles (AFAPs), starting with the W-48 (see Volumes II and VII).

Even implosion weapons are not entirely immune to the predetonation effects of Pu-240. The amount of Pu-240 in plutonium increases in proportion to the length of time the plutonium is "cooked" in reactors; as the amount of plutonium to be produced at the Hanford Engineer Works in Washington state rose after the end of World War II, the resultant yields of implosion weapons made from it were expected to drop, due to predetonation caused by the presence of increasing amounts of Pu-240.⁵¹

On the other hand, in order to minimize the Pu-240 content, the uranium fuel elements could be left in the reactors for only a relatively short period before they were removed and the plutonium extracted. As a result, the lower the Pu-240 content, the higher the cost for the useful plutonium produced.⁵² Conversely, the higher the plutonium-240 content, the smaller the cost per unit mass of Pu-239, but the larger the mass required for criticality. Pu-240 also has adverse effects on the one-point safety of plutonium-fueled weapons.⁵³

For a production reactor of a given type, the Pu-240 content of the plutonium produced depended upon the neutron density (or reactor power) and the exposure time. Consequently, the quality of plutonium at one time was described in terms of megawatt-days (MWD) of exposure in the reactor per ton of

⁴⁹CRITICAL ASSEMBLY: A TECHNICAL HISTORY OF LOS ALAMOS DURING THE OPPENHEIMER YEARS, 1943-1945, Lillian Hoddeson, Paul W. Henriksen, Roger A. Meade, Catherine Westfall, et. al., Cambridge University Press, New York, 1993 (hereafter cited as "Hoddeson, et. al."), p. 418.

⁵⁰Glasstone and Redman, WASH-1037 (Revised), p. 35.

⁵¹Memorandum dated 30 July 1945 to the Chief of Staff from Major General Leslie R. Groves, USA.

⁵²Glasstone and Redman, WASH-1037 (Revised), p. 17.

⁵³Glasstone and Redman, WASH-1037 (Revised), p. 18.

uranium fuel, i.e., in MWD/T units. One MWD of reactor energy produced roughly one gram of plutonium.⁵⁴

This unit is not very precise because the power of the reactor is not uniform throughout its entire volume, and the Pu-240 content of the fuel elements varied with the location of the element within the reactor. The MWD/T unit also varied with the reactor type, so that it was not the same for the graphite-moderated Hanford reactors as it was for the heavy-water moderated Savannah River plant reactors in South Carolina.

Table I-2 shows the different characteristics of plutonium produced by these reactors; the quantity labeled neutrons/gram-sec shows the spontaneous neutron generation backgrounds of the materials. This table illustrates the decreasing Pu-240 content as the MWD/T declines, and the corresponding drop in background neutron emission caused by the spontaneous decay of Pu-240. This background rate is constant at about 10.5 times the percentage of Pu-240 content.

Table I-2 Characteristics of Production Plutonium

<u>MWD/T</u>		<u>Pu-240</u> <u>(%)</u>	<u>Neutrons/gram-sec</u> <u>(n/g/s)</u>
<u>Hanford</u>	<u>Savannah</u> <u>River</u>		
800	1375	7.1	75
600	1000	5.6	58
400	630	3.85	40

These values had a practical utility: some of the early U.S. fission weapons required very "clean" plutonium with low Pu-240 content; the purest material then used contained only about 1.5% by weight of Pu-240.⁵⁵

Another important aspect of fission explosive principles is the efficiency of the reaction. The most commonly cited measure of fission weapon efficiency, aside from the yield-to-weight ratio, is the weapon's "alpha," or rate of fission build-up, in terms of neutron multiplication rate, before explosive disassembly.⁵⁶ Each fission generation takes about 10 nanoseconds, or one "shake;" this is the time between

⁵⁴NUCLEAR WEAPONS DATABOOK, Volume II: U.S. NUCLEAR WARHEAD PRODUCTION, Thomas B. Cochran, et. al, Natural Resources Defense Council, 1987, p. 63; the authors equate one MWD to between 0.97 and 1.04 grams of plutonium.

⁵⁵Glasstone and Redman, WASH-1037 (Revised), p. 18.

⁵⁶McPhee, p. 163; AF ATOMIC ENERGY PROGRAM, Vol. IV, p. 275.

successive releases of neutrons and nucleus fragments in fission reactions.

An essential condition for a high-energy fission explosion is a very high neutron density, since the rate of fission, and hence the rate of energy release, is proportional to the number of free neutrons per unit volume of the reacting fuel mass.⁵⁷ Alpha is a measure of the neutron density in an exploding fissile assembly.

A high alpha value is important because the faster the rate of fission build-up, the more energy is released before the critical assembly disintegrates. A positive alpha value is indicative of a supercritical array. The alpha of an implosion weapon is not truly constant, but varies with the degree of criticality produced by the implosion and subsequent explosion. The value of alpha is negative before the core reaches criticality; is zero when the core is just critical; and becomes increasingly positive as the core implodes and compresses. Alpha decreases as the system disassembles, and becomes zero again, and then negative.⁵⁸

In fact, there is no true alpha immediately after initiation - the magnitude of the neutron population in the weapon core is then governed by the statistical behavior of individual neutrons. Very soon, however, the total number of neutrons present becomes so large that the statistical behavior of individual particles represents only a small percentage of the overall effect.

At this time, the neutron multiplication rate becomes exponential, with the exponent directly dependent upon the alpha value. Alpha increases until the assembly is in its most critical configuration, then remains more or less constant throughout the so-called "incubation period" between the initiation of fission and the explosion and beginning of disassembly and finally decreases to negative values as the core continues to expand and becomes subcritical.⁵⁹

Documentation of alpha versus time, and particularly the value of alpha during the "incubation period," is an extremely valuable tool -- especially when used in

⁵⁷Glasstone and Redman, WASH-1037 (revised), p. 23.

⁵⁸Glasstone and Redman, WASH-1037 (Revised), pp. 26, 33, 34.

⁵⁹Glasstone and Redman, WASH-1037 (Revised), p. 26. An interesting aspect of fission weapon behavior is that 30% or more of the total yield of a chain-reacting compressed core is released after the core has expanded to a subcritical state. (Glasstone and Redman, WASH-1037, (Revised), p. 30.)

conjunction with values of yield and transit time -- for diagnosing fission weapon behavior.⁶⁰

Transit time indicates relative criticality of the assembled active components when the initiator is crushed, in early weapons, or fired, in later weapons, and is a valuable measure of the efficiency of the high explosives and the degree of compression of the pit. By marking the time of initiation relative to the position of the implosion wave in the nuclear assembly, transit time is an important indicator of predetonation, postdetonation, or normal nuclear detonation.

An abnormally long transit time may indicate postdetonation: the implosion wave has been reflected from the center of the configuration and is disintegrating the critical mass. The relative criticality will therefore be low and total energy release will also be low. Conversely, an abnormally short transit time indicates predetonation: the implosion wave has not yet reached the center of the configuration and the density (and hence criticality) of the assembled components has not yet reached the desired maximum.

Transit times measured in nuclear tests were usually compared with values predicted by computation before the test. For atmospheric tests fired on towers, at least one gamma ray detector was usually placed on the tower immediately adjacent to the test device. Gamma ray detectors to measure transit time were also usually included inside the casing of air-dropped test devices.

The alpha measurement depends directly upon the fact that the number of gamma rays radiated by a fissioning system is proportional to the neutron population in the system, which in turn is proportional to the number of fissions occurring at a given moment. In general, the magnitude of alpha will indicate whether or not the predicted yield of the explosion was achieved.

The alpha for any given fissile core can be predicted from a computer calculation based on the neutronic and hydrodynamic

⁶⁰THE TURQUOISE BOOK, OPERATION IVY, compiled by Maj. Paul G. Galentine, Jr., USAF, et., al., Headquarters, Task Group 132.1, Los Alamos Scientific Laboratory, 15 May 1952, pp. 2.1a.1, 3.2; THE ONO BOOK, OPERATION CASTLE, Appendix I, Annex C, Operation Plan 1-53, Task Group 7.1, Joint Task Force Seven, Headquarters, Task Group 7.1, Los Alamos Scientific Laboratory, 1 October 1953, p. 72. (this is the CASTLE equivalent of the Turquoise Book for Operation IVY); Hoddeson, et. al., p. 355.

characteristics of the core, but there are uncertainties involved and experimental measurements are usually used to validate the prediction. In weapons tests, the alpha measurement is one of the most important diagnostic measurements. After the test, calculations are normalized to agree with the observed values and to allow better future prediction.⁶¹

The higher the alpha, the more efficient the weapon. A high fission build-up rate and high alpha usually result from well-tamped, highly-compressed, and chemically-pure fissile fuels. High core compression results in both shorter fission generation times and shorter mean free neutron paths, both of which increase efficiency and alpha.⁶² In addition, the more highly compressed the core is when the explosion begins, the farther the distance the outer core surface must travel during expansion before the supercritical system become subcritical, thus increasing efficiency.⁶³

Typical unboosted fission weapon alphas range from 25 to 200 per microsecond.⁶⁴ This performance criterion is of interest mainly to illustrate the effects of fusion-boosting on fission systems, as described in "Boosted Fission Principles" later in this chapter. The alpha of a boosted system is generally much higher than the alpha of a similar unboosted system.

There are a number of ways to increase the likelihood of achieving a supercritical mass and a nuclear chain reaction, mainly by minimizing reactions that compete with fission.

Fissile materials should be enriched to the greatest extent possible in the isotope that fissions most efficiently and generates large numbers of neutrons. The fissile material should also be as chemically-pure as possible, to remove impurities that might capture neutrons without fissioning.

⁶¹Glasstone and Redman, WASH-1037 (Revised), p. 24.

⁶²Glasstone and Redman, WASH-1037 (Revised), p. 26.

⁶³Glasstone and Redman, WASH-1037 (Revised), p. 32.

⁶⁴A 1948 report, OPERATION SANDSTONE: NUCLEAR EXPLOSIONS 1948, Scientific Director's Report - Volume I, General Report, states: "For atomic bombs such as those tested in 1948, static conditions were maintained for appreciable times and constant alphas on the order of 10^8 (per second) were measured." This works out to 100 per microsecond; Glasstone and Redman, in WASH-1037 (Revised), p. 26, reiterate this number. It is reasonable to assume that unboosted fission weapon alphas have at least doubled since 1948.

One fairly obvious way to improve the efficiency of a fissile weapon is to make the surface area of the fissile core (or "pit") as small as physically possible for a given amount of material. The surface area needs to be minimized in order to reduce the area from which neutrons can escape without participating in a fission reaction. A property of a sphere is that for a given volume, its surface area is the smallest of all geometric solids. Hence the critical mass of a sphere of fissionable material is less than for any other geometrical form of the same amount of material.

Primarily for this reason, almost all postwar U.S. fission weapon pits have been spherical in shape, to minimize neutron loss during nuclear reactions. These "pits" have evolved from solid hemispherical shapes to much smaller and lighter thin shells of fissile material. Cylindrical critical masses have also been developed for use in thermonuclear weapons as "sparkplug" fusion initiators.⁶⁵ These "sparkplugs" are described in a following section in this chapter.

Surrounding the active material with a good scattering medium, such as beryllium, which reflects escaping neutrons back into the reacting mass markedly increases efficiency. Increasing the density of the active material also reduces the probability of escaping neutrons that might be uselessly lost.

If the neutrons can be moderated, i.e., slowed down, the fission cross section of the reacting material may also be increased. Neutrons may be moderated by the inclusion of light-mass nuclei, such as those of boron atoms, which scatter neutrons well; however, this process takes considerable time and greatly slows the reaction, leading to less-than-ideal energy release which results in low-order explosions.⁶⁶

For a while during the early stages of the U.S. thermonuclear weapons program, some thought was given to creating thermonuclear explosions without using fission detonators. In this scheme, ordinary high explosives (HE) might be used to initiate fusion. Within this geometry, the HE compressed a fusion fuel capsule composed of an outer uranium-238 pusher, a charge of lithium-6 deuteride fusion fuel, and a fissionable sparkplug (either uranium-235 or plutonium). An

⁶⁵See Morland, THE SECRET THAT EXPLODED, p. 92, where he quotes Dr. Herbert York (former director of the Lawrence Livermore nuclear weapons lab) referring to cylindrical critical masses.

⁶⁶PRINCIPLES OF NUCLEAR PHYSICS, Atomic Weapons Training Group, Field Command, DASA, Sandia Base, Albuquerque, New Mexico, 24 March 1960, p. 167.

external neutron generator served as a source of neutrons to initiate fission in the sparkplug.

This technique has probably been considered and perhaps even tested on a small scale by the U.S.⁶⁷

Contemporary thermonuclear weapons, however, still contain fission primaries. Uniform instantaneous compression of the fusion fuel capsule, quickly enough to prevent the loss of energy and fuel elements through the ends of the capsule, and the ignition of a large fuel capsule to generate sufficient x-radiation to compress subsequent stages, which would not require HE compression, may be impossible to attain on large-scale warheads. However, these problems might be overcome on small, low-yield thermonuclear devices.

A modern fission weapon, or the primary of a thermonuclear weapon (the only difference between them is that a "neutron pipeline" to initiate fission in the "sparkplug" links a primary to the adjacent secondary stage),⁶⁸ can be characterized as "spheres within spheres." These concentric shells begin with an outermost metal casing, encompassing an arrangement of high explosive "lenses," the purpose of which is to create an inwardly-focused implosive shock wave to compress the pit.

There are usually up to 100 pounds of explosives, comprising 40 or 50 lenses or detonation points, each capped with its own individual detonator attached to wires running to the firing mechanism. Early postwar nuclear weapons contained several hundred pounds of high explosives. These lenses surround a combination tamper-reflector of uranium, gold, or beryllium, which bounces neutrons back into the exploding core, and which also, by its mass and inertia, resists the outward expansion of the core (the nuclear reactions end when the core expands from its highly-compressed configuration to just a few centimeters in diameter).

This tamper-reflector in turn encloses a small air gap, within which sits the fissile core, suspended by thin wires. The air gap permits the tamper-reflector to gather momentum before slamming into and compressing the fissile core. The core itself may vary in diameter between the size of a tennis ball and a cantaloupe. It is usually composed of uranium-235, plutonium-239, or both (see Volume II).

⁶⁷J. Carson Mark interview, LOS ALAMOS SCIENCE, Vol. 4 No. 7, Winter/Spring 1983, p. 51.

⁶⁸Rosengren, RDA-TR-122100-001-Rev. 1, p. 10.

A neutron generator, which fires neutrons into the compressed core to begin the fission reactions, is external to this concentric assembly of spheres. In addition, if a boosted primary is used, a reservoir of deuterium and tritium gases is linked to the hollow center of the pit, which serves as a fusion chamber (see "Boosted Fission Principles" below).

When the conventional high explosives around the core explode, the entire tamper-reflector-core assembly is compressed to an extremely high density. At a pre-selected time, a burst of neutrons is fired from the external neutron generator, radiating a small part of the imploded mass. These neutrons begin the fission process and the neutron population begins to multiply.

As the implosion continues, the compressed core becomes more supercritical, and the rates of fission, neutron production, and energy production grow. At a certain level of released fission energy, the interior pressure in the core is sufficient to halt the implosion and start an outward explosion.

A highly-pressurized mixture of deuterium and tritium gas, injected before detonation into a "boost chamber" in the hollow center of the uncompressed core, figures prominently in the operation of the fission device. The implosion and explosion of the core compress and heat the D-T gas, with the final compression of the gas being determined by the interior cavity symmetry and by the degree of mixing between the fissile materials in the core (plutonium and uranium) and the D-T gas.

Thermonuclear reactions occur in the heated gas, releasing high energy neutrons that initiate additional fission chains in the compressed fissionable materials. These reactions heat the D-T gas even more, increasing the rate of thermonuclear reactions, which in turn release even more high-energy neutrons.

The rate of explosive energy release from the compressed core continues to rise until explosive disassembly makes the mass subcritical. The D-T reactions, in total, release a relatively small amount of energy, but may cause a many-fold increase in the rate of fission reactions and a great increase in the total explosive energy yield.⁶⁹ The complete fusion of only 12 grams of tritium and deuterium yielded explosive force equivalent to a kiloton of TNT.⁷⁰ This D-T

⁶⁹Rosengren, RDA-TR-122100-001-Rev. 1, p. 13.

⁷⁰WORLD ARMAMENTS AND DISARMAMENT, SIPRI YEARBOOK 1978, Stockholm International Peace Research Institute, Crane,

burn and its resultant enhancement of fission yield is referred to as "boosting," and is described in greater detail below.

Thermonuclear Explosives

The process of fusion, or the union of nuclei, is the opposite of fission (the splitting of nuclei). Partly for this reason, and due largely to opposing electrostatic charges on the uniting particles, thermonuclear reactions can only occur at temperatures in the hundreds of millions of degrees, after the particles have acquired energies and velocities sufficient to overcome these repulsive electrostatic forces.

The only known way to fuse large amounts of light elements on earth (mainly hydrogen isotopes) is by using some of the energy of an exploding fission device. For this reason, all contemporary hydrogen warheads require a fission trigger (or "primary") to get them going (the actual fusion reactions are started by a second fission explosion -- see discussion below).

There are three primary thermonuclear explosives: lithium, deuterium, and tritium. The latter two materials are hydrogen isotopes.

Lithium is element number 7, with an atomic number of three, and a nucleus composed of three protons and from two to six neutrons, forming the isotopes ${}^5_3\text{Li}$ to ${}^9_3\text{Li}$. It has a specific gravity of 0.53, and a density of about 33 lbs. per cubic foot. It is also highly chemically-reactive with water, and is extremely hygroscopic. For these reasons, lithium-bearing weapons must be sealed vacuum-tight and loaded with dessicants. Lithium has applications in nuclear weapons as both a salt, for lithium deuteride production, and as a metal, for tritium production.

The lithium isotope used most frequently in thermonuclear weapons is lithium-6, or ${}^6_3\text{Li}$. Lithium-6 comprises about 7% of naturally-occurring lithium; weapons-grade lithium is nominally enriched to 95% in the lithium-6 isotope. Natural lithium-7 is also useful in thermonuclear reactions.

Lithium containing the Li-6 isotope in concentrations up to 90% by weight is generated by the separation of Li-6 from Li-7, by the counter-current flow of a lithium-mercury amalgam, LiHg, and an aqueous solution of lithium hydroxide, LiOH, in a cascade of packed columns. The amalgam is obtained by the

electrolysis of lithium hydroxide solution with a mercury cathode.

The lighter isotope is concentrated in the amalgam by an exchange reaction, and is recovered by decomposing the amalgam with pure water in the presence of a graphite catalyst. The resultant hydroxide enriched in Li-6 is neutralized with hydrochloric acid, and the remaining aqueous solution of lithium chloride is concentrated by evaporation as the salt crystallizes.

Lithium-6 metal is then separated by electrolysis of molten lithium-6 chloride. For production of various lithium hydrides enriched in Li-6 (such as Li-6D and Li-6T), the metal is reacted at a suitable temperature and pressure with the appropriate hydrogen isotope in gaseous form.⁷¹

Lithium is probably one of the most versatile elements on earth. In addition to its thermonuclear utility, it is used in various compounds in electrolytic cells, storage batteries, cosmetics, ceramics, greases and lubricants, welding flux, and in some metalworking processes to remove gases dissolved in the metal. Lithium carbonate has been employed recently as a highly-effective anti-depressant drug for treating some mental disorders.⁷² Other hygroscopic lithium compounds are used in air conditioning and industrial drying applications.

Molten lithium is a coolant (heat exchanger) for some types of nuclear power or propulsion reactors. Lithium hydride can be used to generate hydrogen gas, while other lithium compounds are found in agricultural applications, bleaching, and in some special optical glasses. Lithium hydride is also used to produce boro-hydrides for jet propulsion, and it has theoretical possibilities as a fuel in hybrid rocket engines which use both solid and liquid propellants.

During the 1950s, production of lithium compounds in the United States increased several times over what it had been at the end of World War II (wartime lithium production rose as lithium compounds were used for hydrogen generation, carbon dioxide "scrubbers" to clean air aboard submarines, and for the production of dry batteries). The U.S. Atomic Energy Commission was the largest consumer of lithium during this period; most of it went directly into thermonuclear weapons, while a small amount was used to produce tritium (by

⁷¹Glasstone and Redman, WASH-1037 (revised), p. 22.

⁷²"A Little Lithium May Be Just What the Doctor Ordered." Wall Street Journal. September 23, 1994, pp. A1, A4.

irradiation of lithium in nuclear reactors) and to conduct research into nuclear reactor design.

The second thermonuclear fuel constituent, deuterium, is a heavy isotope of hydrogen, with a nucleus consisting of a proton and a neutron. Deuterium occurs naturally in all of the earth's water (fresh and salt) in a distribution of about one part in 6,700 (0.015%). Deuterium oxide, or heavy water, is separated from water at special facilities in the U.S.; it is then electrolytically decomposed into oxygen and deuterium gas. Heavy water of about 99.75% purity, or concentration, can be produced.⁷³

The gas is piped into pressurized flasks containing lithium salt; when the flasks are heated, the deuterium combines chemically with the lithium to form lithium deuteride (either lithium-6 or lithium-7 can be used). Lithium deuteride is the main fuel in a thermonuclear weapon.

At least one large-scale U.S. thermonuclear test device, the IVY Mike device, used liquid deuterium refrigerated to -423° F for its fuel (see Volumes III and IV). Gaseous deuterium is also used in the "boosted" fission primaries of most U.S. fusion warheads.

One gram of deuterium, completely transformed to helium, produces an energy yield equivalent to 150 tons of TNT. The specific gravity of gaseous deuterium is 0.0695 grams/cubic centimeter, or 6.95×10^4 grams per cubic meter. Hence one cubic meter of gaseous deuterium, if completely transformed to helium, yields 0.15 kt/gram $\times 6.95 \times 10^4$ grams, or 10.42×10^3 kilotons, or 10.42 megatons.⁷⁴

⁷³Glasstone and Redman, WASH-1037 (Revised), p. 21.

⁷⁴ATOMIC ENERGY COMMISSION THERMONUCLEAR WEAPONS PROGRAM CHRONOLOGY, USAEC Secretariat, January 1955, pp. 6, 13. Another source, "Excerpts from Manhattan District History, Book VIII - Los Alamos Project (Y) Volume 2 - Technical," states (in para. 1.48) that only 85 tons of TNT-equivalent explosive force is produced by the reaction of one gram of deuterium. Glasstone and Redman, in WASH-1037 (Revised), p. 156, state that deuterium alone yields 82.4 KT/kg. An Australian document, "Assessment of Radioactive Fallout Arising from Testing of Nuclear Weapons in the South Pacific and the Probable Effects on the Australian Population," (R. M. Fry, J. L. Symonds, and G. M. Watson, Australian Atomic Energy Commission Research Establishment, Lucas Heights, March 1973, p. 3) states that the complete fusion of one kilogram of deuterium is roughly equivalent to 50 KT of TNT. Part of the reason for these divergent figures may be the

Liquid deuterium has a specific gravity of 0.26 grams/cc, nearly four times that of gaseous deuterium. One liter of liquid deuterium contains 0.26×10^3 grams, and yields approximately 39 kilotons of TNT-equivalent energy. A cubic meter of liquid deuterium, completely converted to helium, yields 39 megatons of TNT-equivalent energy. One pound (454 grams) of liquid deuterium, when completely transformed by fusion to helium, yields energy equivalent to approximately 68 kilotons of TNT, more than twice the energy yield of a pound of Li^6D .

Lithium-6 deuteride is chemically identical to ordinary lithium hydride, a substance composed of lithium-7 and hydrogen, the simplest chemical compound known to man. Lithium-6 deuteride is a whitish, slightly blue powdery light salt formed by heating lithium metal in a closed flask, into which deuterium gas is injected. This powder is pressed into a ceramic with the density of an aspirin tablet and machined and shaped as required for use in a fusion warhead.

Lithium hydride poses an extreme fire risk in the presence of water. Lithium reacts violently with water to release hydrogen gas, which, when mixed with oxygen and subjected to the exothermic chemical reaction, may explode violently. In the absence of water, lithium hydride burns more slowly than wood.

The third fusion fuel, tritium, can be called "superheavy" hydrogen. It is another hydrogen isotope, with an atomic weight of three and a nucleus composed of two neutrons and a proton. Unlike the other two isotopes of hydrogen, tritium is radioactive via beta particle decay with a half-life of 12.3 years. Only traces of tritium exist in nature; most of it is produced artificially, by the irradiation of lithium in nuclear reactors. A small amount of tritium has entered the global environment as a result of atmospheric nuclear testing since 1945.

After irradiation by neutrons for a period, the fuel elements containing the lithium are removed from the reactor and gases, comprising mainly tritium and helium and some hydrogen impurities, are separated from the solid. The tritium is then purified and concentrated by a gaseous diffusion process. The tritium can be maintained in its gaseous form, or compounded chemically with other substances, such as lithium.⁷⁵

large number of reactions in which deuterium can participate and the resultant widely-varying energy levels of the reaction products.

⁷⁵Glasstone and Redman, WASH-1037 (Revised), p. 21.

Like deuterium, tritium can be combined chemically with lithium (either lithium-6 or lithium-7). In some test devices, lithium-6 deuteride and lithium-6 tritide have been combined in a mixture; such a combination is designated with the symbol Li^6DT .

Fusion Explosion Principles

The successful detonation of a thermonuclear weapon requires that a variety of competing processes be delicately balanced under conditions of temperature and pressure that are extreme even compared to conditions found at the center of stars. The basic processes in competition are those which are trying to tear apart the nuclear fuel assembly, and those which seek to hold the assembly together long enough for the weapon to achieve an appreciable fraction of its design yield (explosive force).

These opposing forces give rise to a host of incredible physical phenomena in literally infinitesimal moments of time (usually measured in fractions of microseconds). For example, the x-rays released by the fission trigger in a multistage fusion weapon create an ionized plasma (a superheated, electrostatically-charged gas) with the density of solid lead. A method of establishing these conditions, and of igniting and sustaining arbitrarily-sized thermonuclear weapons, has been implemented in U.S. warheads in a fiendishly clever and ingeniously simple way, as described below.

In the late 1940s, thermonuclear weaponeers in the U.S. were confronted with the problem of how to achieve an arbitrarily-sized, self-sustaining thermonuclear reaction (or "burn") by means of a relatively small fission trigger. This problem consisted of four key elements:

1. The establishment of temperatures, pressures, x-ray and neutron flux to form an environment conducive to thermonuclear "ignition" and "burn."
2. The ignition of a large thermonuclear fuel mass.
3. The perpetuation of fusion reactions through many generations.
4. The maintenance of the integrity of the thermonuclear fuel mass (and its surrounding plasma cloud) long enough to reach the desired explosive yield and efficiency (determined by the fraction of fuel consumed before the fuel assembly blows apart).

These were very formidable problems 45 years ago, as the thermonuclear armorsmiths in the United States raced to find the solutions before their counterparts in the Soviet Union.

A number of other factors entered the picture at that time. For example, it was widely known by then that tritium would fuse with deuterium at a temperature (energy level) an order of magnitude lower than that at which deuterium-deuterium fusion occurs. Clearly, the presence of tritium in the fusion fuel would be enormously helpful (see note 2 in "Fusion Physics" above).

It was also known by the late 1940s that tritium could be formed by irradiating lithium-6, which is found in about 8% of all lithium. In addition, lithium bonds chemically with hydrogen, so lithium-6 deuteride could be used as a solid fuel charge, once a method of starting a fusion "burn" could be discovered.

A few more words about fusion fuels are in order here. For a thermonuclear reaction to occur in a fuel charge composed of lithium-6 deuteride, the fuel must be compressed 15 to 30 times its original (uncompressed) density, which is about 50 lbs. per cubic foot.

This compression is required for two reasons, and immensely helpful for two others. It is needed to (a) increase the probability of fusion, by packing atoms closer together, thereby getting a head start on overcoming the repulsive forces generated by the opposing electrostatic charges on the nuclei; and (b) to increase the rate of fusion: reactions in a dense media always proceed much more rapidly than those in a dilute media.

In addition, the density of the fuel mass is crucial to the maintenance of the thermonuclear reactions. The time needed for a considerable fraction of the total number of lithium, tritium, and deuterium nuclei to participate in the fusion process, is, at a given temperature, inversely proportional to the density of the fuel: the more dense the fuel, the shorter the reaction times, and the greater the chance for a large number of reactions within a given period.

The thermonuclear reaction rate (and energy yield) is directly proportional to the square of the fuel density: if the density is increased by a factor of 20, the reaction rate is increased by a factor of 400. The higher the fuel density, the smaller the amount of fuel required for an equivalent yield.

At normal liquid hydrogen densities, fusion plasma temperature can rise to 60 million degrees; if a D-T mixture should be compressed to 10 times its normal density, this temperature can rise to one hundred million degrees. At normal density, mean fusion reaction time is about a microsecond; when the fuel is compressed, reaction time is

reduced by a factor of 40 to about 0.025 microseconds (25 nanoseconds).⁷⁶

In addition, the energy density of thermonuclear fuels compressed to ultrahigh densities -- even to 10 thousand times solid density -- is small compared to the energy density at thermonuclear ignition temperatures.

Compression also provides two helpful side effects: first, the integrity of the fuel mass is maintained a while longer after ignition. This method of keeping fusion fuels together by compressive force is now known as "inertial confinement fusion," in which the inertia (resistance to change in momentum) of the fuel keeps it united; it is a technique common to both fusion weapons and the generation of fusion energy for peaceful purposes.

Once a fusion reaction has been started in a mass of inwardly-collapsing thermonuclear fuel, energy is released. The motion of the fusionable material in the assembly will not be affected significantly, however, until the energy generated by fusion produces appreciable mechanical forces.

After a short period, the increased temperatures resulting from the fusion process will generate expansion pressures high enough to stop the inward motion of the assembly, and disassembly, an explosion, will begin.⁷⁷

The inertia and momentum of the fuel resist explosive forces. Contemporary inertial confinement fusion explosions, limited to microscopic fuel masses, feature confinement times of less than 10^{-10} seconds, and particle densities in the fuel are typically greater than 10^{25} per cubic centimeter.⁷⁸

In 1952, seven years after development of an atomic trigger, seismographs world-wide registered the first successful test of a multimegaton inertial-confinement device.

Second, at the center of a thermonuclear explosion, the neutron flux is also greatly increased by pre-ignition

⁷⁶"Multiple Neutron Capture in the Mike Fusion Explosion," A. G. W. Cameron, Canadian Journal of Physics, Volume 37, 1959, The National Research Council, Ottawa, Canada, p. 323.

⁷⁷Glasstone and Redman, WASH-1037 (Revised), p. 26.

⁷⁸"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 32; "Fusion," Harold P. Furth, SCIENTIFIC AMERICAN, September 1995, pp. 174, 175.

compression of the fuel. The greater the neutron flux, the higher the probability of successful fusion.⁷⁹

For inertially-confined fusion, burn efficiency -- the percentage of nuclei that fuse -- is proportional to fuel density multiplied by confinement time. This product is called the confinement parameter; it has dimensions of seconds/cubic centimeter. Newton's laws and thermal velocity limit the confinement time, but this time can be lengthened by compressing the fuel to a high density. For deuterium-tritium fusion, the confinement parameter is on the order of 10^{14} to 10^{15} seconds per cubic centimeter.⁸⁰

Densities up to one kilogram/cubic centimeter (62,430 lbs./cubic foot, about 88 times the density of lead) are energetically accessible via isentropic (constant-entropy) compression because the Fermi energy -- the minimum energy of compressed matter -- is much less than the thermal energy of the fuel at ignition temperatures.

Because the scaling parameter for the burn efficiency of the fuel is the product of the mass of the fuel and the square of its compression, for a given efficiency value, just a 40% increase in density decreases the fuel mass and primary x-ray flux by a factor of two. This means that a given thermonuclear yield can be attained with a smaller primary and less fusionable fuel, if the fuel is compressed sufficiently.

A powerful force works against compression. Preheating, the undesirable untimely transfer of energy from the primary to the secondary, is caused mainly by neutron "leakage" from the primary before the secondary has been optimally compressed. "Hot," or superthermal, electrons and neutrons add energy to the interior of the thermonuclear fuel capsule and make compression less efficient.⁸¹

As mentioned earlier, the only known practical method of igniting an uncontrolled fusion reaction (or explosion) is with the aid of an atomic, or fission, explosive. As with

⁷⁹"The H-Bomb: Who Really Gave Away the Secret?," Daniel Hirsch and William G. Mathews, BULLETIN OF THE ATOMIC SCIENTISTS, Volume 46 No. 1, January/February 1990, p. 29.

⁸⁰"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 32; "Fusion," Harold P. Furth, SCIENTIFIC AMERICAN, September 1995, p. 174.

⁸¹"The Feasibility of Inertial-Confinement Fusion," John H. Nuckolls, PHYSICS TODAY, September 1982, pp. 25-28.

fission explosives, compression of the thermonuclear fuel mass, before ignition, is the key to a successful ignition and high yield.

If the fuel were not compressed, the atoms in it would be scattered by the ignition of the fission "sparkplug" long before they could have an opportunity to participate in a thermonuclear reaction. Ideally, the fuel is not only highly compressed, but is also collapsing inwards upon itself, giving its atoms an inwardly-directed momentum to help resist the outwardly-directed explosive forces generated by the fusion reactions.

To complicate matters, this compression has to occur before the fuel mass is appreciably heated. It is always easier to compress a substance at a low temperature, i.e., low atomic particle kinetic energy level, than at a high temperature: the higher the temperature of a material, the greater the motion and kinetic energy of its atoms, and the better it can withstand compressive forces. Hence, the thermonuclear fuel has to be compressed while still "cold" -- yet it must be heated in order to be ignited! This apparent contradiction needs to be solved in the weapon design, or fusion will not occur.

To further complicate matters, only a few microseconds are available to start fusion reactions and for them to rise to an appreciable degree before the fragments and shock wave from the exploding primary destroy the integrity of the thermonuclear fuel charges. Whatever must be done to begin fusion must happen very quickly.

The solution to the problems cited above, as developed in the U.S., is known as the "Teller-Ulam configuration." It is named after its inventors, Drs. Edward Teller (a physicist) and Stanislaw Ulam (a mathematician). It was more than a matter of physical positioning of components -- it was a whole concept of arrangement and repetition of arrangements which solved the problem of igniting an arbitrarily-sized thermonuclear explosion by means of a relatively-small fission trigger.

When this configuration is used, the only limits on explosive yield are the military requirement, and the lifting capabilities of the delivery vehicle. The thermonuclear explosive power is theoretically unlimited, bound only by the physical strength of the weapon casing, which acts as a big blast containment chamber, confining the explosive forces primarily by means of its massiveness relative to the mass of the fuel charge.

The fusion reactions proceed as long as possible, until they reach a state of equilibrium in which the energy loss, in the

form of radiated ionizing energy and atomic and subatomic particles and rays, to the atmosphere in the vicinity of the explosion is equaled or exceeded by the rate of energy production, or until the reacting fuel mass blows itself apart. Obviously, some sort of tamper around the fuel mass, within the weapon casing, is very useful in delaying the physical disintegration of the fuel charge. The casing itself acts as a secondary tamper for the entire fission trigger-fusion fuel charge reaction.

The history of the discovery of the Teller-Ulam configuration is told in Volume III. The "secret" of the design is the arrangement of elements within the fusion fuel charge, and the relation of the fusion fuel charge(s) to the fission trigger (or "primary"). The fusion fuel assembly, which can be repeated in stages, each stage setting off the next, is as follows.

Whereas a fission primary is designed as "spheres within spheres," a fusion fuel capsule can be characterized as "cylinders within cylinders."

At the center of the assembly is a thin subcritical cylindrical rod of highly-enriched weapons-grade uranium-235 (oralloy) or plutonium, surrounded by the fusion fuel charge, which can be from six to 24 inches in diameter. The cylindrical fuel charge is in turn wrapped with a thin foil or sheet of ordinary, depleted, or enriched uranium metal (tuballoy, depletalloy, or oralloy). This uranium wrapper is called a "pusher;" it can be up to an inch thick.

This uranium mass can be quite heavy, due to the high density of uranium. Any uranium in a weapon contributes significantly to its overall weight. Uranium has a density of 1,183 lbs. per cubic foot, or about 0.7 lbs. (11.2 ounces) per cubic inch. Table 1-3 below lists the weight of a hollow tube of uranium five feet long and about one meter (39.37") in outside diameter:

Table I-3 Uranium Thickness-to-Weight

<u>Jacket Thickness</u>	<u>Jacket Weight</u>
1/4"	1,262 lbs.
1/2"	2,509 lbs.
3/4"	3,740 lbs.
1"	4,954 lbs.
1 1/4"	6,152 lbs.
1 1/2"	7,334 lbs.
1 3/4"	8,500 lbs.
2"	9,650 lbs.

The first fusion fuel stage, the "secondary," is separated from the fission primary by a few inches, with a thick lead or uranium radiation shield between them. This shield keeps the fusion fuel charge from being preheated by radiation from the primary; the "pusher" surrounding each fusion fuel stage protects the fusion fuel within it from x-radiation. The radiation shield is pierced by a hollow tube linking the primary to the end of the sparkplug in the secondary nearest to the primary.

Subsequent fusion stages, the "tertiary" and "quaternary," if used, are separated from each other by similar radiation shields and linked by similar neutron transport tubes. These tubes conduct neutrons to initiate fission in the compressed sparkplug in each stage.⁸²

⁸²This description of thermonuclear weapons components and explosive processes is based on several sources: Howard Morland's descriptions and diagrams in the November and December 1979 issues of THE PROGRESSIVE magazine (Vol. 43 Nos. 11 and 12); an article entitled "The Humanitarian H-Bomb," by Dr. Ralph E. Lapp, on pages 261 - 264 of BULLETIN OF THE ATOMIC SCIENTISTS, Vol. 12, (September) 1956; descriptions of possible weapons designs presented in Dr. Friedwardt Winterberg's THE PHYSICAL PRINCIPLES OF THERMONUCLEAR EXPLOSIVE DEVICES (Fusion Energy Foundation, New York, New York, 1981; Dr. Winterberg has a marvelously inventive imagination); descriptions of thermonuclear processes in BORN SECRET: THE H-BOMB, THE PROGRESSIVE CASE, AND NATIONAL SECURITY (A. DeVolpi, et. al., Pergamon Press, New York, New York, 1981); an article entitled "Fusion Devices, Explosive," by Dr. Edward Teller in ENCYCLOPEDIA OF PHYSICAL SCIENCE AND TECHNOLOGY, Volume 5, pp. 723-726, Academic Press, Inc., San Diego, California, 1987; and comments on pages 47 and 48 of the Joint Reply Brief of Appellants Knoll, Day and Morland, Public Filing dated September 26, 1979, in the United States Court of Appeals for the Seventh Circuit, Case Number 79-1428, 79-1664. The latter document corrects Morland's mistaken understanding of "radiation pressure" (i.e., radiation implosion) and the role of neutrons in the fusion-initiation process.

An additional description of thermonuclear fuels, weapons components, and explosive processes is given on pages 26 to 28 of NUCLEAR WEAPONS DATABOOK: Volume I, U.S. Forces and Capabilities, by Cochran and Arkin. The inclusion of "sparkplugs" in U.S. thermonuclear weapons is confirmed by an affidavit filed by Dr. Ray Kidder in the PROGRESSIVE case:

UCRL-4725 discloses very detailed information describing the principles of operation, performance and design

specifications of both relatively sophisticated fission bombs and early successful hydrogen bombs. Whereas the details described in the Morland (article) would not be of much help to the would-be H-bomb designer, the information disclosed in UCRL-4725 would be extremely helpful. A vitally important secret of H-bomb design is totally absent in the Morland article. This secret is clearly revealed in UCRL-4725. (In-Camera Affidavit III of Ray E. Kidder in Civil Action No. 79-C-98, United States of America, Plaintiff, v. The PROGRESSIVE Magazine, Erwin Knoll, Samuel H. Day, Jr., and Howard Morland, Defendants, in United States District Court for the Western District of Wisconsin, filed June 4, 1979. Author's emphasis added.)

The "vitally important" design secret that Dr. Kidder refers to is the so-called "sparkplug" that did not appear in Morland's original H-bomb design in the November 1979 issue of THE PROGRESSIVE, but which was included in his "Mark II" corrected design in the December 1979 issue of THE PROGRESSIVE, and which was discussed at some length in his 1981 book, THE SECRET THAT EXPLODED (Random House, New York). This previously in-camera affidavit by Dr. Kidder, with its description of the significance of UCRL-4725 and the "sparkplug," was released by the Departments of Energy and Justice in September 1980, at a time when copies of UCRL-4725 were known to be circulating in the public domain. Morland stated in his book that he saw UCRL-4725 in May 1979, after his original bomb design had been impounded by DOE and before his revised design was drawn up after the government dropped the case in September 1979.

The inclusion of a "sparkplug" in U.S. thermonuclear weapons from their inception was confirmed by no less a person than Dr. Edward Teller himself, in the course of a sickbed interview with Dr. George A. Keyworth in 1979, during the PROGRESSIVE CASE:

There was no question in my mind how the super should be made. We needed a primary, we needed a secondary, we needed a sparkplug. The secondary had to have cylindrical symmetry because that was the symmetry natural to the process. The energy transfer had to go by radiation in order to make it as close

to simultaneous as possible. All this was clear to me. (Interview of Dr. Teller by Dr. Keyworth cited in EDWARD TELLER: GIANT OF THE GOLDEN AGE OF PHYSICS, Stanley A. Blumberg and Louis G. Panos, Charles Scribner's Sons, New York, 1990, p. 130.)

A cutaway diagram of a thermonuclear fuel stage appears on page 302 of ARMS CONTROL: READINGS FROM SCIENTIFIC AMERICAN (W. H. Freeman Company, San Francisco, 1974) in an article entitled "The Great Test Ban Debate," by Dr. Herbert F. York (a former director of the Lawrence Livermore nuclear weapons lab), reprinted from SCIENTIFIC AMERICAN, November 1972. If it is interpreted as a lateral cross section of a cylinder, and not as a cross section of a sphere, this drawing shows the correct configuration of a fusion fuel charge: sparkplug in the center, encased by a layer of lithium deuteride wrapped with uranium-238.

When queried by the author about the significance and origins of this drawing, Dr. York replied in a letter dated 18 November 1986 in which he stated:

The sketch you enclose was in no way intended to be a sketch or configuration of a real hydrogen bomb...The sketch was designed to make this (thermonuclear) sequence clear, not to provide a drawing, or even rough sketch, of an actual device. I fear the caption did not make this absolutely clear.

Dr. York's statement notwithstanding, the sketch -- if interpreted properly -- does show a fusion stage cutaway (either by intent or coincidence).

In addition, the use of "soft" (long-wavelength) x-rays to compress fusion fuels was discussed in the lead article on particle beam weapons in AVIATION WEEK & SPACE TECHNOLOGY magazine in 1977, two years before the PROGRESSIVE case:

His idea startled the U.S. physics community by its magnitude -- transforming laser and electron beams to soft X-rays to compress fusion fuels at low energy levels. ("Particle Beam Weapons," reprinted from AVIATION WEEK & SPACE TECHNOLOGY, May 2, 1977).

"Soft" x-rays are among the primary outputs of exploding

fission weapons:

Within about one microsecond after the explosion, some 70 to 80 percent of the explosion energy ... is emitted as primary thermal radiation, most of which consists of soft X rays. (THE EFFECTS OF NUCLEAR WEAPONS, Samuel Glasstone and Philip J. Dolan, prepared by the United States Departments of Defense and Energy, U.S. Government Printing Office, 1977, p. 24).

The so-called "three concepts" at issue in the PROGRESSIVE case were identified by the U.S. Departments of Justice and Energy in September 1980:

The following items are basic features of the design or operation of the U.S. type thermonuclear weapon and are classified as Secret Restricted Data:

- a. Separate stages -- a physically separate fission explosive (stage) and a capsule (stage) of thermonuclear fuel, centered at separate points.
- b. Radiation coupling -- channeling (ducting) of thermoradiation from the first stage to ignite the second stage.
- c. Compression -- implosion of the thermonuclear fuel capsule prior to ignition to achieve maximum yield. (Brief of the Appellant, THE PROGRESSIVE, Inc., filed June 15, 1979, in the United States Court of Appeal for the Seventh Circuit, case no. 79-1428, pp. 19, 20.)

The third concept above, compression, was revealed in the context of the Operation IVY "Mike" thermonuclear shot in 1959, just seven years after IVY and 20 years before the PROGRESSIVE case:

It seems likely that some compression will have occurred since the D-T mixture was squeezed between the uranium mantle and the expanding gas from the fission explosion. ("Multiple Neutron Capture in the Mike Fusion Explosion," A. G. W. Cameron, Canadian Journal of Physics,

The explosive sequence in a multistage weapon is as follows: the fission trigger, or primary, which itself may be a fusion-boosted device, explodes and emits a very large pulse of photons and "soft" long wavelength x-rays, traveling at the speed of light, about 30 times faster than any unimpeded debris or nuclear particles also released in the explosion.

This "blackbody" radiation flows almost instantaneously through the weapon casing (also called the "radiation channel"). The trigger itself may be configured so as to emit this radiation primarily in one direction, rather than omnidirectionally; in addition, there may be focusing elements present between the trigger and first fusion stage, and between subsequent fusion stages to further concentrate and direct the x-rays and photons.

Most of this energy is absorbed immediately by the interior lining of the weapon casing which is constructed and machined from special materials; it is then re-radiated as secondary x-rays. Up to 70 to 80% of the absorbed energy can be

Volume 37, 1959, The National Research Council, Ottawa, Canada, p. 323; emphasis added.)

These "three concepts" are conspicuously-absent from all other previously-classified court documents released in September 1980 at the conclusion of the PROGRESSIVE case; the cited brief is the only place where they appear (probably by accident). These concepts were (finally) officially declassified by the U.S. Department of Energy in September 1980 with the release of the following statement:

In thermonuclear weapons, radiation from a fission explosive can be contained and used to transfer energy to compress and ignite a physically-separate component containing thermonuclear fuel.(Official Declassification Action of the Department of Energy implemented in September 1980 by Duane C. Sewell, Assistant Secretary of Energy for Defense Programs.)

These concepts had been declassified in the Soviet Union by 1975, at least in the context of inertial confinement fusion research. In the U.S., the Department of Energy had always previously argued that the application of the "three concepts" to weapons research was inseparable from their application to controlled fusion research, and so must remain classified eternally (or at least until 1979).

converted to x-rays.⁸³ This process is called "bremsstrahlung," the secondary photon radiation produced by the deceleration of charged particles passing through matter.

In addition, a small portion of the inner casing is vaporized and turned into a plasma. Although this plasma is a very hot, very energetic ionized gas, so much is created that it has the density of solid lead. By filling the volume between the inner casing and the outer surface of the secondary stage tamper, this plasma helps confine the expanding secondary (after its ignition) and keeps the explosive assembly together for a few more microseconds.

In addition to the plasma, the inner surface of the casing re-radiates secondary x-rays, some of which are absorbed by the outer surface of the fusion stage tamper. This absorption causes the outer surface to ablate, or "blow off;" the reactive force from the ablation -- in accordance with Newton's Third Law of Motion, which states that to every action, there is an equal and opposite reaction -- acts as a circumferential "piston," producing a rapid and high compression of the tuballoy or oralloy foil, the lithium-6 and lithium-7 deuteride fuel charge, and the innermost "sparkplug" oralloy or plutonium rod.⁸⁴

When the fusion fuel capsule has been compressed, neutrons of several different energy levels from the primary come into the sparkplug, through the tube in the radiation shield separating the primary from the secondary. At this point, the sparkplug begins to fission, starting at the end closest to the primary. This fission reaction, occurring in the middle of the highly-compressed fusion fuel charge, initiates the main fusion explosion in the secondary. The heat, i.e., particle velocity, of the fissioning sparkplug is sufficient to start tritium-breeding via lithium transformation caused by neutrons released in the new fission process.

The newborn tritium is spawned with enough energy to fuse with deuterium in the lithium deuteride fuel charge (there are many other thermonuclear reactions which can also occur - see equations in "Fusion Physics" above). Thus the tritium is bred in situ, and no additional tritium is required as a

⁸³"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 34.

⁸⁴"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, pp. 7, 8.

separate fuel constituent. This method of igniting the compressed fuel solves both of the problems cited earlier -- the fuel is compressed "cold," then ignited after being "heated" by the sparkplug.

The ignited thermonuclear fuel burns by means of chain reactions: neutrons from D-D reactions and from the fissioning "sparkplug" interact with lithium to form tritium. The tritium, in turn, undergoes thermonuclear reactions with the deuterium, releasing more neutrons, which produce more tritium and more fusion.⁸⁵

At the beginning of the reaction, there are two distinct regions in the thermonuclear fuel mass: a central hot spot immediately adjacent to the fissioning "sparkplug," and a dense main fuel region, called the "cold fuel pusher." Fusion begins in the central hot spot, and a thermonuclear burn front propagates rapidly outward into the main fuel mass.⁸⁶

In general, only the charged particle reaction products (the helium and hydrogen nuclei) are available to heat the fuel because most of the neutrons escape the plasma. For example, in the D-T reaction which generates an alpha particle (a helium nucleus) and a neutron, the neutron flies out with about four times as much kinetic energy as the alpha particle.⁸⁷

The high-energy neutrons bred by the fusion reactions fly out radially and strike the compressed tuballoy or or alloy surrounding the charge. The uranium also begins to fission; soon the compressed fusion fuel is trapped between two fission explosions (uranium-238 has a higher fission threshold than uranium-235, and requires faster, fusion-bred neutrons to "ignite" it).

The result of all these fission and fusion reactions is a synergistic exchange of energies and particles, increasing the rate and frequency of all the subatomic collisions and unions. Not only does the U-238 contribute directly to total

⁸⁵Rosengren, RDA-TR-122100-001-Rev. 1, p. 15.

⁸⁶"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 32.

⁸⁷"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No., September 1992, p. 33.

explosive yield by fissioning, it also supplies heat and neutrons to increase the burn-up of thermonuclear fuels.⁸⁸

Since the fissionable and fusionable materials within the fusion stages are compressed to very high densities, the resultant reactions are much more efficient than similar reactions occurring in a boosted primary.⁸⁹

Uranium plays a major role in most thermonuclear weapons designs. Thermonuclear explosives can be used as mechanisms for fissioning uranium and plutonium in large quantities. This was especially true for early U.S. thermonuclear devices between 1952 and 1954, 52% to 90% of whose total yields were generated by fission reactions (see Volumes III and IV). Fusion is a method of obtaining fast high energy neutrons to fission ordinary unenriched uranium, as well as a means of overcoming the limits on critical masses, and a technique for improving yield-to-weight and yield-to-volume ratios.⁹⁰

As shown in Table I-1, the fission of uranium isotopes and plutonium by 14 MeV neutrons generates, on average, nearly twice as many neutrons as are released when the same isotopes are struck by 0.5 MeV neutrons.

Devices tested at the Pacific Proving Grounds during the REDWING test series in mid-1956 had total fission yield percentages ranging from 5% to 87%. Those weapons at the low end of this range were the so-called "clean" bombs much touted at U.S. Atomic Energy Commission press conferences. The explosion of the natural uranium pusher boosts the overall yield by 30% to 50%.

The fission yield can be increased another 50% by replacing the tuballoy (unenriched uranium-238) pusher with oralloy (enriched uranium-235). Oralloy can be fissioned by neutrons with several different kinetic energies -- both "slow" and "fast" neutrons -- while tuballoy can be fissioned by only "fast" neutrons.⁹¹ For this reason, most contemporary high-

⁸⁸"Discussion," Appendix "A" to AEC 778, Atomic Energy Commission, Radioactive Contamination, Note by the Secretary, December 14, 1954, p. 4.

⁸⁹"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 8.

⁹⁰Letter dated 16 May 1983 from Jerry Fass to author.

⁹¹"U.S. To Resume Uranium Production for Weapons," Robert Alvarez and Debra Sherman, BULLETIN OF THE ATOMIC SCIENTISTS,

yield U.S. thermonuclear warheads use large amounts of uranium enriched to varying degrees in U-235.

This infernal holocaust proceeds in a roughly end-burning fashion; the x-rays, photons, and neutrons of the exploding secondary are directed into the next stage, the tertiary (if present). During all these explosions and reactions, the outer weapon casing acts like a bottle, keeping the radiation and explosions contained within it .

Because the casing is extremely massive relative to the fusion packages, it expands quite slowly compared to the rate of compression and expansion of the fusion materials. It eventually fragments and vaporizes, when its inertia can no longer withstand the expansive forces of the explosions bursting within it. By that time, all the stages have ignited and burned to an appreciable extent.

This staging process can be repeated indefinitely, leading to the so-called "doomsday" bombs salted with cobalt to maximize the quantity, lethality, and radioactive longevity of the post-blast fallout. As Dr. Strangelove so aptly stated in the movie of the same name, if you just want to bury bombs, as opposed to carrying them to their targets, there is no limit to their size.

This arrangement of fission and fusion fuel charges presents several advantages and disadvantages. The major advantage is that any number of stages can be linked, for any desired explosive size (yield). This particular arrangement is also especially efficient and practical. The primary disadvantages are finding a means of directing the radiation from each stage to compress the following stage, and configuring or lining the interior of the weapon casing to guarantee a symmetric compression of the fusion fuel-sparkplug package in each stage.

In addition, the massive, thick casing required for a high-yield weapon imposes a significant weight penalty on the warhead. This weight is further increased by large amounts of uranium in the pushers, as shown in Table I-3 above.

This arrangement permits fusion efficiencies up to 35 or 40%;⁹² compressions are adequate to squeeze and ignite the

Vol. 41 No. 4, April 1985, p. 29.

⁹²"Nowhere is there a correct description of the type of design used in U.S. weapons. This type is far superior in efficiency and practicality to any other known type of design." (Affidavit of Jack W. Rosengren, filed March 13, 1979, in Civil Action 79-C-98.) This was an unclassified public filing, widely reported in the press at the time during the early

sparkplug in each stage. Fuel densities on the order of 6,243 lbs. per cubic foot, nearly 200 times that of uncompressed Li^6D , may be achieved.⁹³

The staging principle allows weapon yields to be measured in terms of megatons per unit length of the fusion fuel charge. Thickness of the fuel charge is also important, as it affects the degree of compression. The sparkplug itself also makes an appreciable contribution to the yield of each stage.

This glimpse into the fires of hell on earth may seem to be a lengthy process; however, a three-stage thermonuclear weapon

stages of the PROGRESSIVE case in the spring of 1979; the affidavit was written and released in part to stir up public and press outcry against the PROGRESSIVE. Dr. Rosengren's credentials were impressive: he worked at the DOE's Lawrence Livermore Laboratory in the late 1950s and early 1960s, when he helped design the W-47 POLARIS missile warhead and the W-56 MINUTEMAN ICBM warhead. He later became Associate Director for Nuclear Weapons Design at Livermore, then Deputy Director for Science & Technology at the U.S. Defense Nuclear Agency.

Assuming that Dr. Rosengren's assertion was correct, and that the U.S. design is the best of all possible designs (notwithstanding Howard Morland's mistaken original impressions of it), and that Dr. Rosengren's directorship at the DNA probably put him in a very favorable position to become familiar with foreign nuclear warhead designs (and compare them to the U.S. design), then efficiencies in the range that I have cited are extremely probable. It is well-known that most modern thermodynamic appliances, from steam boilers to aircraft and automotive engines, operate in this efficiency range. The United States has been perfecting and improving fusion warheads for more than 40 years, and we are generally acknowledged to lead the rest of the world's thermonuclear powers in the art of nuclear warhead technology, so efficiencies of this order are not unlikely.

⁹³Additional footnote 33' to "The H-Bomb: Who Really Gave Away the Secret," Daniel Hirsch and William G. Mathews, in Soviet Physics Usp. 34 (5) May 1991 (Uspekhi Fizicheskikh Nauk 161, 153-169), translated from the Russian (after originally appearing in English in the Bulletin of the Atomic Scientists, January/February 1990, pp. 22-30. Fuel compression before disintegration or ignition may actually be closer to only 15 to 40 times uncompressed density. Recent small-scale inertial confinement experiments have resulted in fuel densities on the order of 28 times that of uncompressed lithium. ("Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 38.)

will pass through many fission and fusion explosions within a matter of only five to 10 microseconds. To an observer, all of these explosions appear to run together into a single instantaneous blast. Human sensory perception is simply too slow -- by several orders of magnitude -- to discern each separate event in the explosive sequence.

The yield of each stage is directly dependent upon the yield of the stage immediately preceding it starting with the chemical explosion in the primary, the degree of compression of the fusion fuel charge, the yield of the sparkplug in the stage, the thickness (diameter) of the tuballoy or oralloy pusher, and the length of the fusion stage. These many variable have given rise to a wide range of warheads, of differing sizes and yields, currently in the U.S. nuclear stockpile.

Within this complex sequence, each stage releases energy roughly an order of magnitude greater than the preceding stage. If the primary yields a few tens of kilotons, then the secondary will yield several hundred kilotons, and the tertiary (if used), will explode with a force of several megatons.

The 58 MT Soviet blast of October 30, 1961 -- the largest nuclear test ever announced -- was probably a test of a three-stage device with a primary that yielded several hundred kilotons. One or more of the thermonuclear fuel capsules of this Soviet device were reportedly encased in lead, rather than uranium, in order to reduce the overall yield which otherwise would have approached 100 megatons.

Most contemporary U.S. thermonuclear warheads have a fission trigger which yields from 20 to 60 kilotons; modern ICBM and cruise missile warheads are two-stage devices with yields in the hundreds of kilotons. Only a few bombs currently in the U.S. stockpile have yields in the multimegaton range.

The nemesis of thermonuclear weapon designers is Rayleigh-Taylor fluid instabilities. This is a physical phenomena in which strong eddy currents build up at the boundaries of materials of differing densities moving through each other, causing loss of energy via unwanted transfer to random motion; the instabilities are strongest when light materials are accelerated against heavier materials.

When a dense fluid is accelerated by a lighter one, hydrodynamic theory states that any initial perturbation at the interface between the fluids will grow without limit. The resultant eddy currents cause the boundary between the two materials to be unstable and turbulent, leading to unpredictable blending. Taylor instability was a significant

factor in the theoretical consideration of the performance of certain early U.S. thermonuclear weapons designs.

In the design whose operation is described above, there can be two occurrences of hydrodynamic instability. The first occurs during the ablation of the outer layer of the uranium tamper resulting from the impact of the plasma generated by the x-rays from the primary being re-radiated by the inner casing liner. The hot ablated material from the tamper acts as a low-density fluid, while the dense unablated remainder of the tamper acts as a heavy fluid.

The second occurrence takes place during stagnation when the inward-moving tamper is decelerated by the lower-density, high-temperature fusion "hot spot" created by the fissioning sparkplug igniting the center layer of the main thermonuclear fuel charge. The Taylor instability can cause the cold main fuel charge to mix with the "hot spot," preventing ignition.⁹⁴

There may also be another lesser instability around the dense fissioning sparkplug, at its boundary with the innermost area of fusion reactions.

Focusing

The radiation from the exploding primary and subsequent stages must be focused and distributed throughout the weapon casing (the "radiation channel") in order to properly compress the fuel mass in a following stage before it is ignited by the fissioning sparkplug at its center.

Ideally, the compressive radiation wave travels down the length of the cylindrical stage, keeping pace with the igniting sparkplug. This is not easily accomplished, and it was done successfully by the U.S. only after many full-scale thermonuclear experiments. These compression tests were difficult to observe and instrument, and their results were often obscured or obliterated by other effects of the explosion.

At first glance, an obvious initial step toward ideal stage compression is the use of an elliptical radiation and shock wave focusing chamber. One of the properties of an ellipse is that rays emanating from one point within the ellipse (one

⁹⁴"Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 36; "Creating Microsphere Targets for Inertial Confinement Fusion Experiments," ENERGY & TECHNOLOGY REVIEW, UCRL-52000-95-4, April 1995, Lawrence Livermore National Laboratory, Livermore, California, pp. 4, 5.

focus) are reflected to an opposite point (another focus) within the ellipse.

For this reason, it is likely that most modern thermonuclear devices and weapons feature a similar focusing mechanism, at least at the end of the casing nearest the primary. Even though this chamber exists for only a fraction of a millisecond after the primary detonates, it should suffice to properly compress and ignite the secondary. For multistage weapons, each successive stage is located at the focus of another elliptical chamber, in the same way that the primary is located relative to the secondary.

A technique called "indirect drive" is used in inertial confinement fusion experiments and may be applied to thermonuclear weapons. "Indirect drive" achieves highly uniform irradiation of the fusion capsule by suspending the capsule within a spherical or cylindrical cavity lined with a foil made of gold or some other heavy metal. A beam of radiation enters an aperture in the cavity and vaporizes the metallic lining, generating x-rays that converge on the fusion target and compress it to ignition temperature and pressure.⁹⁵

Another arrangement to distribute radiation quickly throughout the interior of the weapon casing is to place the primary stage at the focus of a parabolic reflecting mirror. A property of a paraboloid is that rays emanating from its focus are all radiated parallel to one another. This helps distribute radiation rapidly and uniformly.

In addition, there may be physical or chemical "baffles" in the thermonuclear fuel masses or along the interior of the weapon casing lining to direct the radiation and photons from the exploding primary to achieve a more uniform distribution throughout the fuel masses. A "baffle" would serve its purpose for only a few nanoseconds before being vaporized.⁹⁶

⁹⁵"Fusion's Future," Science and the Citizen, SCIENTIFIC AMERICAN, February 1989, p. 28; "Flash Point," Science and the Citizen, SCIENTIFIC AMERICAN, April 1989, p. 18; "Progress Toward Ignition and Burn Propagation in Inertial Confinement Fusion," John D. Lindl, Robert L. McCrory and E. Michael Campbell, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 34; "Energy from Inertial Fusion," William J. Hogan, Roger Bangerter, and Gerald L. Kulcinski, PHYSICS TODAY, Vol. 45 No. 9, September 1992, p. 48. In Howard Morland's article in the November 1979 issue of THE PROGRESSIVE (p. 15), Morland recounts how a student at the University of Alabama told him that there were "radiation reflectors" in U.S. H-bombs.

Boosted Fission Principles

One way to substantially improve the yield of a fission explosive is by the technique of "boosting," or the creation of a relatively-small fusion reaction in the center of the fissile core. This method is employed in most small fission weapons in the U.S. stockpile and in all fusion weapon triggers.⁹⁷

When a fission weapon is detonated, the center of the core is subjected to extreme pressures and temperatures resulting from fission reactions and the implosion of the pit caused by the explosion of the chemical high explosives surrounding it. The pressure is typically on the order of gigabars and the temperature is in the tens of millions of degrees.

⁹⁶Howard Morland's "exploding foam" H-bomb design included a tapered, carrot-shaped fusion fuel capsule (initially without "sparkplug"). The taper was due to Morland's belief that the end of the capsule closest to the primary would be hotter, so it had to be thicker. According to Morland, DOE offered at one point late in the PROGRESSIVE case to permit publication of his article if he removed the taper from the secondary (he refused to do so, and the case dragged on). Whether or not the secondary is in fact tapered is beyond the scope of this book and not highly relevant to this discussion of thermonuclear explosive principles.

⁹⁷See, for example, "The Making of the H-Bomb," TIME Magazine, 12 April 1954; Morland, BORN SECRET, pp. 87, 88; "The H-Bomb Secret," Howard Morland, THE PROGRESSIVE, November 1979, pp. 14-23; and papers in the Dwight D. Eisenhower Library, Abilene, Kansas, Papers as President of the United States, 1953 - 1961, diary series, page dated October 25, 1957 ("President (met) ... with British regarding ... boosting principle in small weapons"). Ironically, the editor of this information service, although aware that the U.S. had experimented with fusion boosting in the 1950s, did not know for certain that boosting was used in U.S. thermonuclear warheads until the government released the affidavit of Thomas R. Pickering of the National Security Council Ad-Hoc Working Group on (Nuclear) Nonproliferation in March 1979, during the early stages of the PROGRESSIVE case. Dr. Pickering was later named U.S. ambassador to the United Nations and is now U.S. ambassador to India.

A recent letter dated February 27, 1987, from George H. Miller, Associate Director for Defense Systems, Lawrence Livermore National Laboratory, to Senator Edward M. Kennedy, states "...boosted fission triggers .. are a fundamental component of most U.S. [nuclear] weapons."

These conditions, at the center of the compressed pit, are sufficient to initiate fusion reactions, if fuel is present. Fusion-bred neutrons, because of their high energies (on the order of 14 MeV), cause more neutrons to be released when they collide with fissionable nuclei. This is one of the synergistic ways that fission and fusion reactions interact with one another.

This interplay of reactions causes the initial fission chain reaction to multiply much more rapidly than it otherwise would, creating more fission reactions, which in turn forces even more fusion to occur because of the high temperatures, i.e., kinetic energies of the fission neutrons. Hence more nuclei will be split before the pit disintegrates, greatly increasing both the efficiency and yield-to-weight ratio of a given amount of fissile material.

Boosted fission weapons are usually more efficient, lighter (because of the need for less fissile material for the same yield), and more powerful by a factor of two to ten than unboosted fission devices. Boosted warheads generally have yields in the range of a few hundred kilotons, overlapping the low end of thermonuclear weapon yields. An efficiently-boosted fission weapon could -- theoretically -- approach one megaton in yield.⁹⁸

This improvement in efficiency can be seen most easily by comparing the alpha, or rate of fission build-up, in an unboosted weapon to that in a boosted weapon. As noted earlier, the alpha for a pure fission device can range from 25 to 200 per microsecond. For a fusion-boosted weapon, the alpha can be on the order of 1,000 per microsecond -- an improvement by a factor of five to 40.

This number is based largely on the 30:1 ratio of thermonuclear neutrons generated to fission neutrons and on the much-higher kinetic energies of the fusion neutrons. Such a gain is far beyond any that could be achieved by a purely mechanical or physical rearrangement or refinement of fission weapon design.

Enough neutrons are produced by the complete fusion of just three grams of tritium to fission 240 grams of plutonium or weapons-grade uranium, producing a yield of four kilotons from a fusion yield only one-tenth as large.⁹⁹ The amount of

⁹⁸"Known Soviet Nuclear Explosions, 1949 - 1985, Preliminary List," NWD 86-3, Nuclear Weapons Databook Working Paper No. 3, Jeffrey I. Sands, Robert S. Norris, and Thomas B. Cochran, Natural Resources Defense Council, Washington, D.C., February 1986, p. 8.

fission energy required to initiate fusion in an imploded core can be as low as just 300 tons of HE explosive equivalent.¹⁰⁰

There are a limited number of choices for the fusion fuel at the center of the pit. Lithium-6 deuteride is an immediate possibility, but it is not the best choice because of the absorptive effect of the lithium on the fission neutrons: neutrons which otherwise would enhance the fission reactions are required to breed tritium from the lithium.

Liquid deuterium and liquid tritium are unfeasible, due to the radioactive decay of tritium, and the requirement for pressurization and cooling equipment. The only remaining choice is a mixture of deuterium and tritium in a gaseous state. Even this choice requires periodic refreshment of the tritium, due to the short (12.3 years) half-life of the isotope.

Thus any gas that is stored permanently in the fission warhead (actually in a sealed reservoir inside the weapon) requires occasional replacement, causing difficulties in weapon storage and assembly. Another problem with this arrangement is the requirement for a hollow core, and the loss of a logical and convenient location for the fission initiator, or "urchin," which provides neutrons to start the fission reactions after it is crushed by the inwardly-directed implosive shock wave.

The solution to these design problems in fusion-boosted U.S. warheads is the employment of a gas-boosting system, which injects a pressurized D-T mixture into the warhead pit as part of the weapon arming sequence before the high explosives around the pit are fired.¹⁰¹ Boosting does not occur until the core has been appreciably compressed and fissioned; conditions for the thermonuclear boosting reaction are established only after considerable fission yield has already been generated.¹⁰² Substantial "burning" of the deuterium-

⁹⁹"A Little Tritium Goes a Long Way," David Albright and Theodore B. Taylor, BULLETIN OF THE ATOMIC SCIENTIST, Vol. 44 No. 1, January-February 1988, p. 39.

¹⁰⁰Paper presented by Christopher E. Paine, Natural Resources Defense Council, at the 27th JAIF Annual Conference, Hiroshima, Japan, April 13-15, 1994, p. 10.

¹⁰¹See "Errata," Howard Morland, THE PROGRESSIVE, Vol. 43 No. 12, December 1979. This is Morland's "Mark II" H-bomb design.

¹⁰²HYDRONUCLEAR EXPERIMENTS, LA-10902-MS/UC-2, Robert N. Thorn and Donald R. Westervelt, Los Alamos National Laboratory, Los Alamos, New Mexico, February 1987, p. 7.

tritium gas does not occur until a few hundred tons of TNT equivalent fission energy has been released.¹⁰³ In this manner, the mixture is used to maximum advantage.

The gas reservoir and pressurization systems are external to the weapon core. Pressures in the gas boosting system are typically on the order of a few thousand pounds per square inch; usually only a few grams of deuterium and tritium are injected. Injection of the D-T gas mixture into the pit is often a final step in arming a boosted nuclear weapon.

As is the case with most boosted weapons, the fusion reactions usually do not contribute either directly or substantially to the warhead yield, and only enhance the fission process by neutron multiplication. This enhancement, however, is considerable: boosted yield can be five to ten times unboosted yield.

The amount of thermonuclear fuel has a large effect on the overall fission yield. It provides a sudden injection of large numbers of high-energy 14 MeV neutrons into the core. The total energy-production rate is abruptly increased by the large numbers of fresh neutrons and by the effect of their high energy, which produces a correspondingly high number of new neutrons per fission -- on the average, 4.9 neutrons vs. an average of 2.9 neutrons produced by fissions caused by fission-energy-spectrum neutrons.

For a typical boosted primary, each D-T 14 MeV boost neutron produces about 0.6 direct fissions in the compressed core, and considering succeeding neutron chains, leads to five to 10 fissions. The boost process can increase the overall primary yield by a factor of three to five, which means it has a major determining effect on total yield even though the total thermonuclear boost fusion yield is insignificant.¹⁰⁴

Boosted weapons are almost always initiated by external neutron sources. The replacement of core-internal fission initiators by core-external neutron generators is described in Volume II.

Among relatively little-understood phenomena within nuclear weapons, boosting is paramount. The physical process that is by far the most difficult to predict reliably in nuclear

¹⁰³"The Role of Hydronuclear Tests and Other Low-Yield Nuclear Explosions and Their Status Under a Comprehensive Test Ban," Thomas A. Cochran and Christopher E. Paine, Natural Resources Defense Council, Washington, D.C., March 1995, p. 6.

¹⁰⁴Rosengren, RDA-TR-122100-001-Rev. 1, p. 150.

weapon operation is boosting. Even after 40 years boost physics are not completely understood and the weapons labs at Los Alamos and Livermore have embarked on a continuing low-level program to study fusion reactions in the centers of imploding fission cores.¹⁰⁵ Experience has shown that the D-T ignition and burn are very sensitively dependent upon the conditions of temperature, density, and admixture of foreign substances in the thermonuclear material.¹⁰⁶

Boosting has been described as "a very subtle process"¹⁰⁷ in that a variety of subtle and sensitive effects can determine whether or not there will be successful boosting. The details on intermixing of materials into the boost gas are not known; hence, the required detailed behavior of those materials and other materials involved in the implosion of the boost cavity are not known and cannot be specified.¹⁰⁸

Nuclear weapon scientists have never been able to directly calculate and predict boost performance from first principles. Instead, they have had to rely on a mixture of theory and empirical knowledge. There are basic uncertainties about the conditions that exist during boosting.

Because of the limitations on experimental information and because of the complex nature of the boost process, there is no reliable computer program modeling of boosting processes. It has usually been modeled with simplified treatments, with various adjustment to provide agreement with past test results for closely-related device designs.

There has been little attempt to model the boosting process from first principles, and weapon designers are still hard pressed to reliably estimate the performance of any new

¹⁰⁵UCRL-53822, pp. 11, 32, 34, 35.

¹⁰⁶Rosengren, RDA-TR-122100-001-Rev. 1, p. 150. In a public address at the Massachusetts Institute of Technology on November 14, 1990, Dr. Hans Bethe stated that "namely, you need x kilotons to make the booster work; if you have x-1 kilotons in the primary explosion, the booster just doesn't work, and then nothing will work" (the latter part of his statement refers to the performance of subsequent fusion stages).

¹⁰⁷"An Exchange on Stockpile Confidence," correspondence by John D. Immelle, Paul S. Brown, and Steve Fetter, INTERNATIONAL SECURITY, Volume 13 No. 1, Summer 1988, p. 200.

¹⁰⁸Rosengren, RDA-TR-122100-001-Rev. 1, p. 93.

boosted design that varies significantly from previously-tested configurations.¹⁰⁹

Summary

These are the physical and design principles used in U.S. fission and fusion explosives. They have led to the development of weapons a million times more powerful than the largest conventional high explosive bombs used in World War II. Volume II describes how fission principles were efficiently implemented in postwar fission warheads; subsequent volumes explain how fusion principles were incorporated successfully and efficiently into postwar U.S. thermonuclear weapons.

¹⁰⁹Rosengren, RDA-TR-122100-001-Rev. 1, pp. 155, 156.

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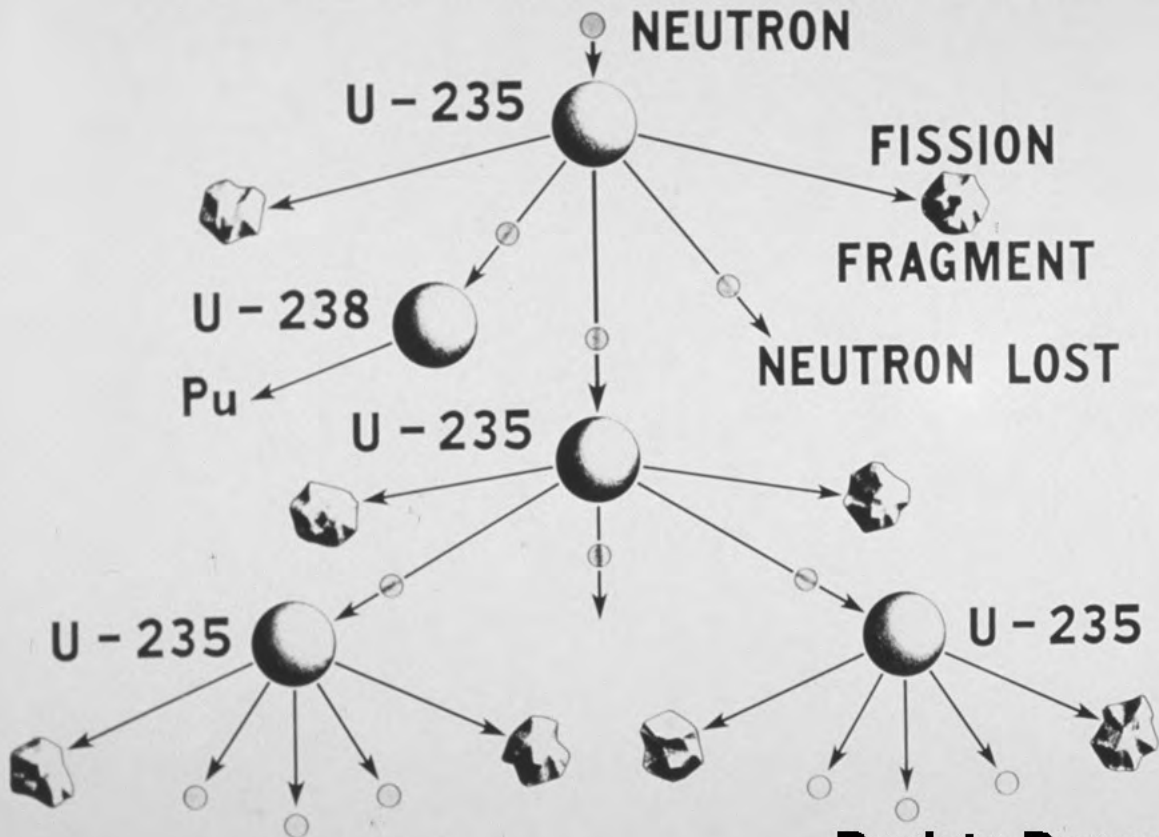
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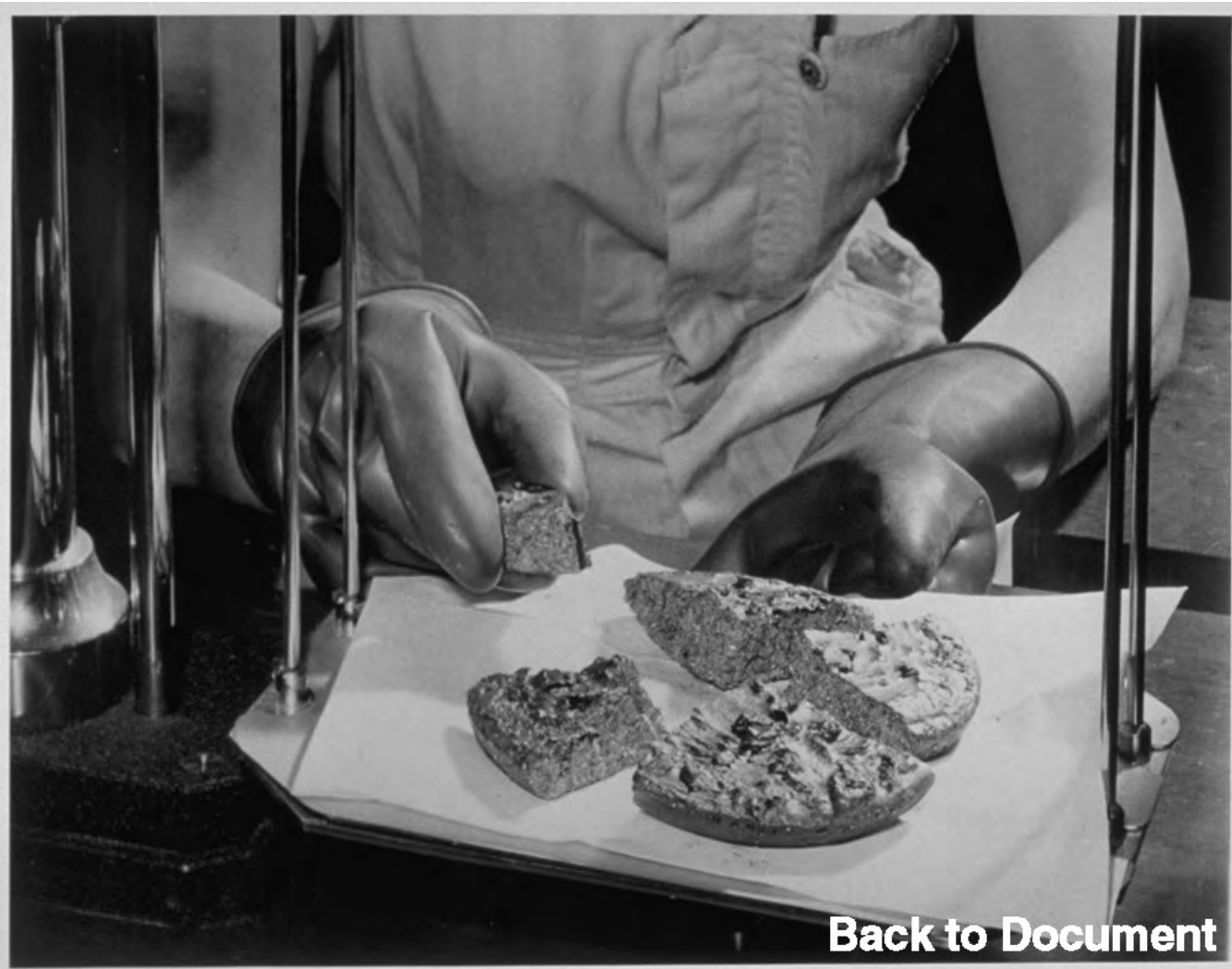
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